

Equity Research Prompt Library

LaTeX Translation and Dependency Map

financial-services-plugins repository

Generated on 2026-03-05 09:31

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1 Repository Structure Overview

1.1 Folder Tree

Root directory: equity-research

- commands
- skills
 - commands/catalysts.md
 - commands/earnings-preview.md
 - commands/earnings.md
 - commands/initiate.md
 - commands/model-update.md
 - commands/morning-note.md
 - commands/screen.md
 - commands/sector.md
 - commands/thesis.md
 - skills/catalyst-calendar
 - skills/earnings-analysis
 - skills/earnings-preview
 - skills/idea-generation
 - skills/initiating-coverage
 - skills/model-update
 - skills/morning-note
 - skills/sector-overview
 - skills/thesis-tracker
 - skills/catalyst-calendar/SKILL.md
 - skills/earnings-analysis/references
 - skills/earnings-analysis/SKILL.md
 - skills/earnings-preview/SKILL.md
 - skills/idea-generation/SKILL.md
 - skills/initiating-coverage/assets
 - skills/initiating-coverage/references
 - skills/initiating-coverage/SKILL.md
 - skills/model-update/SKILL.md
 - skills/morning-note/SKILL.md

- skills/sector-overview/SKILL.md
- skills/thesis-tracker/SKILL.md
- skills/earnings-analysis/references/best-practices.md
- skills/earnings-analysis/references/report-structure.md
- skills/earnings-analysis/references/workflow.md
- skills/initiating-coverage/assets/quality-checklist.md
- skills/initiating-coverage/assets/report-template.md
- skills/initiating-coverage/references/task1-company-research.md
- skills/initiating-coverage/references/task2-financial-modeling.md
- skills/initiating-coverage/references/task3-valuation.md
- skills/initiating-coverage/references/task4-chart-generation.md
- skills/initiating-coverage/references/task5-report-assembly.md
- skills/initiating-coverage/references/valuation-methodologies.md

2 Prompt Dependency Structure

2.1 Command-to-Skill Dependencies

Command Prompt	Loaded Skill Prompt
commands/ catalysts.md	skills/ catalyst-calendar/ SKILL.md
commands/ earnings-preview.md	skills/ earnings-preview/ SKILL.md
commands/ earnings.md	skills/ earnings-analysis/ SKILL.md
commands/ initiate.md	skills/ initiating-coverage/ SKILL.md
commands/ model-update.md	skills/ model-update/ SKILL.md
commands/ morning-note.md	skills/ morning-note/ SKILL.md
commands/ screen.md	skills/ idea-generation/ SKILL.md
commands/ sector.md	skills/ sector-overview/ SKILL.md
commands/ thesis.md	skills/ thesis-tracker/ SKILL.md

2.2 Skill-to-Resource Dependencies

Skill Prompt	Referenced Prompt File
skills/ earnings-analysis/ SKILL.md	skills/ earnings-analysis/ references/ best-practices.md
skills/ earnings-analysis/ SKILL.md	skills/ earnings-analysis/ references/ report-structure.md
skills/ earnings-analysis/ SKILL.md	skills/ earnings-analysis/ references/ workflow.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ assets/ quality-checklist.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ assets/ report-template.md

skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ references/ task1-company-research.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ references/ task2-financial-modeling.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ references/ task3-valuation.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ references/ task4-chart-generation.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ references/ task5-report-assembly.md
skills/ initiating-coverage/ SKILL.md	skills/ initiating-coverage/ references/ valuation-methodologies.md

2.3 Initiating Coverage Task Dependency Chain

Task	Name	Prerequisites	Output
Task 1	Company Research	Company name/ ticker	6-8K word document
Task 2	Financial Modeling	10-K or financials access	Excel model (6 tabs)
Task 3	Valuation Analysis	Financial model (Task 2)	Valuation + price target
Task 4	Chart Generation	Tasks 1, 2, 3 + external data	25-35 PNG/ JPG charts
Task 5	Report Assembly	ALL previous tasks (1-4)	30-50 page DOCX report

3 Translated Prompt Corpus

This section translates every markdown prompt under the equity-research directory into LaTeX-native structure, preserving hierarchy and dependency flow.

3.1 Directory: commands

3.2 Command: catalysts.md

File Metadata

Field	Value
description	View or update the catalyst calendar
argument-hint	[timeframe, e.g. 'next 2 weeks']

Load the *catalyst-calendar* skill to build or review upcoming catalysts across the coverage universe.

If a timeframe is provided, use it. Otherwise default to the next 2 weeks.

3.3 Command: earnings-preview.md

File Metadata

Field	Value
description	Build a pre-earnings preview with scenarios
argument-hint	[company ticker]

Load the *earnings-preview* skill and build a pre-earnings analysis with consensus estimates, key metrics to watch, and bull/base/bear scenarios.

If a ticker is provided, use it. Otherwise ask the user which company is reporting.

3.4 Command: earnings.md

File Metadata

Field	Value
description	Analyze quarterly earnings and create an earnings update report
argument-hint	[company name or ticker] [quarter, e.g. Q3 2024]

Earnings Analysis Command

Create a professional equity research earnings update report analyzing quarterly results.

Workflow

Step 1: Gather Information

Parse the input for:

- Company name or ticker
- Quarter (e.g., Q3 2024, Q2 FY25)

If not provided, ask:

- “What company’s earnings would you like to analyze?”
- “Which quarter? (e.g., Q3 2024)”

Step 2: Verify Timeliness

CRITICAL: Before proceeding, verify you have the latest data:

1. Search for “[Company] latest earnings results [current year]”
2. Verify the earnings release is within the last 3 months
3. Confirm transcript date matches release date

If data is stale, inform the user and search for the latest.

Step 3: Load Earnings Analysis Skill

Use *skill*: “*earnings-analysis*” to create the report:

1. **Data Collection** (search for latest):
 - Earnings release (press release)
 - 10-Q filing from SEC EDGAR
 - Earnings call transcript
 - Investor presentation/supplemental materials
 - Consensus estimates (Bloomberg/FactSet)
2. **Beat/Miss Analysis:**
 - Revenue vs consensus: Beat/Miss by \$X or X%
 - EPS vs consensus: Beat/Miss by \$X or X%
 - Key segment performance vs expectations
 - Explain WHY results differed
3. **Key Metrics Analysis:**
 - Revenue breakdown by segment/geography
 - Margin trends (gross, operating, net)

- Guidance: raised/maintained/lowered
- Updated forward estimates

4. **Generate Charts** (8-12):

- Quarterly revenue progression
- Quarterly EPS progression
- Margin trends
- Revenue by segment
- Beat/miss summary
- Estimate revisions
- Valuation charts

5. **Create Report** (8-12 pages):

- Page 1: Summary with rating and price target
- Pages 2-3: Detailed results analysis
- Pages 4-5: Key metrics & guidance
- Pages 6-7: Updated investment thesis
- Pages 8-10: Valuation & estimates
- Sources section with clickable hyperlinks

Step 4: Deliver Output

Provide:

1. **DOCX report** - 8-12 page earnings update
2. **Summary** highlighting:
 - Beat/miss on key metrics
 - Guidance changes
 - Thesis impact (positive/negative/neutral)

Report Structure Reference

Structured Template

- PAGE 1: EARNINGS SUMMARY
- [Company] Q3 2024 Earnings Update
- Rating: BUY | Price Target: \$XXX (from \$XXX)
- KEY TAKEAWAYS
 - Revenue beat by X% on strong [segment] performance
 - EPS beat by \$X.XX driven by margin expansion

- • FY guidance raised to \$X.XX-\$X.XX (from \$X.XX-\$X.XX)
- • Thesis intact; maintain BUY rating
- RESULTS SNAPSHOT
- Metric Actual Consensus Beat/Miss
- Revenue \$X.XXB \$X.XXB +X.X%
- EPS \$X.XX \$X.XX +\$X.XX
- Gross Margin XX.X% XX.X% +XXbps
- PAGES 2-3: DETAILED RESULTS
- - Segment-by-segment analysis
- - Geographic breakdown
- - Key drivers of beat/miss
- PAGES 4-5: METRICS & GUIDANCE
- - Margin analysis
- - Full-year guidance comparison
- - Updated quarterly estimates
- PAGES 6-7: THESIS UPDATE
- - What's changed
- - Risks and catalysts
- - Investment recommendation
- PAGES 8-10: VALUATION
- - Updated DCF/comps if material
- - Price target justification
- - Scenario analysis
- SOURCES SECTION (with clickable hyperlinks):
- - Earnings Release: [hyperlink]
- - Form 10-Q: [EDGAR hyperlink]
- - Earnings Call Transcript: [hyperlink]
- - Consensus estimates: Bloomberg as of [date]

Quality Checklist

Before delivery:

- ☐ Earnings data is from latest quarter (not stale)

- ☐ Beat/miss quantified with specific numbers
 - ☐ All charts embedded (8-12 total)
 - ☐ Sources section with clickable hyperlinks
 - ☐ Every figure/table has source citation
 - ☐ Guidance changes clearly documented
 - ☐ Rating and price target stated upfront
 - ☐ 8-12 pages, 3,000-5,000 words
-

3.5 Command: initiate.md

File Metadata

Field	Value
description	Create an initiating coverage report
argument-hint	[company ticker]

Load the *initiating-coverage* skill and begin the 5-task workflow to create an institutional-quality initiation report. If a ticker is provided, use it. Otherwise ask the user which company to initiate on.

3.6 Command: model-update.md

File Metadata

Field	Value
description	Update a financial model with new data
argument-hint	[company ticker]

Load the *model-update* skill and plug in new earnings, guidance, or revised assumptions. If a ticker is provided, use it. Otherwise ask the user which model to update and what changed.

3.7 Command: morning-note.md

File Metadata

Field	Value
description	Draft a morning meeting note

argument-hint

Load the *morning-note* skill and draft a concise morning note covering overnight developments, earnings reactions, and trade ideas across the coverage universe.

3.8 Command: screen.md

File Metadata

Field	Value
description	Run a stock screen or generate investment ideas
argument-hint	[screen criteria, e.g. 'undervalued midcap tech']

Load the *idea-generation* skill and run quantitative screens or thematic sweeps to surface new investment ideas. If criteria are provided, use them. Otherwise ask the user what they're looking for (long/short, sector, style, theme).

3.9 Command: sector.md

File Metadata

Field	Value
description	Create a sector overview report
argument-hint	[sector or industry]

Load the *sector-overview* skill and create an industry landscape report covering market sizing, competitive dynamics, and investment implications.

If a sector is provided, use it. Otherwise ask the user which industry to cover.

3.10 Command: thesis.md

File Metadata

Field	Value
description	Create or update an investment thesis
argument-hint	[company ticker]

Load the *thesis-tracker* skill to create a new thesis or update an existing one with new data points.

If a ticker is provided, use it. Otherwise ask the user which position to review.

3.11 Directory: skills

3.12 Skill: catalyst-calendar

Catalyst Calendar

description: Build and maintain a calendar of upcoming catalysts across a coverage universe — earnings dates, conferences, product launches, regulatory decisions, and macro events. Helps prioritize attention and position ahead of events. Triggers on “catalyst calendar”, “upcoming events”, “what’s coming up”, “earnings calendar”, “event calendar”, or “catalyst tracker”.

Workflow

Step 1: Define Coverage Universe

- List of companies to track (tickers or names)
- Sector / industry focus
- Include macro events? (Fed meetings, economic data, regulatory deadlines)
- Time horizon (next 2 weeks, month, quarter)

Step 2: Gather Catalysts

For each company, identify upcoming events:

Earnings & Financial Events

- Quarterly earnings date and time (pre/post market)
- Annual shareholder meeting
- Investor day / analyst day
- Capital markets day
- Debt maturity / refinancing dates

Corporate Events

- Product launches or announcements
- FDA approvals / regulatory decisions
- Contract renewals or expirations
- M&A milestones (close dates, regulatory approvals)
- Management transitions
- Insider trading windows (lockup expirations)

Industry Events

- Major conferences (dates, which companies presenting)

- Trade shows and expos
- Regulatory comment periods or rulings
- Industry data releases (monthly sales, traffic, etc.)

Macro Events

- Fed meetings (FOMC dates)
- Jobs report, CPI, GDP releases
- Central bank decisions (ECB, BOJ, etc.)
- Geopolitical events with market impact

Step 3: Calendar View

Date	Event	Company/ Sector	Type	Impact (H/ M/ L)	Our Position- ing	Notes
			Earnings/ Corp/ Industry/ Macro		Long/ Short/ Neutral	

Step 4: Weekly Preview

Each week, generate a forward-looking summary:

This Week's Key Events:

1. [Day]: [Company] Q[X] earnings — consensus [\$X EPS], our estimate [\$X], key focus: [metric]
2. [Day]: [Event] — why it matters for [stocks]
3. [Day]: [Macro release] — expectations and positioning

Next Week Preview:

- Early heads-up on important events coming

Position Implications:

- Events that could move specific positions
- Any pre-positioning recommended
- Risk management ahead of binary events

Step 5: Output

- Excel workbook with calendar view and sortable columns
- Weekly preview email/note (markdown)
- Optional: integration with Google Calendar

Important Notes

- Earnings dates shift — verify against company IR pages and Bloomberg/FactSet closer to the date
 - Pre-announce risk: track companies with a history of pre-announcing (positive or negative)
 - Conference attendance lists are valuable — which companies are presenting and which are conspicuously absent?
 - Some catalysts are recurring (monthly industry data) — build a template and auto-populate
 - Color-code by impact level: Red = high impact, Yellow = moderate, Green = routine
 - Archive past catalysts with the actual outcome — builds pattern recognition over time
-

3.13 Skill: earnings-analysis

File Metadata

Field	Value
name	earnings-analysis
description	Create professional equity research earnings update reports (8-12 pages, 3,000-5,000 words) analyzing quarterly results for companies already under coverage. Fast-turnaround format focusing on beat/ miss analysis, key metrics, updated estimates, and revised thesis. Includes 1-3 summary tables and 8-12 charts. Use when user requests “earnings update”, “quarterly update”, “earnings analysis”, “Q1/ Q2/ Q3/ Q4 results”, or post-earnings report.

Equity Research Earnings Update

Create professional **EARNINGS UPDATE REPORTS** analyzing quarterly results for companies already under coverage, following institutional standards (JPMorgan, Goldman Sachs, Morgan Stanley format).

Key Characteristics:

- **Length:** 8-12 pages
- **Word Count:** 3,000-5,000 words
- **Tables:** 1-3 summary tables (NOT comprehensive)
- **Figures:** 8-12 charts
- **Turnaround:** 1-2 days (within 24-48 hours of earnings)
- **Audience:** Clients already familiar with the company
- **Focus:** What’s NEW - beat/miss, updated estimates, thesis impact
- **Font:** Times New Roman throughout (unless user specifies otherwise)

When to Use

Use when the user requests:

- “Create an earnings update for [Company] Q3 2024”
- “Analyze [Company]’s quarterly results”
- “Post-earnings report for [Company]”
- “Q1/Q2/Q3/Q4 update for [Company]”

Do NOT use if:

- User requests “initiation report” → Use different skill
- User requests “flash note” or “quick take” → Different format
- Company is not already covered → Need initiation first

Critical Requirements

1. Speed & Timeliness

- Publish within 24-48 hours of earnings release
- Focus on NEW information only
- Don’t rehash company background extensively

2. Beat/Miss Analysis

- Lead with whether company beat or missed estimates
- Quantify variances (e.g., “Revenue beat by \$120M or 3%”)
- Explain WHY results differed from expectations

3. Summary Format

- Keep tables to 1-3 (summary only, not comprehensive)
- No full P&L/Cash Flow/Balance Sheet (just key metrics)
- Assume reader has seen initiation report

4. Citations & Source Attribution ★★ ★ MANDATORY

CRITICAL: Properly cite all data with SPECIFIC sources and CLICKABLE HYPERLINKS.

Include specific citations WITH CLICKABLE LINKS in every figure and table:

Structured Template

- Source: Q3 2024 10-Q filed November 8, 2024; Company earnings release

- [Hyperlink “10-Q” to: <https://www.sec.gov/cgi-bin/viewer?accession=...>]
- [Hyperlink “earnings release” to: <https://investor.company.com/news/q3-2024>]

HOW HYPERLINKS SHOULD APPEAR IN WORD:

- Document names appear as blue, underlined clickable links
- Reader can Ctrl+Click to open source directly
- Not plain text URLs - formatted hyperlinks with display text

REQUIRED SOURCES LIST:

Cite in every earnings update:

- ☒ Earnings release (with date and URL)
- ☒ 10-Q filing (with filing date and EDGAR link)
- ☒ Earnings call transcript (with date)
- ☒ Investor presentation/supplemental materials (if available)
- ☒ Consensus estimates source (Bloomberg/FactSet/etc. with date)
- ☒ Prior guidance (from previous quarter’s materials)

REFERENCE SECTION WITH CLICKABLE HYPERLINKS:

Include “Sources” section at end of report:

Structured Template

- SOURCES & REFERENCES
- Earnings Materials (Q3 2024):
 - Earnings Release (November 7, 2024)
 - [Hyperlink entire line to: <https://investor.company.com/news/q3-2024-earnings>]
 - Form 10-Q (Filed November 8, 2024)
 - [Hyperlink to: <https://www.sec.gov/cgi-bin/viewer?accession=...>]
 - Earnings Call Transcript (November 7, 2024)
 - [Hyperlink to: <https://seekingalpha.com/article/...>]
 - Investor Presentation (November 7, 2024)
 - [Hyperlink to: <https://investor.company.com/presentations/q3-2024.pdf>]

VERIFICATION CHECKLIST:

- ☐ Every figure has source with specific document and date
- ☐ Every table has source with document reference
- ☐ Beat/miss analysis cites consensus source with date

- ☐ Guidance changes cite current and prior guidance sources
- ☐ Key statistics have footnotes
- ☐ Sources section lists all materials with URLs
- ☐ ALL URLs are CLICKABLE HYPERLINKS (not plain text)
- ☐ All SEC filings hyperlinked to EDGAR viewer

5. Updated Estimates

- Update forward estimates based on results
- Show old vs. new estimates clearly
- Explain what changed and why

High-Level Workflow

The earnings update process follows 5 phases:

Phase 1: Data Collection (30-60 minutes)

 **CRITICAL: TRAINING DATA IS OUTDATED** 

BEFORE STARTING - COMPLETE THESE 4 STEPS IN ORDER:

1. **CHECK TODAY'S DATE** - Write down the current date
2. **SEARCH FOR LATEST** - Use web search: "[Company] latest earnings results"
3. **VERIFY THE DATE** - Confirm earnings release is within last 3 months
4. **CHECK TRANSCRIPT DATE** - Verify transcript date matches release date

COMMON MISTAKE: Using outdated earnings calls from training data instead of searching for the latest.

REQUIREMENTS:

- ☒ Search for latest earnings - do NOT rely on training data
- ☒ Write down today's date and the release date found
- ☒ Verify release date is within 3 months of today
- ☒ Verify transcript date matches release date
- ☒ If dates don't match or are old (>3 months), search again

See @@TOKEN0@@ for detailed search procedures and verification steps.

Phase 2: Analysis (2-3 hours)

- Beat/miss analysis for each key metric
- Segment/geographic/product breakdown

- Margin and guidance analysis
- Update financial model and estimates

See @@TOKEN0@@ for detailed analysis framework.

Phase 3: Chart Generation (1-2 hours)

Create 8-12 charts focusing on quarterly trends and what's new:

- Quarterly revenue progression
- Quarterly EPS progression
- Quarterly margin trends
- Revenue by segment/geography
- Key operating metrics
- Beat/miss summary
- Estimate revisions
- Valuation charts

See @@TOKEN0@@ for chart specifications.

Phase 4: Report Creation (2-3 hours)

Create 8-12 page DOCX report with specific structure.

See @@TOKEN0@@ for complete page-by-page templates and formatting requirements.

High-level structure:

- Page 1: Earnings summary with rating and price target
- Pages 2-3: Detailed results analysis
- Pages 4-5: Key metrics & guidance
- Pages 6-7: Updated investment thesis
- Pages 8-10: Valuation & estimates
- Pages 11-12: Appendix (optional)

Phase 5: Quality Check & Delivery (30 minutes)

Verify content, formatting, accuracy, and timeliness before delivery.

See @@TOKEN0@@ for quality checklist and common mistakes to avoid.

Output Specification

Primary Deliverable: DOCX report (8-12 pages) **File Name:** *[Company]_Q[Quarter]_[Year]_Earnings_Update.docx* **Example:** *Nike_Q2_FY24_Earnings_Update.docx*

Contents:

- Page 1: Summary with rating, price target, key takeaways
- Pages 2-3: Detailed results analysis
- Pages 4-5: Key metrics and guidance
- Pages 6-7: Updated thesis assessment
- Pages 8-10: Valuation and estimates
- Pages 11-12: Appendix (optional)
- 8-12 embedded charts
- 1-3 summary tables
- Complete sources section with clickable hyperlinks

Optional Deliverable: XLS model update (optional for earnings updates)

Key Differences from Initiation Report

Aspect	Earnings Update	Initiation Report
Length	8-12 pages	30-50 pages
Words	3,000-5,000	10,000-15,000
Tables	1-3 summary	12-20 comprehensive
Figures	8-12	25-35
Turnaround	1-2 days	3-6 weeks
Scope	Quarterly results	Complete company
Focus	What's NEW	Everything
Company Background	Brief mention	6-10 pages
XLS Model	Optional	Required

Resources

references/workflow.md

Detailed Phase 1-5 instructions with step-by-step procedures for data collection, analysis, chart generation, and report creation.

references/report-structure.md

Complete page-by-page templates, table formats, and formatting requirements for the DOCX report.

references/best-practices.md

Examples of good/bad headlines, tips for success, common mistakes to avoid, and comprehensive quality checklist.

Dependencies

Required:

- Python (matplotlib, pandas, seaborn) for chart generation
- DOCX skill for report creation

Optional:

- XLS skill for model updates (not required for earnings updates)
-

3.14 Skill Resource: skills/earnings-analysis/references/best-practices.md

Best Practices, Examples, and Quality Guidelines

This document provides examples, tips for success, common mistakes to avoid, and comprehensive quality checklists.

Example Headlines

Good Earnings Update Headlines:

- “Nike Q2 FY24: DTC Strength Offsets Wholesale Weakness - Maintaining OW, PT \$95”
- “Tesla Q3’24: Cybertruck Ramp Ahead of Plan - Raising Estimates, PT to \$285”
- “LVMH Q4’24: Fashion & Leather Resilient, Wines Weak - In-Line, Reiterating Buy”
- “Apple Q1 FY24: Services Beat, iPhone Miss - Mixed Quarter, Lowering PT to \$185”

Bad Headlines (Avoid):

- “Nike Quarterly Update” (too generic, no takeaway)
- “Company Reports Earnings” (states obvious, no analysis)
- “Q3 Results Analysis” (no company name, no view)

Tips for Success

1. **Speed matters:** Published 24-48hrs post-earnings, not days later
2. **Lead with conclusion:** Beat or miss? Up or down estimates?
3. **Quantify everything:** “Strong” means nothing, “\$150M beat on \$1.2B revenue” is clear

4. **Focus on drivers:** Don't just say "revenue beat", explain WHY
5. **Show the work:** Old estimates → New estimates with reasons
6. **Update price target if material:** If estimates change >5%, usually PT changes too
7. **Acknowledge the call:** Reference management commentary, don't just analyze the press release
8. **Compare to peers:** If similar companies reported, note relative performance
9. **Be concise:** This is NOT a comprehensive report, stay focused on quarterly results
10. **Chart the trends:** Quarterly progression charts are most valuable

Common Mistakes to Avoid

- ✗ **Too comprehensive:** Don't write an initiation-length report for quarterly results
- ✗ **Missing beat/miss:** Lead with whether results beat or missed expectations
- ✗ **Not updating estimates:** Must provide updated forward estimates
- ✗ **Vague language:** "Strong performance" without quantification
- ✗ **Ignoring guidance:** If company guides, analyze it thoroughly
- ✗ **Too slow:** Publishing 5+ days after earnings loses relevance
- ✗ **Rehashing basics:** Don't spend 3 pages explaining what the company does
- ✗ **Missing price target update:** If estimates changed materially, PT should too
- ✗ **No investment impact:** Must connect results to thesis and rating
- ✗ **Missing citations:** Every number needs a source with clickable hyperlinks
- ✗ **Plain text URLs:** All URLs must be formatted as clickable hyperlinks

Comprehensive Quality Control Checklist

Before delivering earnings update, verify all items below:

Content & Analysis Checklist

Beat/Miss Analysis:

- ☐ Beat/miss analysis leads the report
- ☐ Specific variances quantified (e.g., "beat by \$120M or 3%")
- ☐ Explanation of WHY results differed from expectations
- ☐ Analysis of each key metric (revenue, EPS, margins, etc.)

Metrics & Performance:

- ☐ All key metrics discussed with YoY comparisons
- ☐ QoQ comparisons included where relevant

- ☐ Segment/geographic/product breakdowns provided
- ☐ Operating metrics analyzed (customers, ARPU, units, etc.)

Guidance & Estimates:

- ☐ Guidance changes analyzed and quantified (if provided)
- ☐ If no guidance, this is explicitly noted
- ☐ Updated estimates provided for current year
- ☐ Updated estimates provided for next year
- ☐ Old vs. new estimates clearly shown
- ☐ Explanation of what changed and why

Valuation & Rating:

- ☐ Price target updated (if warranted by results)
- ☐ If PT unchanged, explicitly maintained
- ☐ Valuation methodology explained
- ☐ Rating confirmed or changed with clear rationale
- ☐ Investment thesis assessed and updated if needed

Format & Length Checklist

Overall Structure:

- ☐ Report is 8-12 pages (not shorter, not longer)
- ☐ Page 1 has earnings summary format
- ☐ Page 1 has “EARNINGS UPDATE” in title (NOT “Initiating Coverage”)
- ☐ Event-driven title (e.g., “Strong Q3 Results...”)

Tables:

- ☐ 1-3 summary tables included (NOT comprehensive tables)
- ☐ All tables have clear column headers
- ☐ All tables have header row shading
- ☐ All tables have source lines at bottom
- ☐ Estimates table shows old vs. new with change column

Charts:

- ☐ 8-12 charts embedded throughout document
- ☐ All charts have “Figure X - [Title]” caption above

- ☐ All charts have “Source: [Source]” line below
- ☐ Charts focus on quarterly trends
- ☐ Charts highlight changes (beat/miss, revisions)
- ☐ Charts use professional styling

Citations & Sources Checklist ★★★ **MANDATORY**

Figure & Table Citations:

- ☐ Every figure has specific source with document name and date
- ☐ Every table has specific source with document reference
- ☐ Source citations include page numbers or slide numbers where applicable

Beat/Miss Citations:

- ☐ Beat/miss analysis cites consensus source (Bloomberg, FactSet, etc.)
- ☐ Consensus source includes “as of” date (pre-earnings close)
- ☐ Company reported results cited to earnings release or 10-Q

Guidance Citations:

- ☐ Current guidance cited to earnings call transcript or release
- ☐ Prior guidance cited to previous quarter’s materials
- ☐ Both current and prior guidance sources hyperlinked

Statistics & Metrics:

- ☐ Key statistics have footnotes with sources
- ☐ Footnotes reference specific documents and page/slide numbers
- ☐ Management quotes cite speaker name and source document

Hyperlinks: ★★★ **CRITICAL**

- ☐ ALL URLs are CLICKABLE HYPERLINKS (not plain text)
- ☐ Hyperlinks formatted with meaningful display text
- ☐ Blue, underlined hyperlink formatting in Word document
- ☐ Hyperlinks tested and working (Ctrl+Click opens correct page)
- ☐ All SEC filings hyperlinked to EDGAR viewer
- ☐ All earnings materials hyperlinked (release, transcript, presentation)
- ☐ Prior quarter materials hyperlinked for comparison
- ☐ No raw URLs displayed anywhere in document

Sources Section:

- ☐ “Sources & References” section included at end of report
- ☐ Section lists all earnings materials with dates
- ☐ All materials have clickable hyperlinks
- ☐ Consensus data sources listed (even if no link for subscription data)
- ☐ Prior period references included

Accuracy Checklist**Numerical Accuracy:**

- ☐ Numbers match company’s reported results exactly
- ☐ Math checks out in all calculations
- ☐ Estimate changes calculated correctly
- ☐ Valuation math is accurate
- ☐ Charts match text descriptions

Factual Accuracy:

- ☐ No typos in ticker symbol
- ☐ No typos in company name
- ☐ Dates are current and accurate
- ☐ Quarter/year references are correct
- ☐ Year notation correct (A for actual, E for estimate)

Timeliness Checklist**Publication Timing:**

- ☐ Report published within 24-48 hours of earnings release
- ☐ If later than 48 hours, acknowledged as “delayed reaction”
- ☒ **VERIFIED all data is from LATEST quarter by searching for recent earnings**
- ☒ **Did NOT rely on knowledge cutoff - actively searched for current data**
- ☐ Consensus estimates are pre-earnings (not post-earnings)
- ☐ No outdated information included
- ☐ Earnings release date is within last 1-3 months (not 6+ months old)

Writing Style Checklist

Clarity & Directness:

- ☐ Lead with numbers (“Revenue grew 15% to \$1.2B” not “Strong revenue”)
- ☐ Use “vs.” not “versus”
- ☐ Be direct and concise throughout
- ☐ Focus on what’s NEW (not rehashing company basics)
- ☐ Avoid vague language (“strong performance” without quantification)

Professional Standards:

- ☐ Institutional tone maintained
- ☐ Consistent terminology throughout
- ☐ No informal language
- ☐ Proper financial notation

Pre-Delivery Final Check

Run through this quick final check before sending report to user:

5-Minute Final Review:

1. **Page 1:** Rating clear? Price target updated? Key takeaways compelling?
2. **Numbers:** Do reported results match company’s press release exactly?
3. **Citations:** Spot check 3-4 figures/tables - all have sources with clickable hyperlinks?
4. **Estimates:** Old vs. new clearly shown? Changes explained?
5. **Charts:** All 8-12 embedded? All numbered and captioned?
6. **Length:** Is it 8-12 pages (not 6, not 15)?
7. **Hyperlinks:** Test 3-4 hyperlinks - do they work with Ctrl+Click?
8. **Timeliness:** Is this being published within 48 hours of earnings?

If all items check out, the report is ready for delivery.

Summary Delivery Format

When delivering the completed report to the user, provide this summary:

Structured Template

- [Company] Q[X] [Year] Earnings Update Complete
- Results: [BEAT / INLINE / MISS]

- - Revenue: \$X.XB ([beat/missed] by \$XXM or X%)
 - - EPS: \$X.XX ([beat/missed] by \$X.XX)
 - Key Takeaways:
 - ■ [Takeaway 1]
 - ■ [Takeaway 2]
 - ■ [Takeaway 3]
 - Updated Estimates:
 - - FY[Year]E Revenue: \$XX.XB (prior: \$XX.XB, [+/-]X%)
 - - FY[Year]E EPS: \$X.XX (prior: \$X.XX, [+/-]X%)
 - Rating: [MAINTAINED / RAISED / LOWERED] [RATING]
 - Price Target: \$XXX (prior: \$XXX) - [+/-]XX% upside
 - Deliverables:
 - ✓ 8-12 page earnings update report (DOCX)
 - ✓ 8-12 embedded charts
 - ✓ Updated estimates with old/new comparison
 - ✓ Complete sources section with clickable hyperlinks
 - ✓ [Optional: Updated XLS financial model]
 - File: [Company]_Q[X]_[Year]_Earnings_Update.docx
-

3.15 Skill Resource: [skills/earnings-analysis/references/report-structure.md](#)

Report Structure and Templates

This document provides complete page-by-page templates and formatting requirements for the earnings update DOCX report.

Complete Report Structure

REPORT STRUCTURE:

PAGE 1: EARNINGS SUMMARY

Top Section - Header:

Structured Template

- [COMPANY NAME] ([TICKER])
- [QUARTER] [YEAR] EARNINGS UPDATE

- [Current Date]
- Rating: [MAINTAIN/RAISE/LOWER] [RATING]
- Price (as of [date]): \$XX.XX
- Price Target: [OLD → NEW if changed, or MAINTAIN \$XXX]

Top Section - Quick Summary Box:

Structured Template

- EARNINGS SUMMARY
- Q[X] [YEAR] RESULTS: [BEAT / INLINE / MISS]
- Reported Est Variance
- Revenue \$X,XXX \$X,XXX +\$XXX (+X%)
- EPS (Adj) \$X.XX \$X.XX +\$X.XX (+X%)
- Key Takeaways:
 - ■ [Takeaway 1 - one sentence]
 - ■ [Takeaway 2 - one sentence]
 - ■ [Takeaway 3 - one sentence]

Main Content - Investment Impact (3-4 bullets):

Use ■ character with **bold headers** and paragraph-length explanations:

Structured Template

- ■ **Results beat on strong [segment/geography/product], maintaining positive momentum**
- Q[X] revenue of \$X.XB exceeded our \$X.XB estimate by X% and consensus by X%,
- driven primarily by [specific driver]. [Segment] revenue grew X% YoY (vs. our
- X% estimate), while [segment] grew X% (vs. X% estimate). Management highlighted
- [specific products/initiatives] as key growth drivers and maintained confident
- tone on outlook. The beat demonstrates [thesis point], reinforcing our positive
- view.
- ■ **Margins expanded XXbps YoY despite [headwind], showcasing operational leverage**
- [Detailed margin analysis paragraph...]
- ■ **Guidance raised / maintained / lowered - implies [interpretation]**
- [Detailed guidance analysis paragraph...]
- ■ **Maintaining [RATING] with [raised/unchanged] \$XXX price target**
- [Investment conclusion paragraph...]

Bottom Section - Updated Estimates Table:

Structured Template

- UPDATED FINANCIAL ESTIMATES
 - FY2024E (OLD) FY2024E (NEW) Change FY2025E (NEW)
 - Revenue (\$M) XX,XXX XX,XXX +X% XX,XXX
 - Revenue Growth (%) X.X% X.X% +XXbps X.X%
 - Gross Margin (%) XX.X% XX.X% +XXbps XX.X%
 - EBITDA (\$M) X,XXX X,XXX +X% X,XXX
 - EBITDA Margin (%) XX.X% XX.X% +XXbps XX.X%
 - EPS (Adjusted) (\$) X.XX X.XX +X% X.XX
 - P/E (x) XX.Xx XX.Xx -X% XX.Xx
 - Note: “E” = Estimate. Old estimates from [prior report date].
 - Source: Company data, [Firm Name] estimates.
-

PAGES 2-3: DETAILED RESULTS ANALYSIS

Break down results by:

Revenue Analysis (1 page)

- Total revenue beat/miss explanation
- Segment/geographic/product breakdown
- YoY and sequential trends
- Comparison to guidance (if provided)

Table: Quarterly Revenue Progression

Structured Template

- Q[X-3] Q[X-2] Q[X-1] Q[X] YoY Chg QoQ Chg
- Total Revenue (\$M) X,XXX X,XXX X,XXX X,XXX +X% +X%
- [Segment A] (\$M) XXX XXX XXX XXX +X% +X%
- [Segment B] (\$M) XXX XXX XXX XXX +X% +X%
- [Segment C] (\$M) XXX XXX XXX XXX +X% +X%
- Note: Q[X] = [Quarter] [Year]
- Source: Company reports, [Firm Name] analysis

Profitability Analysis (1 page)

- Gross margin analysis (drivers, trends)

- Operating margin analysis
- Below-the-line items (interest, tax, etc.)
- EPS reconciliation (adjusted vs. GAAP)

Table: Margin Analysis

Structured Template

- Q[X-3] Q[X-2] Q[X-1] Q[X] YoY Chg
- Gross Margin (%) XX.X% XX.X% XX.X% XX.X% +XXbps
- Operating Margin (%) XX.X% XX.X% XX.X% XX.X% +XXbps
- Net Margin (%) XX.X% XX.X% XX.X% XX.X% +XXbps
- Key Drivers:
- + [Positive driver 1]
- + [Positive driver 2]
- - [Negative driver 1]
- - [Negative driver 2]

Embed 2-3 charts on these pages:

- Chart 1: Quarterly revenue progression
 - Chart 2: Quarterly EPS progression
 - Chart 3: Margin trends
-

PAGES 4-5: KEY METRICS & GUIDANCE

Business Metrics (1 page)

- Customer count, ARPU, units, store count, etc.
- Whatever metrics company emphasizes
- Comparison to expectations
- Trends and outlook

Table: Key Operating Metrics

Structured Template

- Q[X-3] Q[X-2] Q[X-1] Q[X] YoY Chg Our Est Var
- [Metric 1] XXX XXX XXX XXX +X% XXX +X%
- [Metric 2] XXX XXX XXX XXX +X% XXX +X%
- [Metric 3] XXX XXX XXX XXX +X% XXX +X%

- Source: Company reports

Guidance & Outlook (1 page)

- What guidance was provided (if any)
- Comparison to prior guidance
- Comparison to Street estimates
- Our assessment of achievability
- Key assumptions

If guidance provided:

Structured Template

- MANAGEMENT GUIDANCE vs. ESTIMATES
- New Guidance Old Guidance Change Street
- FY2024E Revenue \$XX-XXB \$XX-XXB Raised \$XX.XB
- FY2024E EPS \$X.XX-X.XX \$X.XX-X.XX Raised \$X.XX
- Our Take: [Brief assessment of guidance]

Embed 2-3 charts:

- Chart 4: Key metrics trends
 - Chart 5: Guidance vs. Street comparison
 - Chart 6: Revenue by segment/geography
-

PAGES 6-7: UPDATED INVESTMENT THESIS

Thesis Impact Assessment (1-2 pages)

For each key thesis pillar, assess impact of results:

Structured Template

- ■ **Thesis Pillar 1: [Original thesis statement]**
- Status: [STRENGTHENED / UNCHANGED / WEAKENED]
- Q[X] results [supported / challenged] this thesis pillar because [specific evidence from results]. [Detailed analysis of 150-200 words explaining how results impact this specific thesis element.]
- ■ **Thesis Pillar 2: [Original thesis statement]**
- [Similar analysis]

- ■ **Thesis Pillar 3: [Original thesis statement]**
- [Similar analysis]

Risks Update (0.5 pages)

- Any new risks identified?
- Have existing risks been mitigated or worsened?
- Brief assessment

Embed 1-2 charts:

- Chart 7: Valuation vs. historical
 - Chart 8: Estimate revision comparison
-

PAGES 8-10: VALUATION & ESTIMATES

Updated Valuation (1-2 pages)

DCF Update:

Structured Template

- Updated DCF inputs based on Q[X] results:
- - Revenue growth FY24E: X.X% → X.X% (raised/lowered)
- - EBIT margin FY24E: XX.X% → XX.X%
- - Terminal growth: X.X% (unchanged)
- - WACC: X.X% (unchanged)
- Updated DCF fair value: \$XXX (prior: \$XXX)

Comparable Companies:

Structured Template

- [Company] trades at XX.Xx NTM P/E vs. peer median of XX.Xx (-X% discount).
- Given [rationale], we believe [premium/discount/inline] valuation is warranted.

Price Target Methodology:

Structured Template

- Our \$XXX price target (prior: \$XXX) is based on:
- - XX% DCF
- - XX% NTM P/E of XX.Xx (vs. peers at XX.Xx)
- - XX% EV/EBITDA
- Implied upside: +XX% from current price of \$XXX

Updated Estimates Detail

Provide updated estimates for at least current year and next year:

Structured Template

- DETAILED ESTIMATE UPDATES
- FY2024E FY2025E
- Old New Change New Estimate
- Revenue (\$B) XX.X XX.X +X.X% XX.X
- [Segment A] XX.X XX.X +X.X% XX.X
- [Segment B] XX.X XX.X +X.X% XX.X
- Gross Profit (\$B) XX.X XX.X +X.X% XX.X
- Gross Margin (%) XX.X% XX.X% +XXbps XX.X%
- EBITDA (\$B) X.X X.X +X.X% X.X
- EBITDA Margin (%) XX.X% XX.X% +XXbps XX.X%
- Operating Income X.X X.X +X.X% X.X
- Op Margin (%) XX.X% XX.X% +XXbps XX.X%
- Net Income (\$B) X.X X.X +X.X% X.X
- EPS - Adjusted (\$) X.XX X.XX +X.X% X.XX
- EPS - GAAP (\$) X.XX X.XX +X.X% X.XX
- P/E (x) XX.Xx XX.Xx XX.Xx
- EV/EBITDA (x) XX.Xx XX.Xx XX.Xx
- Source: [Firm Name] estimates

Embed 1-2 charts:

- Chart 9: P/E or EV/EBITDA bands
 - Chart 10: Price target walk (old → new)
-

PAGES 11-12: APPENDIX (Optional)

Detailed Quarterly Models (if space allows)

- Income statement detail
- Cash flow highlights
- Balance sheet highlights

Call Transcript Highlights (optional)

- Key Q&A excerpts
- Notable management quotes

Peer Comparison (if peers have reported)

- How results compare to competitors
- Market share implications

Embed final charts:

- Chart 11: Peer comparison
 - Chart 12: Additional supporting charts
-

FORMATTING REQUIREMENTS

1. Page 1 Requirements

- Clear rating (MAINTAIN OUTPERFORM, RAISE TO BUY, etc.)
- Updated price target prominently displayed
- Summary table with old/new estimates
- 3-4 paragraph-length bullets with ■ character

2. All Tables Requirements

- Source line at bottom
- Clear column headers
- Shading for header rows

3. All Charts Requirements

- “Figure X - [Title]” caption above
- “Source: [Source]” line below
- Professional styling

4. Year Notation

- Use A for actual (Q3’24A)
- Use E for estimate (Q4’24E)

5. Writing Style

- Lead with numbers (“Revenue grew 15% to \$1.2B” not “Strong revenue growth”)
- Use “vs.” not “versus”
- Be direct and concise
- Focus on what’s NEW

6. Hyperlink Requirements

- ALL URLs must be clickable hyperlinks in Word
- Blue, underlined text that opens on Ctrl+Click
- Display text meaningful (not raw URL)
- Every source citation should have clickable link where applicable
- No plain text URLs - always format as hyperlinks

Citation Examples for Specific Content

For Beat/Miss Analysis:

Structured Template

- Revenue of \$2.45B beat consensus of \$2.39B by \$60M (2.5%)¹
- ¹ Bloomberg consensus as of market close November 6, 2024; Company earnings release November 7, 2024
- [Hyperlink “earnings release” to: <https://investor.company.com/news/q3-2024-earnings>]

For Guidance:

Structured Template

- Management raised FY2024 revenue guidance to \$9.8-10.0B from prior \$9.5-9.7B²
- ² Q3 2024 Earnings Call, November 7, 2024, CFO prepared remarks
- [Hyperlink “Earnings Call” to: <https://seekingalpha.com/article/...>]
- Prior guidance from Q2 earnings call August 8, 2024
- [Hyperlink “Q2 earnings call” to August transcript]

For Key Metrics:

Structured Template

- Enterprise customers grew 23% YoY to 845, with net revenue retention at 128%³
- ³ Q3 2024 10-Q, page 23

- [Hyperlink “10-Q” to: <https://www.sec.gov/cgi-bin/viewer?accession=...>]
 - Q3 2024 Investor Presentation slide 8
 - [Hyperlink “Investor Presentation” to PDF]
-

3.16 Skill Resource: [skills/earnings-analysis/references/workflow.md](#)

Detailed Workflow for Earnings Updates

This document provides detailed step-by-step instructions for each phase of the earnings update process.

⚠️⚠️⚠️ CRITICAL WARNING: ALWAYS USE THE LATEST EARNINGS DATA ⚠️⚠️⚠️

STOP AND READ THIS FIRST:

Training data is OUTDATED. Actively search for and retrieve the MOST RECENT earnings materials. Using outdated earnings data is the #1 mistake in earnings analysis.

BEFORE STARTING:

1. **CHECK TODAY’S DATE** - Write down the current date
2. **SEARCH FOR LATEST** - Use web search to find the most recent earnings
3. **VERIFY THE DATE** - Confirm the earnings release is within the last 3 months
4. **IF OLDER THAN 3 MONTHS** - Wrong quarter obtained, search again

Phase 1: Earnings Data Collection (30-60 minutes)

Step 1: Identify the Latest Earnings Period

CRITICAL: ALWAYS SEARCH FOR THE LATEST EARNINGS - DO NOT RELY ON KNOWLEDGE CUTOFF. CRITICAL: NEVER USE EARNINGS DATA FROM TRAINING - IT IS OUTDATED.

Step 1a: Search for Latest Earnings Release

🔴 ACTIVELY SEARCH - training data is outdated. 🔴

MANDATORY STEP 1: CHECK TODAY’S DATE

- **Write down today’s date explicitly:** [Month] [Day], [Year]
- **Use this to verify** that any earnings found are within 3 months
- **Example:** “Today is October 29, 2024”

MANDATORY STEP 2: SEARCH FOR “LATEST EARNINGS”

- **Use web search** with queries like:
 - *[Company name] latest earnings results*
 - *[Company name] most recent quarterly earnings*

- *[Ticker symbol] earnings latest quarter*
- **OR search company investor relations site:**
 - Go to *investor.[company].com* or *[company].com/investors*
 - Navigate to “Press Releases”, “News”, or “Earnings” section
 - **Sort by date to find MOST RECENT release**
 - Look for keywords: “earnings”, “results”, “financial results”, “quarterly results”

MANDATORY STEP 3: VERIFY THE RELEASE DATE

- **Look at the date of the earnings release found**
- **Calculate:** Is this date within the last 3 months from today?
- **If YES** → Proceed to next step
- **If NO (older than 3 months)** → 🚫 **WRONG QUARTER** - Search again for more recent

❌ COMMON MISTAKES TO AVOID:

- ❌ Using earnings data from training without searching
- ❌ Assuming “Q3 2024” is latest based on expectations
- ❌ Grabbing the first earnings release found without checking the date
- ❌ Not comparing the release date to today’s date
- ❌ Proceeding when the release is 4+ months old

✅ CORRECT APPROACH:

- ✅ Check today’s date first
- ✅ Search explicitly for “latest” or “most recent”
- ✅ Read the actual release date on the materials
- ✅ Confirm release date is within 3 months of today
- ✅ If unsure, search again with different terms

MANDATORY STEP 4: IDENTIFY THE QUARTER

- **Read the title/headline** to identify the quarter (Q1, Q2, Q3, Q4 or fiscal quarter)
- **Read the release date** on the document itself
- **Verify both the quarter name AND the date are recent**

1. Alternative search methods if IR site is unclear:

- Web search: *[Company name] latest earnings results*
- Web search: *[Company name] most recent quarterly earnings*
- Web search: *[Ticker symbol] earnings latest quarter*

- SEC EDGAR: Search for company and look at most recent 10-Q or 10-K filing date

Example searches that find latest data:

- “Nike latest earnings results” → Returns most recent quarter reported
- “AAPL most recent quarterly earnings” → Shows latest Apple earnings
- “Tesla Q3 2024 earnings” → Results confirm Q3 2024 exists

Step 1b: Understand Company’s Fiscal Calendar

After identifying the latest quarter from search, understand the company’s fiscal year to interpret it correctly:

Common fiscal year patterns:

- **Calendar year (CY):** Q1=Jan-Mar, Q2=Apr-Jun, Q3=Jul-Sep, Q4=Oct-Dec
- **Nike fiscal:** Q1=Jun-Aug, Q2=Sep-Nov, Q3=Dec-Feb, Q4=Mar-May (May fiscal year-end)
- **Apple fiscal:** Q1=Oct-Dec, Q2=Jan-Mar, Q3=Apr-Jun, Q4=Jul-Sep (September fiscal year-end)
- **Walmart fiscal:** Q1=Feb-Apr, Q2=May-Jul, Q3=Aug-Oct, Q4=Nov-Jan (January fiscal year-end)

Many companies state their fiscal year in the earnings release header. Search *[company] fiscal year calendar* if needed.

Step 1c: MANDATORY VERIFICATION - Verify Latest Data Obtained

STOP - DO NOT PROCEED until verifying ALL of these:

- ☐  **Today’s date written down:** [Month] [Day], [Year]
- ☐  **Actively searched** using “latest earnings” or “most recent earnings”
- ☐  **Earnings release date found:** [Month] [Day], [Year]
- ☐  **Verified release is within 3 months of today** (do the math!)
- ☐  **Did NOT assume** the quarter based on today’s date alone
- ☐  **Can see the actual press release** confirming the quarter/period
- ☐  **Opened and read** the actual earnings materials (not just assumed they exist)

RED FLAGS - If ANY of these are true, WRONG quarter obtained:

-  Release date is more than 90 days old
-  Relying on expectations rather than what was FOUND by searching
-  Have not actually SEEN a press release or filing confirming this quarter exists
-  Used data from training without searching
-  Cannot state the exact release date
-  Release date found is from 2023 or earlier (when today is 2024+)

IF ANY RED FLAGS PRESENT: STOP and search again. Do not proceed with outdated data.

Step 1c: Handle Naming Variations

Companies use different terminology - recognize these patterns:

Quarter terminology:

- “Q1 2024”, “Q1 FY24”, “First Quarter 2024”, “1Q24”
- “Third Quarter Fiscal 2024”, “Q3 FY2024”, “3Q FY24”

Earnings release titles:

- “[Company] Reports Q3 2024 Results”
- “[Company] Announces Third Quarter Fiscal 2024 Financial Results”
- “[Company] Q3 Revenue Grew 15% Year-over-Year”

SEC filing searches:

- Company name may differ from common name (e.g., “Meta Platforms, Inc.” vs “Facebook”)
- Search by ticker symbol to find filings reliably
- Look for most recent 10-Q (quarterly) or 10-K (annual if Q4)

Step 2: Gather Earnings Materials

After SEARCHING FOR and confirming the latest quarter, collect the following:

⚠ IMPORTANT: SEARCH for and ACCESS actual documents - do not rely on training data.

Primary Materials (REQUIRED):

- **Earnings press release** - Usually on company investor relations site under “Press Releases” or “News”
 - Navigate to IR site and find the actual press release
 - Search patterns: “[Company name] latest earnings”, “[Company name] Q[X] [Year] earnings results”
 - Look for PDF or HTML version
 - **Verify the date matches what was found in Step 1** (should be within last 1-3 months)
 - **Read the actual document** to confirm the quarter and get reported numbers
- **10-Q or 10-K filing** - On SEC EDGAR (sec.gov/edgar/searchedgar/companysearch.html)
 - Search by ticker symbol
 - For quarters 1-3: Look for most recent 10-Q
 - For Q4: Look for 10-K (annual report)
 - Note: May be filed 1-5 days after earnings release
 - Direct link format: [https://www.sec.gov/cgi-bin/viewer?accession=\[accession-number\]](https://www.sec.gov/cgi-bin/viewer?accession=[accession-number])
- **Earnings call transcript** - 🚩 **VERIFY THE DATE ON THE TRANSCRIPT** 🚩
 - **Search for:** “[Company] latest earnings call transcript” or “[Company] Q[X] [Year] earnings call transcript”

- **Sources:**
 - * Company IR site (some post transcripts directly)
 - * Seeking Alpha: Search “[Company] [latest quarter] earnings call transcript”
 - * AlphaStreet, Motley Fool (alternative sources)
- **CRITICAL DATE CHECK:**
 - *  **Before using ANY transcript, verify the date on the transcript itself**
 - *  **The transcript date MUST match the earnings release date from Step 1**
 - *  **If transcript says “Q2 2023” but release was “Q3 2024”, WRONG transcript obtained**
 - *  **Common mistake:** Grabbing an old transcript without checking the date
- If transcript not yet available, listen to webcast replay or note to wait for transcript

Supplemental Materials (if available):

- **Investor presentation/slides** - Often posted on IR site alongside press release
 - Usually titled “Q[X] [Year] Earnings Presentation” or “Investor Presentation”
 - PDF format with slides management presented during earnings call
- **Supplemental data file** - Some companies provide Excel files with detailed metrics
 - Look for “Supplemental Financial Information” or “Investor Data Sheet”

Reference Materials (for comparison):

- **Prior quarter results** - For QoQ comparison
 - From prior quarter’s earnings release (90 days ago)
- **Prior year same quarter** - For YoY comparison
 - From same quarter last year (4 quarters ago)
- **Prior estimates** - If this company was previously covered
 - From last earnings update or initiation report
 - Check what was estimated for this quarter’s metrics
- **Consensus estimates** - From Bloomberg, FactSet, Refinitiv, or Yahoo Finance
 - **CRITICAL:** Use estimates from BEFORE earnings release
 - Look for “as of [date before earnings]” to ensure pre-announcement consensus
 - Needed for beat/miss analysis

MANDATORY VERIFICATION before proceeding to Step 3:

DATES - Verify ALL dates match:

- ☐  **Today’s date written down:** _____
- ☐  **Earnings release date:** _____ (MUST be within 3 months of today)
- ☐  **Earnings call transcript date:** _____ (MUST match release date ± 1 day)

- ☒ **10-Q/10-K filing date:** _____ (MUST be same quarter as release)
- ☒ **ALL materials show SAME quarter** (e.g., all say “Q3 2024”, not mixed quarters)

SEARCH & ACCESS - Verify active search completed:

- ☒ **SEARCHED** for “latest earnings” (not assumed based on current date)
- ☒ **ACCESSED** actual earnings press release and read it
- ☒ **OPENED** actual earnings call transcript and verified date
- ☒ **CONFIRMED** this is the MOST RECENT quarter by checking dates
- ☒ Have full financial results (revenue, EPS, margins, etc.) from actual release
- ☒ Have pre-earnings consensus estimates with source date

🚩 RED FLAGS - STOP if ANY of these are true:

- 🚩 Did NOT actually search for or access the earnings materials
- 🚩 Working from memory or training data instead of current documents
- 🚩 The earnings release date is more than 90 days old
- 🚩 Cannot state the EXACT DATE of the earnings release
- 🚩 The transcript date does NOT match the release date
- 🚩 Materials show different quarters (e.g., release says Q3 but transcript says Q2)
- 🚩 Grabbed the first result without verifying the date

Step 3: Extract Key Metrics

Create a structured summary:

Structured Template

- REPORTED RESULTS vs. ESTIMATES:
- Reported Our Est Consensus Beat/(Miss)
- Revenue \$X,XXX \$X,XXX \$X,XXX \$XX (X%)
- Gross Margin XX.X% XX.X% XX.X% XXbps
- EBITDA \$XXX \$XXX \$XXX \$XX (X%)
- Operating Profit \$XXX \$XXX \$XXX \$XX (X%)
- EPS (Adjusted) \$X.XX \$X.XX \$X.XX \$X.XX
- EPS (GAAP) \$X.XX \$X.XX \$X.XX \$X.XX
- KEY BUSINESS METRICS:
- [Metric 1] XXX XXX XXX +X% YoY
- [Metric 2] XXX XXX XXX +X% YoY
- [Metric 3] XXX XXX XXX +X% YoY

Step 4: Identify Key Themes from Call

Listen to or read earnings call transcript and note:

- Management's tone (confident, cautious, defensive?)
- Key topics emphasized (product launches, geographic trends, competition)
- Questions from analysts (what are investors concerned about?)
- Guidance provided (raised, lowered, maintained, introduced?)
- Any surprises or unexpected commentary

Phase 2: Analysis (2-3 hours)

Step 5: Beat/Miss Analysis

For EACH key metric that beat or missed, explain:

If BEAT:

- What drove the outperformance?
- Was it one-time or sustainable?
- Did management guide higher going forward?
- How does this impact our thesis?

If MISS:

- What went wrong?
- Was it company-specific or industry-wide?
- Is management taking corrective action?
- How does this impact our thesis?

Example Format:

Structured Template

- ■ **Revenue Beat by 3% Driven by Strong DTC Performance**
- Revenue of \$13.5B exceeded our estimate of \$13.1B by \$400M (3%) and consensus
- of \$13.2B by \$300M (2%). The outperformance was driven primarily by Direct-to-
- Consumer channels, which grew 18% YoY (vs. our 12% estimate), offsetting
- weaker-than-expected wholesale (-5% vs. flat estimate). Management cited strong
- digital demand and successful product launches (Pegasus 40 running shoe, new
- Jordan colorways) as key drivers. DTC now represents 42% of total revenue vs.
- 38% a year ago, demonstrating successful channel shift strategy.

Step 6: Segment/Geographic/Product Analysis

Analyze performance by:

- Business segment (if multi-segment company)
- Geography (North America, Europe, China, etc.)
- Product category
- Channel (retail, wholesale, e-commerce)

Identify:

- What outperformed expectations?
- What underperformed?
- Trends vs. prior quarters
- Management commentary on outlook for each area

Step 7: Margin Analysis

Analyze profitability:

- Gross margin: up or down? why?
- Operating margin: up or down? why?
- Key drivers (pricing, mix, costs, leverage)
- Outlook going forward

Step 8: Guidance Analysis

If company provided guidance:

- Compare new guidance to prior guidance
- Compare to internal estimates and Street estimates
- Assess credibility (does company have track record of sandbagging? beating?)
- Identify key assumptions behind guidance

If company did NOT provide guidance:

- Note this explicitly
- Provide independent outlook based on results and commentary

Step 9: Update Financial Model

Update estimates for:

- Current year (remaining quarters)

- Next year
- Potentially year after

Show clearly:

Structured Template

- UPDATED ESTIMATES:
- Old Est New Est Change Reason
- FY2024E Revenue \$XX.XB \$XX.XB +X.X% [Brief reason]
- FY2024E EBITDA \$X.XB \$X.XB +X.X% [Brief reason]
- FY2024E EPS \$X.XX \$X.XX +X.X% [Brief reason]
- FY2025E Revenue \$XX.XB \$XX.XB +X.X% [Brief reason]
- FY2025E EBITDA \$X.XB \$X.XB +X.X% [Brief reason]
- FY2025E EPS \$X.XX \$X.XX +X.X% [Brief reason]

Step 10: Update Valuation & Price Target

Based on updated estimates:

- Recalculate DCF (use updated cash flows)
- Update comparable company multiples (if peer group has reported)
- Determine new fair value
- Decide if price target changes

Price Target Decision:

- If estimates changed significantly (>5%) → Usually change price target
- If estimates changed marginally (<5%) → May maintain price target
- If thesis strengthened/weakened → May change even without estimate change

Step 11: Assess Rating Impact

Decide whether to change rating:

- If results significantly better than expected + guidance raised → Consider upgrade
- If results significantly worse + guidance cut → Consider downgrade
- If inline or mixed → Usually maintain rating

Consider:

- Stock reaction (up/down/flat?)
- Valuation (expensive/cheap relative to new estimates?)
- Risk/reward (asymmetry shifted?)

Phase 3: Chart Generation (1-2 hours)

Step 12: Generate 8-12 Charts

Create charts focusing on QUARTERLY TRENDS and WHAT'S NEW.

REQUIRED CHARTS (8-12 total):

1. **Quarterly Revenue Progression** (Bar chart)
 - Last 8-12 quarters
 - Show beat/miss vs. estimates each quarter
 - Highlight current quarter
2. **Quarterly EPS Progression** (Bar chart)
 - Last 8-12 quarters
 - Show beat/miss vs. estimates
 - Adjusted and GAAP
3. **Quarterly Margin Trend** (Line chart)
 - Gross margin, EBIT margin, net margin
 - Last 8-12 quarters
 - Show trajectory
4. **Revenue by Segment/Geography** (Stacked bar OR table)
 - Current quarter vs. YoY
 - Growth rates by segment
5. **Key Operating Metrics** (Multi-line chart)
 - Customer count, ARPU, units sold, etc. (whatever is relevant)
 - Last 8-12 quarters
6. **Beat/Miss Summary** (Waterfall or table)
 - Show components of beat/miss
 - What drove variance from estimates
7. **Estimate Revision Chart** (Before/after comparison)
 - Old FY estimates vs. new FY estimates
 - Bar chart showing change
8. **Valuation Chart** (P/E or EV/EBITDA multiple)
 - Historical multiple range
 - Current multiple
 - Fair value multiple

OPTIONAL CHARTS (if space allows):

- Peer comparison (if peers have reported)
- Guidance vs. Street comparison
- Cash flow metrics
- Balance sheet highlights (if notable)

Chart Style Guidelines:

- Focus on TRENDS (quarterly progression)
- Highlight CHANGES (beat/miss, estimate revisions)
- Keep simple and clear (this is a fast-turnaround report)

Phase 4: Report Creation (2-3 hours)

Step 13: Create DOCX Report

Use DOCX skill to create 8-12 page report.

See [report-structure.md](#) for complete page-by-page templates and formatting requirements.

Key Steps:

1. Create Page 1 with earnings summary and quick takeaways
2. Add detailed results analysis (Pages 2-3)
3. Include key metrics and guidance (Pages 4-5)
4. Update investment thesis (Pages 6-7)
5. Provide valuation and estimates (Pages 8-10)
6. Add appendix if needed (Pages 11-12)
7. Embed all 8-12 charts throughout
8. Add 1-3 summary tables
9. Include complete sources section with clickable hyperlinks

Step 14: Optional - Update XLS Model

If a full financial model exists for this company (from initiation), update it with:

- Actual Q[X] results
- Revised estimates for future quarters
- Updated valuation

Note: For earnings updates, a full XLS file is OPTIONAL (not required like in initiation reports). The DOCX report is the primary deliverable.

If creating XLS, include:

- Quarterly model tab
- Updated annual projections
- Revised DCF
- Updated comps analysis

Phase 5: Quality Check & Delivery (30 minutes)

Step 15: Quality Checklist

Before publishing, verify:

Content:

- ☐ Beat/miss clearly stated and quantified
- ☐ Key drivers explained (not just “strong performance”)
- ☐ Updated estimates provided (old vs. new shown)
- ☐ Price target updated or explicitly maintained
- ☐ Rating confirmed or changed with rationale
- ☐ Guidance analyzed (if provided)
- ☐ Thesis impact assessed

Formatting:

- ☐ Page 1 has summary box and key bullets
- ☐ All tables have source lines
- ☐ All figures numbered and captioned
- ☐ Estimates table shows old vs. new
- ☐ 8-12 charts embedded throughout
- ☐ Report is 8-12 pages (not too long, not too short)

Accuracy:

- ☐ Numbers match company’s reported results exactly
- ☐ Math checks out (estimates, valuation)
- ☐ No typos in ticker, company name, numbers
- ☐ Charts match text descriptions
- ☐ Date is current

Citations: ★ MANDATORY

- ☐ Every figure has specific source with document and date

- ☐ Every table has specific source with document reference
- ☐ Beat/miss analysis cites consensus source with date
- ☐ Guidance changes cite current and prior guidance sources
- ☐ Key statistics have footnotes with specific page/slide references
- ☐ Sources section lists all materials with URLs
- ☐ ALL URLs are CLICKABLE HYPERLINKS (not plain text)
- ☐ Hyperlinks tested and working (Ctrl+Click opens correct page)
- ☐ All SEC filings hyperlinked to EDGAR viewer
- ☐ All earnings materials hyperlinked (release, transcript, presentation)
- ☐ Prior guidance hyperlinked to prior quarter's materials
- ☐ No raw URLs displayed - all formatted as clickable links
- ☐ Earnings call quotes cite specific speaker and approximate timestamp

Timeliness:

- ☐ Report published within 24-48 hours of earnings release
- ☐ All data is from LATEST quarter
- ☐ Consensus estimates are pre-earnings (not post-earnings)

Step 16: Deliver Report

Provide user with:

1. **DOCX file:** *[Company]_Q[X]_[Year]_Earnings_Update.docx*
2. **Chart files:** All PNG/JPG charts (for reference)
3. **Optional XLS:** Updated financial model if maintained

Brief summary for user:

Structured Template

- [Company] Q[X] [Year] Earnings Update Complete
- Results: [BEAT / INLINE / MISS]
- - Revenue: \$X.XB ([beat/missed] by \$XXM or X%)
- - EPS: \$X.XX ([beat/missed] by \$X.XX)
- Key Takeaways:
 - ■ [Takeaway 1]
 - ■ [Takeaway 2]
 - ■ [Takeaway 3]

- Updated Estimates:
 - - FY[Year]E Revenue: \$XX.XB (prior: \$XX.XB, [+/-]X%)
 - - FY[Year]E EPS: \$X.XX (prior: \$X.XX, [+/-]X%)
 - Rating: [MAINTAINED / RAISED / LOWERED] [RATING]
 - Price Target: \$XXX (prior: \$XXX) - [+/-]XX% upside
 - Deliverable: 8-12 page earnings update report with updated estimates and valuation.
-

3.17 Skill: earnings-preview

Earnings Preview

description: Build pre-earnings analysis with estimate models, scenario frameworks, and key metrics to watch. Use before a company reports quarterly earnings to prepare positioning notes, set up bull/bear scenarios, and identify what will move the stock. Triggers on “earnings preview”, “what to watch for [company] earnings”, “pre-earnings”, “earnings setup”, or “preview Q[X] for [company]”.

Workflow

Step 1: Gather Context

- Identify the company and reporting quarter
- Pull consensus estimates via web search (revenue, EPS, key segment metrics)
- Find the earnings date and time (pre-market vs. after-hours)
- Review the company’s prior quarter earnings call for any guidance or commentary

Step 2: Key Metrics Framework

Build a “what to watch” framework specific to the company:

Financial Metrics:

- Revenue vs. consensus (total and by segment)
- EPS vs. consensus
- Margins (gross, operating, net) — expanding or contracting?
- Free cash flow
- Forward guidance vs. consensus

Operational Metrics (sector-specific):

- Tech/SaaS: ARR, net retention, RPO, customer count
- Retail: Same-store sales, traffic, basket size

- Industrials: Backlog, book-to-bill, price vs. volume
- Financials: NIM, credit quality, loan growth, fee income
- Healthcare: Scripts, patient volumes, pipeline updates

Step 3: Scenario Analysis

Build 3 scenarios with stock price implications:

Scenario	Revenue	EPS	Key Driver	Stock Reaction
Bull				
Base				
Bear				

For each scenario:

- What would need to happen operationally
- What management commentary would signal this
- Historical context — how has the stock moved on similar prints?

Step 4: Catalyst Checklist

Identify the 3-5 things that will determine the stock's reaction:

1. [Metric] vs. [consensus/whisper number] — why it matters
2. [Guidance item] — what the buy-side expects to hear
3. [Narrative shift] — any strategic changes, M&A, restructuring

Step 5: Output

One-page earnings preview with:

- Company, quarter, earnings date
- Consensus estimates table
- Key metrics to watch (ranked by importance)
- Bull/base/bear scenario table
- Catalyst checklist
- Trading setup: recent stock performance, implied move from options

Important Notes

- Consensus estimates change — always note the source and date of estimates

- “Whisper numbers” from buy-side surveys are often more relevant than published consensus
 - Historical earnings reactions help calibrate expectations (search for “[company] earnings reaction history”)
 - Options-implied move tells you what the market expects — compare to your scenarios
-

3.18 Skill: idea-generation

Idea Generation

description: Systematic stock screening and investment idea sourcing. Combines quantitative screens, thematic research, and pattern recognition to surface new long and short ideas. Use when looking for new ideas, running screens, or conducting thematic sweeps. Triggers on “idea generation”, “stock screen”, “find ideas”, “what looks interesting”, “screen for”, “new ideas”, or “pitch me something”.

Workflow

Step 1: Define Search Criteria

Ask the user for parameters:

- **Direction:** Long ideas, short ideas, or both
- **Market cap:** Large, mid, small, micro
- **Sector:** Specific sector or cross-sector
- **Style:** Value, growth, quality, special situation, event-driven
- **Geography:** US, international, global
- **Theme:** Any specific thematic angle (AI, reshoring, aging demographics, etc.)

Step 2: Quantitative Screens

Run screens based on the style:

Value Screen

- P/E below sector median
- EV/EBITDA below historical average
- Free cash flow yield >5%
- Price/book below 1.5x
- Insider buying in last 90 days
- Dividend yield above market average

Growth Screen

- Revenue growth >15% YoY

- Earnings growth >20% YoY
- Revenue acceleration (growth rate increasing)
- Expanding margins
- High return on invested capital (>15%)
- Strong net retention (>110% for SaaS)

Quality Screen

- Consistent revenue growth (5+ years)
- Stable or expanding margins
- ROE >15%
- Low debt/equity
- High free cash flow conversion
- Insider ownership >5%

Short Screen

- Declining revenue or decelerating growth
- Margin compression
- Rising receivables / inventory vs. sales
- Insider selling
- Valuation premium to peers without justification
- High short interest with deteriorating fundamentals
- Accounting red flags (auditor changes, restatements)

Special Situation Screen

- Recent IPOs / SPACs with lockup expirations
- Spin-offs in last 12 months
- Companies emerging from restructuring
- Activist involvement
- Management changes at underperforming companies

Step 3: Thematic Sweep

For thematic ideas, research the theme and identify beneficiaries:

1. Define the thesis (e.g., “AI infrastructure spending accelerates through 2026”)
2. Map the value chain — who benefits directly vs. indirectly?

3. Identify pure-play vs. diversified exposure
4. Assess which names are already “priced in” vs. under-appreciated
5. Look for second-order beneficiaries that the market hasn’t connected to the theme

Step 4: Idea Presentation

For each idea that passes the screen, present:

[Company Name] — [Long/Short] — [One-Line Thesis]

Metric	Value	vs. Peers
Market cap		
EV/ EBITDA (NTM)		
P/ E (NTM)		
Revenue growth		
EBITDA margin		
FCF yield		

Thesis (3-5 bullets):

- Why this is mispriced
- What the market is missing
- Catalyst to realize value

Key Risks:

- What would make this wrong

Suggested Next Steps:

- Build full model? Deep-dive diligence? Expert call?

Step 5: Output

- Shortlist of 5-10 ideas with one-page summaries
- Screening criteria and methodology documented
- Comparison table across all ideas
- Prioritized list: which ideas to research first

Important Notes

- Screens surface candidates, not conclusions — every screen output needs fundamental work
- The best ideas often come from intersections (e.g., quality company at value price due to temporary headwind)

- Avoid crowded trades — check ownership data, short interest, and how many analysts cover the name
 - Contrarian ideas need a catalyst — being early without a catalyst is the same as being wrong
 - Track idea hit rates over time — which screens and approaches produce the best ideas?
 - Short ideas need higher conviction — timing is harder and risk is asymmetric
-

3.19 Skill: initiating-coverage

File Metadata

Field	Value
name	initiating-coverage
description	Create institutional-quality equity research initiation reports through a 5-task workflow. Tasks must be executed individually with verified prerequisites - (1) company research, (2) financial modeling, (3) valuation analysis, (4) chart generation, (5) final report assembly. Each task produces specific deliverables (markdown docs, Excel models, charts, or DOCX reports). Tasks 3-5 have dependencies on earlier tasks.

Initiating Coverage

Create institutional-quality equity research initiation reports through a structured 5-task workflow. Each task must be executed separately with verified inputs.

Overview

This skill produces comprehensive first-time coverage reports following institutional standards (JPMorgan, Goldman Sachs, Morgan Stanley format). Tasks are executed individually, each verifying prerequisites before proceeding.

Default Font: Times New Roman throughout all documents (unless user specifies otherwise).

CRITICAL: One Task at a Time

THIS SKILL OPERATES IN SINGLE-TASK MODE ONLY.

If User Requests Full Pipeline

When user requests:

- “Create a coverage initiation report for [Company]”
- “Write an initiation report for [Company]”
- “Do the entire equity research process for [Company]”

- “Complete all 5 tasks for [Company]”
- Any request that implies running multiple tasks or the entire workflow

REQUIRED RESPONSE:

1. **Ask which specific task to perform:** ““ I can help you create an equity research initiation report for [Company]. This involves 5 separate tasks that need to be completed individually:
 - (a) Company Research - Research business, management, industry
 - (b) Financial Modeling - Build projection model
 - (c) Valuation Analysis - DCF and comparable companies
 - (d) Chart Generation - Create 25-35 charts
 - (e) Report Assembly - Compile final report Which task would you like to start with? ““
2. **When user explicitly requests all tasks together:** “ *I understand you’d like to complete the entire initiation report pipeline. Currently, this skill supports executing one task at a time, which allows for better quality control and review at each stage. We’re working on a seamless end-to-end workflow that will make this process more automated, but for now, we’ll need to complete each task separately. Would you like to start with Task 1 (Company Research)? “* “
3. **Never automatically assume which task to start** - always ask user to confirm.
4. **Never execute multiple tasks in sequence** - complete one task, deliver outputs, then wait for next user request.

Task Execution Rules

-  Execute exactly ONE task per user request
-  Always verify prerequisites before starting a task
-  Deliver task outputs and confirm completion
-  Wait for user to explicitly request the next task
-  Never chain multiple tasks together automatically
-  Never assume user wants to proceed to next task
-  Never execute Tasks 3-5 without verifying required inputs exist

Deliverables Policy: NO SHORTCUTS

DELIVER ONLY THE SPECIFIED OUTPUTS. DO NOT CREATE EXTRA DOCUMENTS.

Each task specifies exact deliverables. Do NOT create:

-  “Completion summaries”
-  “Executive summaries”
-  “Quick reference guides”
-  “Next steps documents”

- ❌ “Task completion reports”
- ❌ Any other “helpful” documentation not explicitly specified

Why: These extras waste context and are not part of the professional workflow.

What TO deliver:

- ✅ Task 1: Research document (.md) — **NOTHING ELSE**
- ✅ Task 2: Financial model (.xlsx) — **NOTHING ELSE**
- ✅ Task 3: Valuation analysis (.md) + Excel tabs added to Task 2 file — **NOTHING ELSE**
- ✅ Task 4: Charts zip file (.zip) — **NOTHING ELSE**
- ✅ Task 5: Final report (.docx) — **NOTHING ELSE**

If a deliverable is not listed above, DO NOT CREATE IT.

Task Selection

Select which task to execute:

Task	Name	Prerequisites	Output
1	Company Research	Company name/ ticker	6-8K word document
2	Financial Modeling	10-K or financials access	Excel model (6 tabs)
3	Valuation Analysis	Financial model (Task 2)	Valuation + price target
4	Chart Generation	Tasks 1, 2, 3 + external data	25-35 PNG/ JPG charts
5	Report Assembly	ALL previous tasks (1-4)	30-50 page DOCX report

How to Use This Skill

User Request Patterns and Responses

Pattern 1: User specifies a specific task

Structured Template

- User: “Use initiating-coverage, Task 1 for Tesla”
- Response: ✅ Execute Task 1 immediately

Pattern 2: User asks for “initiation report” or “full pipeline”

Structured Template

- User: “Create a coverage initiation report for Tesla”
- Response: ❌ DO NOT start any task automatically
- ✅ Ask which task to start with (see template above)

Pattern 3: User wants to do “all tasks” or “entire workflow”

Structured Template

- User: “I want to complete all 5 tasks for Tesla”
- Response: **✗** DO NOT chain tasks together
- **✓** Explain one-at-a-time limitation (see template above)
- **✓** Ask if they want to start with Task 1

Correct Usage Examples

Executing a single task:

Structured Template

- “Use initiating-coverage skill, Task 1 for Tesla”
- “Do Task 2 of initiating-coverage for Tesla”
- “Run Task 3 for Tesla using the initiating-coverage skill”

Completing full report (requires 5 separate requests):

Structured Template

- Request 1: “Do Task 1 for Tesla” → Complete → Deliver outputs
- Request 2: “Do Task 2 for Tesla” → Complete → Deliver outputs
- Request 3: “Do Task 3 for Tesla” → Complete → Deliver outputs
- Request 4: “Do Task 4 for Tesla” → Complete → Deliver outputs
- Request 5: “Do Task 5 for Tesla” → Complete → Deliver outputs

Task Execution Order

For a complete initiation report, tasks must be executed in separate user requests following this order:

Structured Template

- Request 1: Task 1 - Company Research (independent)
- ↓ [User reviews outputs and requests next task]
- Request 2: Task 2 - Financial Modeling (independent)
- ↓ [User reviews outputs and requests next task]
- Request 3: Task 3 - Valuation Analysis (requires Task 2 output)
- ↓ [User reviews outputs and requests next task]
- Request 4: Task 4 - Chart Generation (requires Tasks 2 & 3 outputs)
- ↓ [User reviews outputs and requests next task]
- Request 5: Task 5 - Report Assembly (requires ALL previous task outputs)

Note: Tasks 1 and 2 can be run in any order. Tasks 3-5 have strict dependencies and must verify inputs before proceeding.

Task 1: Company Research

Purpose: Research company's business, management, competitive position, industry, and risks.

Prerequisites:  None (fully independent)

- Company name or ticker symbol

Process:

1. Verify company name/ticker provided
2. Load detailed instructions from references/task1-company-research.md
3. Execute qualitative research workflow
4. Deliver research document

Output: Company Research Document (6,000-8,000 words)

- Company overview & history
- Management bios (300-400 words × 3-4 execs)
- Products & services analysis
- Industry overview
- Competitive analysis (5-10 competitors)
- TAM sizing
- Risk assessment (8-12 risks)

File name: *[Company]_Research_Document_[Date].md*

 **DELIVER ONLY THIS 1 FILE. NO completion summaries, no extra documents.**

 **DO NOT TAKE SHORTCUTS:**

-  Write full 6,000-8,000 words (not summaries)
-  Complete 300-400 word bios for ALL 3-4 executives
-  Analyze ALL 5-10 competitors thoroughly
-  Cover all 8-12 risks across 4 categories
-  Do not abbreviate sections to save time
-  Do not skip any required sections

Verification before proceeding: None required for this task.

Task 2: Financial Modeling

Purpose: Extract historical financials and build comprehensive Excel financial model with projections and scenarios.

Prerequisites: ⚠️ Verify before starting

- **Required:** Access to company financial data
 - For public companies: Latest 10-K from SEC EDGAR
 - For private companies: Financial statements or available estimates
 - OR: Pre-extracted historical financials provided by user
- **Optional:** Company research (Task 1) for business context

Input Verification:

Structured Template

- BEFORE STARTING - Select approach:
- Option A: Extract financials (most common)
 - - ☐ Have access to 10-K or financial statements?
 - - ☐ Ready to extract 3-5 years of data?
- Option B: User provided pre-extracted financials
 - - ☐ Historical financials file received?
 - - ☐ Contains income statement, cash flow, balance sheet (3-5 years)?
- Optional:
 - - ☐ Company research (Task 1) complete for context?

Process:

1. Verify access to financial data
2. Load detailed instructions from references/task2-financial-modeling.md
3. **Step 1:** Extract historical financials (if needed)
4. **Step 2+:** Build projection model with 6 essential tabs
5. Deliver Excel model

Output: Excel Financial Model (.xlsx)

- 6 essential tabs:
 1. **Revenue Model** - Product breakdown (20-30 rows) + Geography breakdown (15-20 rows)
 2. **Income Statement** - Full P&L with 40-50 line items, historical (3-5 years) + projected (5 years)
 3. **Cash Flow Statement** - Operating/Investing/Financing activities, historical + projected
 4. **Balance Sheet** - Assets/Liabilities/Equity, historical + projected

5. **Scenarios** - Bull/Base/Bear comparison table
6. **DCF Inputs** - Prepared for Task 3 valuation

File name: *[Company]_Financial_Model_[Date].xlsx*

⚠ DELIVER ONLY THIS 1 FILE. NO completion summaries, no extra documents.

⚠ DO NOT TAKE SHORTCUTS:

- ☒ If extracting financials: Extract ALL line items from 3 financial statements (3-5 years)
- ☒ Build ALL 6 projection tabs completely with full detail
- ☒ Create detailed revenue model with 20-30 product rows AND 15-20 geography rows
- ☒ Build complete income statement with 40-50 line items (not abbreviated)
- ☒ Include full cash flow statement and balance sheet with all line items
- ☒ Complete ALL three scenarios (Bull/Base/Bear) with different parameters
- ☒ Do not create simplified/abbreviated versions
- ☒ Do not skip any of the 6 essential tabs
- ☒ Do not skip historical financials extraction if needed

Verification before proceeding to Task 3:

- ☐ Historical financials extracted (if needed) or provided
 - ☐ Excel file created and can be opened
 - ☐ Model has all 6 essential tabs (Revenue Model, Income Statement, Cash Flow, Balance Sheet, Scenarios, DCF Inputs)
 - ☐ Historical data (3-5 years) incorporated
 - ☐ Projections complete (5 years forward)
 - ☐ Scenarios complete (Bull/Base/Bear)
-

Task 3: Valuation Analysis

Purpose: Perform comprehensive valuation using DCF, comparables, and precedent transactions.

Prerequisites: ⚠ Verify before starting

- **Required:** Financial model from Task 2
 - Projected income statements
 - Projected cash flows
 - Revenue and EBITDA forecasts
 - DCF inputs (unlevered FCF)

⚠ CRITICAL: DO NOT START THIS TASK UNLESS TASK 2 IS COMPLETE

This task requires the financial model from Task 2. Starting without it will result in incomplete work.

IF TASK 2 IS NOT COMPLETE: Stop immediately and inform the user that Task 2 (Financial Modeling) must be completed first. Do not attempt to proceed or create placeholder valuations.

Input Verification:

Structured Template

- BEFORE STARTING:
- - ☐ Task 2 complete? (Financial model exists)
- - ☐ Model file path/location known?
- - ☐ Can access projected financials from model?
- Required from model:
- - ☐ Projected FCF (5 years)
- - ☐ Revenue projections
- - ☐ EBITDA projections
- - ☐ Terminal year metrics

Process:

1. Verify financial model is accessible
2. Load detailed instructions from references/task3-valuation.md
3. Execute valuation workflow
4. Deliver valuation analysis

Output: Valuation Analysis (4-6 pages + Excel tabs)

- DCF analysis with sensitivity tables
- Comparable companies (5-10 peers with statistical summary)
- Precedent transactions (if applicable)
- Valuation football field
- **Price target:** \$XX.XX
- **Recommendation:** BUY/HOLD/SELL
- **Upside:** XX%
- Key catalysts (3-5)

Files:

- *[Company]_Valuation_Analysis_[Date].md* (written analysis document)
- Excel tabs added to *[Company]_Financial_Model_[Date].xlsx* (from Task 2)

- DCF tab with calculations
- Sensitivity analysis tab
- Comparable companies tab
- Valuation summary tab

⚠ DELIVER ONLY: 1 markdown file + 4 tabs added to existing Excel. NO completion summaries, no extra documents.

⚠ DO NOT TAKE SHORTCUTS:

- ☒ Complete full DCF analysis with sensitivity matrix (not simplified)
- ☒ Analyze ALL 5-10 comparable companies with full data
- ☒ Include statistical summary in comps table (max/75th/median/25th/min)
- ☒ Create complete sensitivity analysis tab with multiple WACC and terminal growth scenarios
- ☒ Write full 4-6 pages of valuation analysis (not abbreviated)
- ☒ Research and justify price target with specific methodology
- ☒ Do not skip comparable company analysis
- ☒ Do not create simplified DCF without sensitivity

Verification before proceeding to Task 4:

- ☐ Price target determined
 - ☐ Valuation uses multiple methods (DCF + Comps minimum)
 - ☐ DCF sensitivity table complete
 - ☐ Comparable companies table includes statistical summary
-

Task 4: Chart Generation

Purpose: Generate 25-35 professional financial charts for the report.

Prerequisites: ⚠ Verify before starting

- **Required:** Company research from Task 1
 - Company history and milestones (for timeline charts)
 - Management team and org structure (for org charts)
 - Product portfolio (for product charts)
 - Customer segmentation (for customer charts)
 - Competitive landscape (for competitive charts)
 - TAM analysis (for market size charts)
- **Required:** Financial model from Task 2 (with Task 3 valuation tabs added)
 - Revenue by product/geography data (Task 2 tabs)

- Margin trends (Task 2 tabs)
- Scenario comparison data (Task 2 tabs)
- DCF sensitivity table (Task 3 tab in same Excel file)
- Comparable companies data (Task 3 tab in same Excel file)
- Valuation ranges (Task 3 tab in same Excel file)
- **Required:** External market data
 - Historical stock price data (Yahoo Finance, Bloomberg, etc.)
 - Historical valuation multiples (for historical trend charts)

⚠ CRITICAL: DO NOT START THIS TASK UNLESS TASKS 1, 2, AND 3 ARE COMPLETE

This task requires outputs from all three previous tasks. Starting without them will result in incomplete charts.

IF ANY OF TASKS 1, 2, OR 3 ARE NOT COMPLETE: Stop immediately and inform the user which tasks need to be completed first. The specific requirements are:

- Task 1: Company research document (for 9 charts)
- Task 2: Financial model with all 6 tabs (for 8 charts)
- Task 3: Valuation tabs added to the model (for 6 charts)
- External data access (for 2 charts)

Do not attempt to create placeholder charts or skip charts due to missing data.

Input Verification:

Structured Template

- BEFORE STARTING:
 - - [] Task 1 complete? (Company research exists)
 - - [] Task 2 complete? (Financial model exists)
 - - [] Task 3 complete? (Valuation analysis exists)
 - - [] Can access external market data sources?
- Required from Task 1:
 - - [] Company history and milestones (for charts 05, 06)
 - - [] Management team structure (for chart 07)
 - - [] Product portfolio details (for chart 08)
 - - [] Customer segmentation data (for chart 09)
 - - [] Competitive landscape analysis (for charts 16, 17, 18)
 - - [] TAM sizing and market data (for chart 15)
- Required from Task 2:
 - - [] Revenue by product (historical + projected) - for chart 03 ★

- - [] Revenue by geography (historical + projected) - for chart 04 ★
- - [] Income statement with margins (for charts 02, 10, 11)
- - [] Cash flow statement (for chart 12)
- - [] Scenario comparison data (for chart 14)
- Required from Task 3:
- - [] DCF sensitivity matrix - for chart 28 ★
- - [] DCF components (for chart 29)
- - [] Comparable companies data (for charts 30, 31)
- - [] Valuation ranges - for chart 32 ★
- Required from External Sources:
- - [] Historical stock price data (for chart 01)
- - [] Historical valuation multiples (for chart 34)

Process:

1. Verify model and valuation outputs are accessible
2. Load detailed instructions from references/task4-chart-generation.md
3. Execute chart generation workflow
4. Package all charts into a zip file
5. Deliver zip file

Output: 25-35 Professional Chart Files (PNG/JPG, 300 DPI) packaged in zip

4 MANDATORY Charts (must be present) ★:

- chart_03: Revenue by product (stacked area)
- chart_04: Revenue by geography (stacked bar)
- chart_28: DCF sensitivity (2-way heatmap)
- chart_32: Valuation football field (horizontal bars)

25 REQUIRED Charts (specific list):

- Investment Summary: chart_01
- Financial Performance: charts 02, 03★, 04★, 10, 11, 12, 14
- Company 101: charts 05, 06, 07, 08, 09, 15, 16
- Competitive/Market: charts 17, 18
- Scenario Analysis: chart 13
- Valuation: charts 28★, 29, 30, 31, 32★, 33, 34

10 OPTIONAL Charts (for 26-35 range):

- charts 19-27, 35 (customer acquisition, unit economics, product roadmap, etc.)

IMPORTANT: Task 5 embeds ALL charts created (25-35) for visual density (1 chart per 200-300 words).

File naming: *chart_01_description.png, chart_02_description.png, etc.*

Deliverable: *[Company]_Charts_[Date].zip* containing all 25-35 chart files + chart_index.txt

⚠ DELIVER ONLY THIS 1 ZIP FILE. NO completion summaries, no separate chart lists, no extra documents.

⚠ DO NOT TAKE SHORTCUTS:

- ☒ Create ALL 25 required charts minimum (specific list provided in task4-chart-generation.md)
- ☒ Include ALL 4 mandatory charts:
 - chart_03: Revenue by product (stacked area) ★
 - chart_04: Revenue by geography (stacked bar) ★
 - chart_28: DCF sensitivity (heatmap) ★
 - chart_32: Valuation football field ★
- ☒ Optional: Add 1-10 more charts to reach 26-35 total for greater visual density
- ☒ Generate professional-quality charts at 300 DPI (not low-res placeholders)
- ☒ Create unique, well-formatted charts for each visualization
- ☒ Package all charts in zip file with chart index
- ☒ Do not create only 10-15 charts (minimum is 25)
- ☒ Do not skip any of the 4 mandatory charts
- ☒ Do not use low-quality/placeholder images

Verification before proceeding to Task 5:

- ☐ Minimum 25 chart files created (required)
- ☐ All 4 mandatory charts present:
 - ☐ chart_03: Revenue by product ★
 - ☐ chart_04: Revenue by geography ★
 - ☐ chart_28: DCF sensitivity ★
 - ☐ chart_32: Valuation football field ★
- ☐ All charts open and display correctly
- ☐ Charts saved at 300 DPI (print quality)
- ☐ Chart index created listing all files with categories
- ☐ All charts packaged in zip file
- ☐ File naming follows convention: chart_##_description.png

Task 5: Report Assembly

Purpose: Write and assemble the comprehensive final DOCX report.

Prerequisites: ⚠️ Verify before starting

- **Required:** Company research from Task 1
 - All 6-8K words of content
 - Management bios
 - Competitive analysis
 - Risk assessment
- **Required:** Financial model from Task 2
 - Excel workbook
 - All projections and scenarios
- **Required:** Valuation analysis from Task 3
 - Price target and recommendation
 - DCF, comps, precedent transactions
 - All valuation data
- **Required:** Chart files from Task 4
 - Zip file containing all 25-35 PNG/JPG files
 - Chart index included in zip

⚠️ CRITICAL: DO NOT START THIS TASK UNLESS ALL TASKS 1-4 ARE COMPLETE

This is the final assembly task. It cannot be completed without all previous work products.

IF ANY OF TASKS 1, 2, 3, OR 4 ARE NOT COMPLETE: Stop immediately and inform the user which tasks need to be completed first. The specific requirements are:

- Task 1: Company research document (6-8K words)
- Task 2: Financial model with all 6 tabs
- Task 3: Valuation analysis with price target and recommendation
- Task 4: Charts zip file with 25-35 charts

Do not attempt to create placeholder content, substitute missing sections, or assemble an incomplete report. The report requires ALL inputs to be publication-ready.

Input Verification:

Structured Template

- BEFORE STARTING - ALL TASKS MUST BE COMPLETE:
- Task 1 Verification:
- - [] Company research document exists? (6-8K words)

- - [] Management bios complete? (300-400 words × 3-4 execs)
- - [] Competitive analysis complete? (5-10 competitors)
- - [] Risk assessment complete? (8-12 risks)
- Task 2 Verification:
 - - [] Financial model exists and can be opened?
 - - [] Model has projections (5 years)?
 - - [] Scenarios exist (Bull/Base/Bear)?
- Task 3 Verification:
 - - [] Valuation analysis complete?
 - - [] Price target determined?
 - - [] Recommendation set? (BUY/HOLD/SELL)
 - - [] DCF and comps complete?
- Task 4 Verification:
 - - [] Chart zip file exists?
 - - [] Can extract/access all 25-35 chart files from zip?
 - - [] All 4 mandatory charts present?
 - - [] Revenue by product (stacked area)
 - - [] Revenue by geography (stacked bar)
 - - [] DCF sensitivity (heatmap)
 - - [] Valuation football field
 - - [] Chart files accessible and can be opened?
- IF ANY VERIFICATION FAILS: Stop and complete missing task first.

Process:

1. **CRITICAL:** Verify ALL prerequisites before starting
2. Load detailed instructions from references/task5-report-assembly.md
3. Execute report assembly workflow using Claude's built-in skills:
 - **Use DOCX skill** to create and manipulate the Word document
 - **Use XLSX skill** to read Excel data from Task 2/3
 - **Use Read tool** to read Task 1 and Task 3 markdown files
 - Read Task 1 .md file → Convert to Word formatting → Insert charts inline
 - Read Task 2 .xlsx file → Extract tables → Write quantitative analysis
 - Read Task 3 .md file + Excel tabs → Copy/adapt valuation analysis
 - Insert Task 4 .png chart files throughout using DOCX skill

- Create text-dense report with charts interspersed every 200-300 words

4. Save and deliver final DOCX report

Key Principles:

- Use Claude's DOCX and XLSX skills (NOT Python libraries)
- Use actual file operations (read .md/.xlsx/.png files, write .docx file)
- Good equity research reports are text-dense with lots of illustrating images (60-80% page coverage, 1+ chart per page)

CRITICAL: GO ALL OUT ON THIS TASK

THIS IS THE FINAL DELIVERABLE. DO NOT TAKE SHORTCUTS.

-  **Use full token budget** - This is the culmination of all previous work
-  **Write every section completely** - Do not summarize or abbreviate
-  **Hit ALL minimum requirements** - 30+ pages, 10,000+ words, 25+ charts, 12+ tables
-  **Be thorough on projection assumptions** - 2,000-3,000 words with product-by-product detail
-  **Be comprehensive on scenarios** - 1,500-2,000 words with specific Bull/Base/Bear parameters
-  **Insert ALL charts from Task 4** - Not just a few, ALL 25-35 charts throughout
-  **Create ALL tables from Task 2/3** - Extract every financial table, don't skip any
-  **Use Task 1 content verbatim** - Copy/paste full Company 101 sections (6-8K words)
-  **Professional quality only** - This must be indistinguishable from JPMorgan/Goldman Sachs research

NEVER:

-  "This section would include..." - WRITE THE ACTUAL SECTION
-  "Charts would be inserted here..." - INSERT THE ACTUAL CHARTS
-  "See financial model for details..." - EXTRACT AND INCLUDE THE DETAILS
-  Skip sections due to length - Every section MUST be complete
-  Abbreviate for token conservation - Use whatever tokens are needed

This is publication-ready institutional research. Spare no effort, tokens, or detail.

Output: Comprehensive Equity Research Report (.docx)

Specifications:

- **Length:** 30-50 pages (MINIMUM 30)
- **Word count:** 10,000-15,000 words (MINIMUM 10,000)
- **Charts:** 25-35 embedded images
- **Tables:** 12-20 comprehensive tables

- **Format:** Professional DOCX with clickable hyperlinks

Structure:

- Page 1: Investment Summary (INITIATING COVERAGE format)
- Pages 2-5: Investment thesis & risks
- Pages 6-17: Company 101
- Pages 18-30: Financial analysis & projections
- Pages 31-40: Valuation analysis
- Pages 41-50: Appendices

File name: *[Company]_Initiation_Report_[Date].docx*

⚠ DELIVER ONLY THIS 1 DOCX FILE. NO executive summaries, no “highlights” documents, no extra files.

Final Verification:

- ☐ Report is 30-50 pages
 - ☐ Word count is 10,000-15,000
 - ☐ 25-35 charts embedded
 - ☐ 12-20 tables included
 - ☐ All citations are clickable hyperlinks
 - ☐ Numbers match financial model exactly
-

Input Verification Protocol

Why Input Verification Matters

Each task has strict prerequisites. Running a task without required inputs will:

- Waste time with incomplete work
- Produce low-quality outputs
- Require rework

Always verify inputs before starting a task.

Verification Checklist by Task

Task 1: ☒ No verification needed (independent)

Task 2: Check historical financials available

Task 3: Check financial model exists and is accessible

Task 4: Check both model (Task 2) AND valuation (Task 3) exist

Task 5: Check ALL previous tasks complete (1, 2, 3, 4)

What to Do if Verification Fails

Structured Template

- IF prerequisites are missing:
- 1. Stop the current task
- 2. Identify which prerequisite task(s) need to be completed
- 3. Execute the missing task(s) first
- 4. Return to current task once prerequisites are met

Example Verification Dialogue

Structured Template

- User: “Use initiating-coverage, Task 5 for Tesla”
 - Claude: “Starting Task 5: Report Assembly for Tesla
 - VERIFICATION - Checking prerequisites:
 - - Task 1 (Company research)... ❌ NOT FOUND
 - - Task 2 (Financial model)... ❌ NOT FOUND
 - - Task 3 (Valuation)... ❌ NOT FOUND
 - - Task 4 (Charts)... ❌ NOT FOUND
 - ⚠️ VERIFICATION FAILED
 - Task 5 requires ALL previous tasks to be complete.
 - To proceed:
 - 1. Complete Task 1: Company Research
 - 2. Complete Task 2: Financial Modeling
 - 3. Complete Task 3: Valuation Analysis
 - 4. Complete Task 4: Chart Generation
 - 5. Then return to Task 5
 - Would you like to start with Task 1 first?
-

Task Reference Files

Detailed instructions for each task are in separate reference files to keep this skill lean:

- **references/task1-company-research.md** - Company research workflow

- **references/task2-financial-modeling.md** - Financial modeling workflow
- **references/task3-valuation.md** - Valuation methodology
 - Also see: [references/valuation-methodologies.md](#) for DCF/comps deep dive
- **references/task4-chart-generation.md** - Chart generation workflow
- **references/task5-report-assembly.md** - Report writing workflow
 - Also see: [assets/report-template.md](#) for report structure
 - Also see: [assets/quality-checklist.md](#) for quality checks

When to load reference files: Load ONLY the reference file associated with the specific task being performed. These files are very large - do not load multiple reference files at once. Read the appropriate task reference file at the start of the task for detailed step-by-step instructions.

Quality Standards

All outputs meet institutional standards from leading investment banks (JPMorgan, Goldman Sachs, Morgan Stanley):

- **Comprehensive:** Meet all minimum requirements
 - **Detailed:** Specific data and examples, not generic statements
 - **Quantified:** Lead with numbers and metrics
 - **Cited:** Proper sources with clickable hyperlinks
 - **Professional:** Institutional-quality formatting
 - **Accurate:** All numbers verified and cross-checked
-

Important Notes

Task Independence

- **Task 1** can run anytime (no dependencies)
- **Task 2** can run anytime (just needs historical data)
- **Tasks 1 & 2** can run in parallel
- **Task 3** requires Task 2
- **Task 4** requires Tasks 2 & 3
- **Task 5** requires Tasks 1, 2, 3, & 4

Session Management

Same session: Outputs automatically available to subsequent tasks

Different sessions: Reference previous task outputs explicitly

Structured Template

- “Use Task 3 with the model from yesterday at [path]”
- “Use Task 5 with the research document at [path]”

File Organization

Recommended structure during workflow:

Structured Template

- ProjectFolder/
- Task1_Research/
- [Company]_Research_Document.md
- Task2_Model/
- [Company]_Financial_Model.xlsx
- Task3_Valuation/
- [Company]_Valuation_Analysis.pdf
- Task4_Charts/
- chart_01.png
- ... (25-35 files)
- Task5_Report/
- [Company]_Initiation_Report.docx

No End-to-End Execution

This skill does **NOT** support running all tasks automatically in sequence. Each task must be explicitly requested and verified.

Why: This ensures:

- Quality control at each stage
 - Ability to review outputs before proceeding
 - Flexibility to pause/resume workflow
 - Clear verification of prerequisites
-

Success Criteria

A successful initiation report workflow should:

1. Complete all 5 tasks in order
2. Pass all input verifications
3. Meet all quality standards
4. Produce all required deliverables
5. Numbers cross-check between outputs
6. Final report is publication-ready

Output quality: Institutional (JPMorgan/Goldman/Morgan Stanley level) **Use case:** First-time comprehensive coverage of a company

3.20 Skill Resource: [skills/initiating-coverage/assets/quality-checklist.md](#)

Quality Control Checklist for Initiation Reports

Before delivering an initiation report, verify all items below are complete.

Critical Minimums - Reports Must Meet These

CRITICAL DO NOT DELIVER IF:

- **✗** DOCX report fewer than 30 pages → INCOMPLETE
- **✗** Fewer than 25 embedded charts → INCOMPLETE
- **✗** Fewer than 12 comprehensive tables → INCOMPLETE
- **✗** Fewer than 10,000 words → INCOMPLETE
- **✗** No XLS financial model → MISSING DELIVERABLE
- **✗** Charts are text descriptions, not actual PNG/JPG files → MAJOR FAILURE

Deliverables Checklist

- ☐ DOCX report file created
- ☐ XLS financial model file created
- ☐ Both files named properly: *[Company]_Initiation_Report_[Date].docx* and *[Company]_Financial_Model_[Date].xlsx*

DOCX Report - Length & Content

Length Verification:

- ☐ Report is 30-50 pages (count pages in final document)

- ☐ Word count is 10,000-15,000 words
- ☐ If under 30 pages: STOP and add more content

Visual Elements:

- ☐ 25-35 charts embedded (count them: _____ charts)
- ☐ All charts are actual PNG/JPG image files (NOT text descriptions)
- ☐ 12-20 comprehensive tables included (count them: _____ tables)
- ☐ Charts and tables interspersed throughout, not grouped at end

Chart Requirements:

- ☐ Revenue by product chart: Stacked Area format ✓
- ☐ Revenue by geography chart: Stacked Bar format ✓
- ☐ DCF sensitivity: 2-way Heat Map with color coding ✓
- ☐ Valuation football field: Horizontal bar chart ✓
- ☐ All other charts are actual image files ✓

Table Requirements:

- ☐ Full Income Statement (40-50 rows) with 5 years historical + 5 years projected
- ☐ Full Cash Flow Statement (30-40 rows)
- ☐ Full Balance Sheet (35-45 rows)
- ☐ Revenue by product table (20-30 rows)
- ☐ Revenue by geography table (15-20 rows)
- ☐ Revenue by channel table (10-15 rows)
- ☐ Comparable companies table with statistical summary (max/75th/median/25th/min)
- ☐ DCF calculation table (30-40 rows)
- ☐ WACC calculation table (8-10 rows)
- ☐ Two sensitivity tables
- ☐ 2-3 additional financial/competitive tables

DOCX Report - Structure

Page 1 Requirements:

- ☐ “INITIATING COVERAGE” header present (NOT “Company Update”)
- ☐ Thesis-focused title (NOT event-driven like “Strong Q4 Results”)
- ☐ Rating box with rating, price, target price, 52-week range, market cap, EV

- ☐ 3-4 paragraph-length bullets with ■ character and bold headers
- ☐ Financial & valuation metrics table with 2-3 years historical, 2 years projected
- ☐ Table shows “A” suffix for actuals, “E” suffix for estimates
- ☐ Source lines on all visuals

Content Sections:

- ☐ Table of Contents (Page 2)
- ☐ Investment Thesis & Risks (3-5 pages)
- ☐ Company Overview (6-12 pages) including:
 - ☐ Company description
 - ☐ History and milestones
 - ☐ Management bios (300-400 words EACH for 3-4 executives)
 - ☐ Products/services detail
 - ☐ Competitive landscape
- ☐ Financial Analysis & Projections (10-15 pages)
- ☐ Valuation Analysis (8-12 pages)
- ☐ Assumptions section (2,000-3,000 words documenting ALL projection assumptions)
- ☐ Scenario Analysis (1,500-2,000 words with Bull/Base/Bear parameters)
- ☐ Appendices including Data Sources & References page

DOCX Report - Formatting

Figure & Table Formatting:

- ☐ Every figure has caption above: “Figure X - [Company] [Descriptive Title]”
- ☐ Every figure has source line below: “Source: [Specific sources with dates]”
- ☐ Sequential figure numbering (Figure 1, 2, 3... no gaps)
- ☐ Every table has header row with shading
- ☐ Every table has source line at bottom
- ☐ All years use “A” for actual, “E” for estimate notation

Professional Formatting:

- ☐ Consistent fonts throughout (Calibri, Arial, or similar)
- ☐ Headers and footers with page numbers
- ☐ Dense layout: 60-80% page coverage, minimal white space
- ☐ Every page has both text AND visuals (charts or tables)
- ☐ Professional business report template used

Citations & Sources ★★★ **CRITICAL**

Source Attribution:

- ☐ Every figure has specific source with document name and date
- ☐ Every table has specific source with document reference
- ☐ Key statistics throughout text have footnotes with sources
- ☐ NOT just generic “Company data” - must be specific

Hyperlinks: ★★★ **MANDATORY**

- ☐ ALL URLs are CLICKABLE HYPERLINKS (not plain text)
- ☐ SEC filings hyperlinked to EDGAR viewer
- ☐ Earnings transcripts hyperlinked (Seeking Alpha or company IR)
- ☐ Press releases hyperlinked to company IR page
- ☐ Presentations hyperlinked to PDF URLs
- ☐ Industry reports hyperlinked (if publicly available)
- ☐ Subscription data (Bloomberg, FactSet) noted as “(subscription required)”
- ☐ No raw URLs displayed anywhere - all formatted as hyperlinks
- ☐ Test 3-5 sample hyperlinks to ensure they work (Ctrl+Click)

Reference Page:

- ☐ “Data Sources & References” page at end of report
- ☐ Lists ALL sources used in report
- ☐ Sources organized by category (SEC Filings, Earnings Transcripts, etc.)
- ☐ Every source has date
- ☐ Every source has clickable hyperlink (where applicable)

XLS Financial Model - Structure

File Structure:

- ☐ 15+ tabs in Excel workbook
- ☐ Tabs include: Executive Summary, Assumptions, Historical Financials, Revenue Model, Operating Expenses, Income Statement, Balance Sheet, Cash Flow, Supporting Schedules, DCF Valuation, Comps Analysis, Precedent Transactions, Scenarios, Sensitivity Analysis, Charts

Formatting:

- ☐ Blue text for hardcoded inputs
- ☐ Black text for formulas

- ☐ Green text for links to other sheets
- ☐ Professional formatting with borders and shading
- ☐ Clear section headers and labels

Model Functionality:

- ☐ All numbers flow (change assumption → entire model updates)
- ☐ DCF links to assumptions and projections
- ☐ No circular references or errors
- ☐ All important cells/ranges are named
- ☐ Sensitivity tables work dynamically

XLS Financial Model - Content

Projections:

- ☐ 3-5 years historical data
- ☐ 5 years forward projections (FY+1 through FY+5)
- ☐ Revenue broken down by product, geography, channel
- ☐ Full P&L with 40-50 line items
- ☐ Full cash flow with 30-40 line items
- ☐ Full balance sheet with 35-45 line items

Valuation:

- ☐ Complete DCF model with all calculations shown
- ☐ WACC calculation with all components
- ☐ Terminal value calculation
- ☐ Comparable companies analysis (5-10 companies)
- ☐ Precedent transactions analysis (5-10 deals)
- ☐ Scenario analysis (Bull/Base/Bear)
- ☐ Two sensitivity tables

Cross-File Consistency

CRITICAL: Numbers must match EXACTLY between DOCX and XLS

- ☐ Revenue numbers match across both files
- ☐ EPS numbers match across both files
- ☐ Margin percentages match across both files

- ☐ Valuation numbers match across both files
- ☐ Price target matches across both files
- ☐ All projected years match across both files

Verification Method: Spot check 10-15 key numbers between DOCX report and XLS model.

Content Quality

Investment Thesis:

- ☐ 3-5 clear thesis pillars
- ☐ Each pillar supported with specific data and quantification
- ☐ Financial impact quantified for each pillar
- ☐ Catalysts identified with timelines

Analysis Depth:

- ☐ Comprehensive business model analysis
- ☐ Detailed competitive assessment
- ☐ 3-5 year financial trends analyzed
- ☐ 8-12 risks identified and quantified
- ☐ Management team analyzed (300-400 words per executive)

Assumptions:

- ☐ 2,000-3,000 words documenting ALL assumptions
- ☐ Revenue growth assumptions by category/geography
- ☐ Margin assumptions with bridge showing drivers
- ☐ Working capital assumptions
- ☐ CapEx assumptions
- ☐ Each assumption has specific quantification

Scenarios:

- ☐ 1,500-2,000 words on scenario analysis
- ☐ Bull case with specific parameters and catalysts
- ☐ Base case with detailed rationale
- ☐ Bear case with specific triggers
- ☐ Probability assessments for each scenario

Writing Quality

Style:

- ☐ Lead with numbers (“Revenue grew 15% to \$1.2B” not “Strong revenue”)
- ☐ Use “vs.” not “versus”
- ☐ Be direct and concise
- ☐ Professional institutional tone throughout
- ☐ No informal language

Accuracy:

- ☐ No typos in ticker symbol
- ☐ No typos in company name
- ☐ All dates accurate
- ☐ All calculations verified
- ☐ Charts match text descriptions
- ☐ All numbers properly formatted (\$ signs, % signs, commas)

Pre-Delivery Final Check

Run through this quick final review:

1. **Deliverables:** Both DOCX and XLS files created ✓
2. **Length:** DOCX is 30-50 pages ✓
3. **Charts:** 25-35 actual PNG/JPG files embedded ✓
4. **Tables:** 12-20 comprehensive tables included ✓
5. **Words:** 10,000-15,000 words ✓
6. **Hyperlinks:** Test 3-5 hyperlinks - all work ✓
7. **Cross-check:** Spot check 10 numbers match between DOCX and XLS ✓
8. **Page 1:** “INITIATING COVERAGE” header present ✓

If ANY item fails, DO NOT DELIVER. Go back and fix.

Actual Count Verification

Before delivery, fill in actual counts:

DOCX Report:

- Page count: _____ pages (MUST BE 30-50)
- Chart count: _____ charts (MUST BE 25-35)

- Table count: _____ tables (MUST BE 12-20)
- Word count: _____ words (MUST BE 10,000-15,000)

XLS Model:

- Tab count: _____ tabs (SHOULD BE 15+)
- Model years: _____ historical + _____ projected

If any count is below minimum, STOP and add content before delivery.

3.21 Skill Resource: [skills/initiating-coverage/assets/report-template.md](#)

Equity Research Initiation Report Template

This template provides the structure for creating a comprehensive equity research initiation report. Use this as a guide when constructing the final report document.

NOTE: The actual report MUST be created using the DOCX skill. DO NOT generate markdown content.

CRITICAL REQUIREMENTS:

1. **Generate 20-30+ chart images** using Python (matplotlib/plotly) BEFORE creating the Word document
2. **Use DOCX skill:** Create professional report with proper styles, headers/footers, and formatting
3. **Embed actual chart images:** Insert the generated PNG/JPG chart files into the Word document at appropriate locations
4. **NO MARKDOWN:** Do not generate markdown content. Use DOCX skill to create a .docx file.

CRITICAL FORMATTING GUIDANCE:

- **MAXIMUM DENSITY:** Every page should be packed with information. Intersperse text, charts, and tables throughout.
 - **NO ORPHANED SECTIONS:** Never have a section header alone or a single chart on its own page.
 - **20-30+ ACTUAL CHART IMAGES:** Generate charts as image files, then embed within text sections using DOCX skill.
-

PAGE 1: INVESTMENT UPDATE (MOST IMPORTANT PAGE)

CRITICAL: Page 1 is NOT a traditional executive summary. It is an **Investment Update** with a specific institutional format used by professional equity research firms.

IMPORTANT STRUCTURAL NOTES:

- This is an “Investment Update” or “Company Update” page, not “Executive Summary”
- Uses a rating box in top left corner
- Features stock price performance chart (Figure 1) prominently

- Contains 3-4 detailed bullet points with ■ character
- Each bullet has **bold topic header** followed by 3-5 sentence explanation
- Bottom section has financial and valuation metrics table
- All charts must have figure numbers (Figure 1, Figure 2, etc.) with source lines

Layout Structure

TOP LEFT - RATING BOX:

Structured Template

- Rating: [OUTPERFORM / NEUTRAL / UNDERWEIGHT / etc.]
- Price ([Date]): \$[XX.XX]
- Target Price: \$[XX.XX]
- 52-Week Range: \$[XX.XX] - \$[XX.XX]
- Market Cap: \$[XX.X]B
- Enterprise Value: \$[XX.X]B

TOP LEFT - RESEARCH ANALYSTS:

Structured Template

- [Name], [Credentials (Ph.D., CFA, M.D., etc.)]
- [Email] | [Phone]
- [Name 2], [Credentials]
- [Email] | [Phone]

TOP RIGHT - STOCK PRICE PERFORMANCE:

Structured Template

- Figure 1 - [Company Name] Stock Price Performance
- [Line chart showing stock price over 12-24 months with benchmark comparison]
- Source: Company data, [Firm Name] estimates.

MAIN CONTENT - GRAY HEADER BAR:

Structured Template

- [OUTPERFORM / NEUTRAL / etc.] RECOMMENDATION / COMPANY UPDATE

MAIN CONTENT - DETAILED BULLETS (3-4 bullets):

Use ■ character for bullets. Each bullet follows this format:

Structured Template

- ■ **[Bold Topic Header capturing main point]**. Regular text explanation providing 3-5 sentences of detail with specific numbers, comparisons, and analysis. Lead with numbers and quantification where possible. Use “vs.” not “versus”. Be specific and concrete.

- ■ [Second Topic Header]. [3-5 sentences of detailed explanation...]
- ■ [Third Topic Header]. [3-5 sentences of detailed explanation...]
- ■ [Fourth Topic Header - Optional]. [3-5 sentences of detailed explanation...]

EXAMPLE BULLET FORMAT:

Structured Template

- ■ **Vertical SaaS leadership and regulatory moat should enable \$50bn+ TAM by 2030.**
- Deep domain expertise in healthcare IT, strong customer retention (95%+ net revenue retention),
- and cross-sell capabilities have driven Acme Health’s market expansion. With the healthcare IT
- market expected to reach \$50bn+ by 2030, Acme Health is well-positioned to capture share given
- its regulatory moat and high switching costs. Management has indicated that 70% of current
- revenue comes from enterprise hospital systems, suggesting strong product-market fit.

BOTTOM SECTION - FINANCIAL AND VALUATION METRICS TABLE:

Structured Template

- [Year-3]A [Year-2]A [Year-1]A [Year]E [Year+1]E
 - Revenue (\$M) [X] [X] [X] [X] [X]
 - Revenue Growth (%) X.X% X.X% X.X% X.X% X.X%
 - Gross Margin (%) X.X% X.X% X.X% X.X% X.X%
 - EBITDA (\$M) [X] [X] [X] [X] [X]
 - EBITDA Margin (%) X.X% X.X% X.X% X.X% X.X%
 - EPS (\$) X.XX X.XX X.XX X.XX X.XX
 - P/E (x) XX.Xx XX.Xx XX.Xx XX.Xx XX.Xx
 - EV/Revenue (x) X.Xx X.Xx X.Xx X.Xx X.Xx
 - EV/EBITDA (x) XX.Xx XX.Xx XX.Xx XX.Xx XX.Xx
 - Note: Use “A” suffix for actual/historical years, “E” suffix for estimated/projected years
 - Source: Company data, [Firm Name] estimates.
-

FIGURE NUMBERING AND FORMATTING STANDARDS

CRITICAL: All charts, graphs, and tables must follow strict figure numbering conventions used in professional equity research.

Figure Numbering Format

Every chart/table must have:

1. **Sequential numbering:** Figure 1, Figure 2, Figure 3, etc. (continue sequentially throughout entire report)

2. **Descriptive title:** “Figure X - [Company] [Specific Metric] [Type of Chart/Analysis]”
3. **Source line** (always at bottom): “Source: Company data, [Firm Name] estimates.”

Examples:

- Figure 1 - [Company] Stock Price Performance
- Figure 2 - [Company] Historical and Projected Revenue Mix by Product
- Figure 3 - [Company] Revenue by Geographic Region
- Figure 4 - [Product Name] Revenue and Price per Patient per Year
- Figure 5 - [Company] Gross Margin Evolution
- Figure 6 - DCF Sensitivity Analysis (\$/share)
- Figure 7 - Valuation Football Field

Caption Format

Structured Template

- Figure X - [Descriptive Title]
- [Chart/Table/Graph content]
- Source: Company data, [Firm Name] estimates.

For tables with multiple data sources:

Structured Template

- Figure X - [Descriptive Title]
- [Table content]
- Source: Company filings, FactSet, [Firm Name] estimates.

Placement Guidelines

- Figures should be numbered in order of appearance in the report
- First figure (Figure 1) is typically the stock price chart or revenue growth trajectory on Page 1
- Each figure must have its caption directly below the visual
- Source line should be in smaller font, italicized, at the very bottom of the figure

PAGE 2: TABLE OF CONTENTS

Structured Template

- Executive Summary.....1
- Investment Thesis & Risks.....3

• Company Overview.....	6
• Business Description & History.....	6
• Management & Ownership.....	8
• Products & Technology.....	9
• Customers & Go-to-Market.....	11
• Growth Outlook & Drivers.....	13
• Financial Analysis & Performance.....	16
• Historical Performance.....	16
• Financial Projections.....	19
• Industry Overview & Competitive Landscape.....	21
• Market Size & TAM.....	21
• Competitive Analysis.....	23
• Industry Trends.....	25
• Valuation Analysis.....	27
• Appendices & Disclosures.....	31

PAGES 3-5: INVESTMENT THESIS & RISKS

LAYOUT PRINCIPLE: Intersperse text with 2-3 charts in this section. Each page should have both text AND graphics. Never have pages with text only or charts only.

Investment Thesis

[Thesis Pillar 1]: [Title - e.g., “Large and Growing TAM”]

[Opening sentence with key statistic]

[Paragraph 1: Market opportunity quantification]

- Current market size
- Growth drivers
- Company’s positioning

[Paragraph 2: Why company will capture share]

- Competitive advantages
- Go-to-market strategy
- Early traction/proof points

[Paragraph 3: Financial impact]

- Revenue opportunity
- Margin profile
- Timeline

[EMBED CHART: TAM Growth Chart] - Stacked area chart showing market size evolution and company's opportunity

[Thesis Pillar 2]: [Title - e.g., "Differentiated Technology/Product"]

[Similar structure - 3 paragraphs covering the opportunity, competitive positioning, and financial impact]

[EMBED CHART: Competitive Positioning Matrix] - 2×2 chart showing company vs. competitors on key dimensions

[Thesis Pillar 3]: [Title - e.g., "Strong Execution and Management"]

[Similar structure]

[Add 2-3 more pillars as needed]

[EMBED CHART: Margin Expansion Pathway] - Waterfall or line chart showing path to margin improvement

Investment Risks

Company-Specific Risks

[Risk 1]: [Title - e.g., "Customer Concentration"] [Description of risk, quantification if possible, mitigating factors. 2-3 sentences.]

[Risk 2]: [Title - e.g., "Execution Risk on Product Roadmap"] [Description. 2-3 sentences.]

[Risk 3-5]: [Additional company-specific risks] [Continue with 3-5 total company risks]

Industry/Market Risks

[Risk 1]: [Title - e.g., "Regulatory Uncertainty"] [Description. 2-3 sentences.]

[Risk 2]: [Title - e.g., "Intense Competition"] [Description. 2-3 sentences.]

[Risk 3-4]: [Additional industry/market risks] [Continue with 2-4 total industry risks]

PAGES 8-19: COMPANY 101

Company Description (1 page)

Overview [3-4 paragraphs describing:

- What the company does (in plain English)
- How it makes money
- Who its customers are
- Geographic presence
- Scale/size metrics]

Business Model Diagram/Visual [Insert visual showing how the company creates value]

Company History (2-3 pages)

The Early Days: [Founding Story Title]

[Paragraph on founding: who, when, why, initial vision]

Timeline of Key Milestones

Structured Template

- [Year]: [Founding event, initial funding]
- [Year]: [Product launch, key milestone]
- [Year]: [Major partnership, funding round]
- [Year]: [Geographic expansion, new product]
- [Year]: [Recent achievement]

[Major Turning Point or Pivot] [If applicable, describe any major strategic shifts]

[Company Name] Today: [Current State Title] [Paragraphs describing current position, recent developments, current strategy]

Management & Ownership (2 pages)

Key Executives

For each executive:

Structured Template

- [Name] - [Title]
- [Bio paragraph including:
 - - Current role and responsibilities
 - - Prior experience and track record
 - - Key accomplishments at company
 - - Education/credentials]

Corporate Structure & Governance

- Entity type (C-Corp, PBC, etc.)
- Board composition
- Special governance features
- [Include governance diagram if applicable]

Ownership Structure [if disclosed]

- Major shareholders and ownership %
- Strategic investors
- Employee ownership

- Insider ownership trends

Core Technology/Products (2-3 pages)

Technology Overview [Description of core technology/platform]

Product Portfolio

For each major product:

Structured Template

- [Product Name]
- Description:
 - [What it does, key features]
- Target Customers:
 - [Who uses it, use cases]
- Pricing Model:
 - [How it's priced, typical contract values]
- Competitive Positioning:
 - [How it compares to alternatives]
- Traction:
 - [Customers, revenue, growth metrics]

Product Roadmap [Future products/features in development]

Customers & Distribution (2-3 pages)

Customer Base

- Total customers: [number]
- Customer segments (Enterprise, Mid-Market, SMB)
- Geographic breakdown
- Customer case studies/testimonials

Go-to-Market Strategy

- Sales channels (direct, partner, etc.)
- Sales cycle and CAC
- Key partnerships for distribution
- Marketing strategy

Customer Economics

- LTV/CAC ratio
 - Net retention rate
 - Churn rates
 - Expansion rates
-

PAGES 20-22: GROWTH OUTLOOK

Growth Framework Overview

Short-term Growth Drivers (1-2 years)

1. [Driver 1]
2. [Driver 2]
3. [Driver 3]

Medium-term Growth Drivers (3-5 years)

1. [Driver 1]
2. [Driver 2]

Detailed Growth Driver Analysis

[Growth Driver 1]: [Title]

Current State: [Baseline metrics, current performance]

Opportunity: [Market size, company positioning, growth potential]

Timeline & Milestones:

- Near-term (1-2 years): [Expected progress]
- Medium-term (3-5 years): [Expected progress]

Risks & Challenges: [What could prevent realization of this opportunity]

[Repeat for each major growth driver]

Financial Projections

Revenue Build-up [Visual showing how revenue grows from current to projected]

Scenario Analysis [Table or chart showing Bear/Base/Bull case projections]

PAGES 21-24: FINANCIAL ANALYSIS & PERFORMANCE

LAYOUT PRINCIPLE: This section should be VERY DENSE with 5-7 charts interspersed with financial tables. Each page should have multiple elements (table + 1-2 charts).

Historical Financial Analysis

Income Statement Highlights (3-5 Year History)

Structured Template

- 2021 2022 2023 2024 LTM
- Revenue (\$M) [X] [X] [X] [X] [X]
- Growth % - X% X% X% X%
- Gross Profit (\$M) [X] [X] [X] [X] [X]
- Margin % X% X% X% X% X%
- EBITDA (\$M) [X] [X] [X] [X] [X]
- Margin % X% X% X% X% X%
- Net Income (\$M) [X] [X] [X] [X] [X]
- Margin % X% X% X% X% X%
- FCF (\$M) [X] [X] [X] [X] [X]

[CHART 1: Revenue Growth Trajectory] Line chart showing historical revenue with annotations for key milestones. Include growth % labels on chart.

[CHART 2: REVENUE BY PRODUCT/SEGMENT] ★ CRITICAL Stacked area chart showing revenue composition by product line or business segment over time. This shows mix shift and which products are driving growth.

Structured Template

- Example segments:
- - Product A Revenue
- - Product B Revenue
- - Product C Revenue
- - Services Revenue

[CHART 3: REVENUE BY GEOGRAPHY] ★ CRITICAL Stacked bar chart showing revenue breakdown by geographic region over time.

Structured Template

- Example regions:
- - North America
- - Europe
- - Asia-Pacific
- - Rest of World

Financial Performance Analysis

[CHART 4: Gross Margin Evolution] Line chart with annotations explaining margin drivers (scale, pricing, mix, etc.)

[CHART 5: Operating Margin Progression] Waterfall chart showing path from gross margin to operating margin, or line chart showing EBITDA margin trend

[CHART 6: Free Cash Flow Generation] Bar + line combo chart: Bars = FCF, Line = FCF margin %

[CHART 7: Key Operating Metrics Dashboard] Multi-panel chart showing 3-4 key metrics:

- Customer count or user growth
- ARPU (Average Revenue Per User) or ACV (Annual Contract Value)
- Customer cohort retention or net revenue retention
- LTV/CAC or magic number or other unit economic metric

Forward Projections (3-5 Years)

Projected Financial Model

Structured Template

- 2025E 2026E 2027E 2028E 2029E
- Revenue (\$M) [X] [X] [X] [X] [X]
- Growth % X% X% X% X% X%
- Gross Profit (\$M) [X] [X] [X] [X] [X]
- Margin % X% X% X% X% X%
- EBITDA (\$M) [X] [X] [X] [X] [X]
- Margin % X% X% X% X% X%
- FCF (\$M) [X] [X] [X] [X] [X]
- FCF Margin % X% X% X% X% X%

Key Assumptions

- Revenue growth drivers and assumptions
- Margin progression assumptions
- CapEx as % of revenue
- Working capital assumptions

Charts:

- Revenue bridge showing growth drivers
- Margin waterfall showing path to profitability/margin expansion
- Free cash flow trajectory

Fundraising & Valuation [For Private Companies]

Fundraising History

Structured Template

- Round Date Amount Valuation Lead Investor(s)
- Seed [Date] \$XM \$XM [Investor]
- Series A [Date] \$XM \$XM [Investor]
- Series B [Date] \$XM \$XM [Investor]
- [etc.]

Valuation Evolution Chart [Visual showing valuation progression over time]

Current Valuation Metrics

- Latest valuation: \$XXbn
 - Implied valuation multiple: XX.Xx
 - Comparison to public peers
-

PAGES 26-31: INDUSTRY OVERVIEW

Industry Definition & Market Size

Industry Overview [2-3 paragraphs on:

- Industry definition and scope
- Current market size
- Historical growth rates
- Key trends and drivers]

Market Size Chart [Visual showing market growth from historical through projected]

Competitive Landscape

Competitive Positioning Matrix [2x2 chart showing company vs. competitors on key dimensions]

Competitive Comparison Table

Structured Template

- Metric [Company] Comp A Comp B Comp C Comp D
- Revenue (\$B) [X] [X] [X] [X] [X]
- Growth % X% X% X% X% X%
- Market Share X% X% X% X% X%

- Gross Margin X% X% X% X% X%
- Key Differentiator [X] [X] [X] [X] [X]

Competitive Analysis Narrative [2-3 paragraphs analyzing:

- Competitive strengths and weaknesses
- Market positioning
- Share gains/losses
- Competitive moats]

Total Addressable Market

TAM Calculation

Structured Template

- Current TAM (2025): \$XXbn
- Projected TAM (2030): \$XXbn
- CAGR: XX%
- Segmentation:
 - - [Segment A]: \$XXbn
 - - [Segment B]: \$XXbn
 - - [Segment C]: \$XXbn

TAM Growth Chart [Visual showing TAM expansion over time by segment]

Company's Market Opportunity

Structured Template

- Total TAM (2030): \$XXbn
- Serviceable TAM: \$XXbn
- Company's Realistic Share: XX%
- Implied Revenue Potential: \$XXbn

Industry Dynamics

Porter's Five Forces Analysis

- Threat of new entrants: [High/Medium/Low] - [Explanation]
- Bargaining power of suppliers: [High/Medium/Low] - [Explanation]
- Bargaining power of buyers: [High/Medium/Low] - [Explanation]
- Threat of substitutes: [High/Medium/Low] - [Explanation]
- Industry rivalry: [High/Medium/Low] - [Explanation]

Key Industry Trends

1. [Trend 1]: [Description and impact]
 2. [Trend 2]: [Description and impact]
 3. [Trend 3]: [Description and impact]
-

PAGES 32-34: VALUATION ANALYSIS

Valuation Methodology Summary

Structured Template

- Valuation Method Weight Implied Value Weighted Value
- DCF Analysis 50% \$XX - \$YY \$ZZ
- Trading Comparables 30% \$XX - \$YY \$ZZ
- Precedent Transactions 20% \$XX - \$YY \$ZZ
- Weighted Average Price Target \$AA - \$BB \$CC

DCF Analysis

Key Assumptions

Structured Template

- Revenue Growth (2025-2029): XX% CAGR
- Terminal Growth Rate: X.X%
- WACC: X.X%
- Terminal Year EBITDA Margin: XX%

Figure X - DCF Sensitivity Analysis (\$/share)

CRITICAL FORMAT: DCF sensitivity must be shown as a 2-way heat map table with color coding.

Structured Template

- Terminal Growth Rate
- WACC 2.0% 2.5% 3.0% 3.5% 4.0%
- 8.0% \$52 \$55 \$58 \$62 \$66
- 9.0% \$48 \$51 \$54 \$57 \$61
- 10.0% \$45 \$47 \$50 \$53 \$56
- 11.0% \$42 \$44 \$47 \$49 \$52
- 12.0% \$39 \$41 \$44 \$46 \$49
- Color coding: Green (higher values) → Yellow (mid) → Red (lower values)

- Source: [Firm Name] estimates.

Scenario Analysis

Structured Template

- Scenario Enterprise Value Equity Value Price/Share
- Bear Case \$XXbn \$XXbn \$XX
- Base Case \$XXbn \$XXbn \$XX
- Bull Case \$XXbn \$XXbn \$XX

Trading Comparables

Figure X - Comparable Companies Analysis

CRITICAL FORMAT: Comp table must have two-part structure with statistical summary.

Part 1: Individual Company Data

Structured Template

- Company Ticker Market EV/Rev EV/Rev EV/EBITDA EV/EBITDA Rev EBITDA
- Cap(\$B) 2024E 2025E 2024E 2025E Growth Margin
- Peer A PERA XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- Peer B PERB XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- Peer C PERC XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- Peer D PERD XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- [Company] COMP XX.X X.Xx X.Xx XX.X XX.X XX% XX%

Part 2: Statistical Summary

Structured Template

- Max XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- 75th Percentile XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- Median XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- 25th Percentile XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- Min XX.X X.Xx X.Xx XX.X XX.X XX% XX%
- Source: FactSet, company filings, [Firm Name] estimates.

Implied Valuation [Calculation showing application of peer multiples to company's metrics]

Precedent Transactions [If Applicable]

Figure X - Precedent Transaction Analysis

Structured Template

- Date Target Acquirer Deal EV/Rev EV/EBITDA Premium
- Value(\$B)
- [MM/YYYY] [Company A] [Buyer A] X.X X.Xx XX.X XX%
- [MM/YYYY] [Company B] [Buyer B] X.X X.Xx XX.X XX%
- [MM/YYYY] [Company C] [Buyer C] X.X X.Xx XX.X XX%
- Median X.Xx XX.X XX%
- Source: Capital IQ, company filings, [Firm Name] estimates.

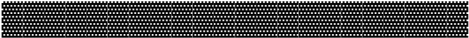
Control Premium Analysis [Discussion of typical premiums in the industry]

Valuation Summary

Figure X - Valuation Football Field

CRITICAL FORMAT: Football field must be a horizontal bar chart showing all valuation methods.

Structured Template

- Valuation Method Low End Range High End
- DCF Analysis \$42  \$58
- Trading Comps (NTM) \$45  \$55
- Precedent Trans. \$48  \$60
- ↑
- Current Price: \$50
- Valuation Range: \$42 \$60
- Color coding: Each method should have distinct color bar
- Add vertical line showing current stock price
- Source: [Firm Name] estimates.

Price Target & Recommendation

Structured Template

- Current Price: \$XX.XX ([Date])
- Price Target: \$YY.YY
- Upside/Downside: ZZ%
- Recommendation: BUY / HOLD / SELL
- Time Horizon: 12 months
- Catalysts:
 - [Near-term catalyst with timeframe]
 - [Medium-term catalyst with timeframe]

- • [Long-term catalyst with timeframe]
-

PAGES 35+: APPENDICES & DISCLOSURES

Appendix A: Detailed Financial Model

[Reference to Excel model]

Appendix B: Management Bios

[Extended bios if not included in main text]

Appendix C: Product Detail

[Additional product information if needed]

Appendix D: Industry Data Sources

[List of sources used for industry analysis]

Required Disclosures

- Analyst certification
 - Important disclosures
 - Company-specific disclosures
 - Legal entity disclosures
 - Other regulatory disclosures
-

GRAPHICS & CHARTS TO INCLUDE

TARGET: 20-30+ charts throughout the report

CRITICAL PRINCIPLE: Charts should be embedded within text sections, NOT grouped on separate pages. Every page (except TOC) should have at least one chart or table.

Page 1 - Executive Summary (3 charts)

1. Revenue/ARR growth trajectory (line chart, historical + projected)
2. Key metrics dashboard (multi-panel chart)
3. Market positioning or margin progression

Pages 3-5 - Investment Thesis & Risks (3 charts)

1. TAM growth and opportunity (stacked area chart)
2. Competitive positioning matrix (2×2 with bubbles)
3. Margin expansion pathway (waterfall or line)

Pages 6-17 - Company 101 (6-8 charts)

1. Business model diagram (flow chart)
2. Company timeline (horizontal timeline)
3. Funding history (bar chart with valuation line)
4. Organization chart
5. Product portfolio matrix
6. Customer segmentation (pie or tree map)
7. Geographic revenue breakdown
8. Customer cohort retention

Pages 18-20 - Growth Outlook (4 charts)

1. Revenue bridge showing drivers (waterfall)
2. Market share evolution (line chart)
3. Product roadmap (timeline)
4. Geographic expansion (map with timeline)

Pages 21-24 - Financials (7 charts) ★ CRITICAL SECTION

1. Revenue growth trajectory (line with annotations)
2. **Revenue by product/segment** (stacked area) ★ MUST HAVE
3. **Revenue by geography** (stacked bar) ★ MUST HAVE
4. Gross margin evolution (line chart)
5. Operating margin progression (waterfall or line)
6. Free cash flow trajectory (bar + line combo)
7. Key operating metrics dashboard (multi-panel)
8. Scenario comparison (grouped bar: Bear/Base/Bull)

Pages 25-30 - Industry Overview (6 charts)

1. Market size evolution (area chart with CAGR)
2. Competitive landscape map (2×2)
3. Market share pie chart
4. Market share evolution over time (line chart)
5. TAM segmentation
6. Industry trend charts

Pages 31-34 - Valuation (5 charts)

1. DCF sensitivity analysis (heat map)
2. DCF waterfall (PV of cash flows → equity value)
3. Trading comps scatter plot (growth vs. multiple)
4. Peer valuation multiples (grouped bar)
5. Valuation football field (range chart)
6. Price target scenarios (bar with upside/downside)

Chart Style Guidelines:

- **Consistent color scheme** throughout (pick 3-5 brand colors)
 - **Professional fonts** (Arial, Calibri, or similar)
 - **Clear labels and legends** on every chart
 - **Source citations** at bottom of each chart
 - **High information density** - use chart space efficiently
 - **Mix of chart types** for visual interest
 - **Annotations** to highlight key insights
 - **Embedded in text** - never standalone pages
 - **Sparklines in tables** where appropriate
-

NOTES ON USING THIS TEMPLATE

1. **PAGE 1 IS CRITICAL:** The executive summary on page 1 must contain ALL key information - fast facts, financial snapshot, 3 charts, valuation summary, thesis, and risks. This is the most important page.
2. **MAXIMUM DENSITY:** Professional equity research is EXTREMELY information-dense. Every page should be packed with interspersed text, charts, and tables. Aim for 60-80% page coverage with minimal white space.

3. **NO ORPHANED SECTIONS:** Never have a section header alone, or a single chart/table on its own page. Always combine elements. Example: DON'T put "Financial Snapshot" on page 6 by itself - integrate it with surrounding content.
 4. **20-30+ CHARTS:** Include extensive graphics throughout, with specific emphasis on:
 - **Revenue by product/segment** (stacked area chart)
 - **Revenue by geography** (stacked bar chart)
 - **Financial performance trends** (multiple charts)
 - Charts should be embedded within text, not grouped separately
 5. **Use DOC Skill:** This outline should be converted to a professional Word document using the DOC skill, with proper formatting, styles, headers/footers, and page numbers
 6. **Intersplice Content:** Text paragraphs should have charts embedded inline. Each page should have 2-4 distinct elements (tables, charts, text blocks).
 7. **Consistent Formatting:** Use consistent styles for headers, body text, tables, and charts throughout. Pick a color scheme and stick to it.
 8. **References:** Include citations and sources for all data points
 9. **Proofread:** Always proofread for accuracy, especially financial data and calculations
 10. **Executive Summary Last:** While it appears on page 1, write this section last after completing the full analysis
 11. **Balance:** Present both positive and negative aspects objectively
 12. **Specific > Generic:** Use specific data and examples rather than generic statements
-

3.22 Skill Resource: [skills/initiating-coverage/references/task1-company-research.md](#)

Task 1: Company Research - Detailed Workflow

This document provides step-by-step instructions for executing Task 1 (Company Research) of the initiating-coverage skill.

Task Overview

Purpose: Research company's business, management, competitive position, industry, and risks.

Prerequisites:  None (fully independent)

- Company name or ticker symbol only

Output: Company Research Document (6,000-8,000 words)

Data Sources to Gather

Primary Sources (Company)

- **SEC Filings (for public companies):**
 - Latest 10-K: Business description, risk factors, MD&A, financials
 - Recent 10-Qs: Quarterly updates
 - DEF 14A (Proxy): Executive compensation, board composition
 - 8-Ks: Material events, acquisitions, management changes
- **Company Website & IR:**
 - Investor presentations
 - Earnings transcripts (last 2-3 quarters)
 - Press releases
 - Product documentation
- **For Private Companies:**
 - Company website and blog
 - Press releases and media coverage
 - LinkedIn for management bios
 - Crunchbase or PitchBook for funding history

Secondary Sources (Industry/Competitive)

- Competitor websites and SEC filings
- Industry research reports (Gartner, Forrester, IDC, etc.)
- News articles and trade publications
- Market research reports
- LinkedIn profiles for key executives

Key Information to Extract

Key Information:

- Company founding date, headquarters, employee count
- Revenue size and growth trajectory (if available)
- Product portfolio and pricing
- Customer segments and case studies
- Management backgrounds and track records
- Competitive landscape and market share
- Industry trends and growth drivers

- Regulatory considerations
 - High-level financial metrics (from 10-K prose, not detailed extraction)
-

Step-by-Step Research Workflow

Step 1: Initial Data Collection

1. Start with company website

- Read About/Company pages
- Review product pages
- Identify customer case studies
- Note key metrics mentioned (employees, customers, etc.)

2. Gather SEC filings (if public)

- Download latest 10-K from SEC EDGAR
- Download most recent 10-Q
- Download latest DEF 14A (proxy statement)
- Note filing dates

3. Read earnings materials

- Latest earnings transcript
- Most recent investor presentation
- Press releases from last 12 months

4. Document basic facts

- Founding date and story
- Headquarters location
- Employee count
- Products/services
- Key customers

Step 2: Business Model Analysis

1. Map revenue streams

- What does the company sell?
- How is it priced? (subscription, transaction, license, etc.)
- Who pays?
- What are typical deal sizes?

2. Understand customer segments

- Enterprise vs. SMB vs. consumer
- Industries served
- Geographic distribution
- Customer concentration (top 10 customers)

3. **Document go-to-market**

- Direct sales vs. channel partners
- Sales cycle length
- Customer acquisition strategy
- Distribution model

4. **Identify unit economics**

- LTV/CAC if available
- Gross margins
- Net revenue retention
- Payback periods

Step 3: Management Research

For each of 3-4 key executives:

1. **Identify key leaders**

- CEO (always required)
- CFO (always required)
- COO, CTO, or other C-suite (2 additional)

2. **Research each executive**

- Find LinkedIn profile
- Review DEF 14A for background
- Search for press interviews
- Note tenure at company

3. **Write 300-400 word bio including:**

- Current role and responsibilities
- Prior roles and companies (last 2-3 positions)
- Key accomplishments and track record
- Education and credentials
- Years of experience in industry
- Time at current company

4. **Assess governance**

- Board composition and independence

- Key board members and their backgrounds
- Insider ownership percentage
- Executive compensation structure

Step 4: Competitive Intelligence

1. Identify 5-10 competitors

- Direct competitors (same products/markets)
- Indirect competitors (substitute solutions)
- Emerging competitors (disruptors)
- Check 10-K for company's own list of competitors

2. Research each competitor

- Visit competitor website
- Review their SEC filings (if public)
- Note key products and positioning
- Identify differentiators
- Estimate market share (if data available)

3. Create competitive framework

- Map on key dimensions (price, features, scale, etc.)
- Identify company's competitive advantages
- Note competitive vulnerabilities
- Assess switching costs and network effects

4. Document competitive insights

- Who are the market leaders?
- Where does this company rank?
- What are unique differentiators?
- What are competitive threats?

Step 5: Industry Analysis

1. Define the industry

- Industry classification (NAICS/SIC)
- Scope and boundaries
- Related/adjacent industries

2. Size the market

- Total addressable market (TAM)
- Serviceable addressable market (SAM)

- Serviceable obtainable market (SOM)
- Current penetration rate

3. **Research growth drivers**

- Historical market growth rate
- Projected growth rate (next 3-5 years)
- Key trends accelerating/decelerating growth
- Technology changes impacting industry

4. **Understand industry structure**

- Fragmented vs. consolidated
- Barriers to entry
- Supplier/buyer power
- Threat of substitutes
- Regulatory environment

Step 6: Risk Assessment

Identify 8-12 risks across four categories. For each risk, write 50-100 words.

Company-Specific Risks (4-6 risks):

- Execution risk (can management deliver?)
- Customer concentration (top customers)
- Key person dependency
- Product/technology obsolescence
- Geographic concentration
- Integration risk (if recent M&A)

Industry/Market Risks (3-4 risks):

- Competitive intensity
- Regulatory changes
- Technology disruption
- Market saturation

Financial Risks (2-3 risks):

- Profitability timeline
- Funding requirements
- Debt levels and covenants
- Cash burn rate (if unprofitable)

Macroeconomic Risks (2-3 risks):

- Economic sensitivity (cyclical vs. defensive)
- Interest rate sensitivity
- Foreign exchange exposure
- Geopolitical factors

For each risk:

- Describe the risk clearly
- Quantify impact if possible
- Note likelihood/severity
- Identify mitigating factors

Step 7: Synthesis and Writing**Write document following this structure:**

1. **Company Overview** (800-1,200 words)
 - What does the company do? (plain English)
 - How do they make money? (business model)
 - Where do they operate? (geographic presence)
 - How large are they? (revenue, employees, customers)
 - Key metrics and scale indicators
2. **Company History** (800-1,200 words)
 - Founding story (who, when, why, where)
 - Timeline of major milestones
 - Strategic pivots or transformations
 - Key acquisitions
 - Recent developments (last 1-2 years)
3. **Management Team** (1,000-1,400 words)
 - 300-400 word bio for each of 3-4 executives
 - Board composition and governance
 - Insider ownership
 - Management track record assessment
4. **Products & Services** (700-1,000 words)
 - Detailed product portfolio
 - Key features and capabilities

- Product differentiation
- Target customers and use cases
- Pricing models and typical deal sizes

5. **Customers & Go-to-Market** (500-700 words)

- Customer segments and profiles
- Distribution channels
- Sales strategy and cycle
- Key partnerships
- Customer case studies

6. **Industry Overview** (800-1,200 words)

- Industry definition and scope
- Market size and structure
- Growth rates (historical and projected)
- Key trends and drivers
- Regulatory environment
- Industry dynamics

7. **Competitive Landscape** (700-1,000 words)

- Analysis of 5-10 key competitors
- Market positioning framework
- Company's competitive advantages
- Competitive vulnerabilities
- Market share analysis

8. **Market Opportunity** (500-700 words)

- TAM sizing and methodology
- Market growth projections
- Company's serviceable market
- Market share opportunity
- Penetration strategy

9. **Risk Assessment** (600-900 words)

- Company-specific risks (4-6)
- Industry/market risks (3-4)
- Financial risks (2-3)
- Macroeconomic risks (2-3)
- Each risk: 50-100 word description

Data Sources Section

- List all sources used
 - Include dates and URLs
 - Organize by source type
-

Quality Standards

Content Depth

- Each section must meet minimum word count targets
- Analysis should be substantive, not just descriptive
- Use specific examples and quantitative data
- Cite sources throughout
- Maintain objectivity and balance

Management Bios

- 300-400 words per executive for 3-4 key executives
- Must include: current role, prior experience, key accomplishments, education
- Provide enough detail to assess track record and capabilities

Competitive Analysis

- Must analyze 5-10 specific competitors
- Include both direct and indirect competitors
- Assess relative positioning on key dimensions
- Identify company's competitive advantages and vulnerabilities
- Use specific data and examples

Risk Assessment

- Must identify 8-12 distinct risks across all four categories
- Each risk needs 50-100 word description
- Quantify impact where possible
- Note mitigating factors
- Cover all four risk categories

Writing Quality

- Professional, analytical tone
 - Lead with key insights
 - Use concrete examples and data
 - Avoid generic statements
 - Proper citations throughout
-

Output Format

Structured Template

- COMPANY RESEARCH REPORT: [Company Name]
- Date: [Date]
- Analyst: [Your name if applicable]
- TABLE OF CONTENTS
- 1. Company Overview
- 2. Company History
- 3. Management Team
- 4. Products & Services
- 5. Customers & Go-to-Market
- 6. Industry Overview
- 7. Competitive Landscape
- 8. Market Opportunity (TAM)
- 9. Risk Assessment
- 1. COMPANY OVERVIEW (800-1,200 words)
- [Content]
- 2. COMPANY HISTORY (800-1,200 words)
- [Content]
- 3. MANAGEMENT TEAM (1,000-1,400 words)
- [Name], [Title]
- [300-400 word bio]
- [Repeat for 3-4 key executives]
- [Governance section]
- 4. PRODUCTS & SERVICES (700-1,000 words)

- [Content]
 - 5. CUSTOMERS & GO-TO-MARKET (500-700 words)
 - [Content]
 - 6. INDUSTRY OVERVIEW (800-1,200 words)
 - [Content]
 - 7. COMPETITIVE LANDSCAPE (700-1,000 words)
 - [Content]
 - 8. MARKET OPPORTUNITY (500-700 words)
 - [Content]
 - 9. RISK ASSESSMENT (600-900 words)
 - Company-Specific Risks:
 - [4-6 risks with descriptions]
 - Industry/Market Risks:
 - [3-4 risks with descriptions]
 - Financial Risks:
 - [2-3 risks with descriptions]
 - Macroeconomic Risks:
 - [2-3 risks with descriptions]
 - DATA SOURCES
 - [List all sources with dates and URLs]
-

Success Criteria

A successful Task 1 completion should deliver:

1. Meet 6,000-8,000 word target (verify word count)
2. Include all 9 required sections with target word counts
3. Provide substantive analysis, not just description
4. Use specific examples and quantitative data
5. Cite all sources properly
6. Enable reader to understand:
 - What the company does and how it makes money
 - Quality and track record of management team
 - Company's competitive position

- Market opportunity size
 - Key risks to consider
-

File Naming Convention

Save the output as:

[Company]_Research_Document_[Date].md

Example: *Tesla_Research_Document_2024-10-27.md*

Next Steps

After completing Task 1, the research document will be used:

- As standalone company analysis
 - As input for Task 2 (Financial Modeling) - provides business context for projections
 - As input for Task 4 (Chart Generation) - provides data for company/competitive charts
 - As foundation for Task 5 (Report Assembly) - Company 101 sections copied verbatim
-

3.23 Skill Resource: [skills/initiating-coverage/references/task2-financial-modeling.md](#)

Task 2: Financial Modeling - Detailed Workflow

This document provides step-by-step instructions for executing Task 2 (Financial Modeling) of the initiating-coverage skill.

Task Overview

Purpose: Extract historical financials and build comprehensive Excel financial model with projections and scenarios.

Prerequisites: ⚠️ Verify before starting

- **Required:** Access to company financial data
 - For public companies: Latest 10-K and recent 10-Qs from SEC EDGAR
 - For private companies: Financial statements or estimates from available sources
 - OR: Pre-extracted historical financials provided by user
- **Optional:** Company research (Task 1) for business context

Output: Excel Financial Model (.xlsx) with 6 essential tabs:

1. Revenue Model
2. Income Statement

3. Cash Flow Statement
 4. Balance Sheet
 5. Scenarios
 6. DCF Inputs
-

Input Verification

BEFORE STARTING - CHECK:

Option A: Extract financials directly (most common)

- ☐ Have access to 10-K filings (public company)?
- ☐ OR have access to financial statements (private company)?
- ☐ Ready to create Excel file for historical extraction?

Option B: User has pre-extracted financials

- ☐ Historical financials file provided? (.xlsx or other format)
- ☐ Contains 3-5 years of income statement, cash flow, balance sheet?
- ☐ Data is clean and ready to use?

Optional Context:

- ☐ Company research (Task 1) complete for business understanding?

IF VERIFICATION FAILS: Stop and obtain access to financial statements (10-K or equivalent) before proceeding.

Model Structure and Formatting

Color Coding (Industry Standard)

- **Blue text:** Hardcoded inputs (user can change)
- **Black text:** Formulas and calculations
- **Green text:** Links to other sheets
- **Red text:** Errors or flags (should be resolved)

Formatting Standards

- Professional borders and shading
- Clear section headers
- Grouped rows for collapsibility

- Named ranges for key inputs/outputs
- No hardcoded numbers in formulas (except constants like 12 months)
- Clear units (\$ thousands, \$ millions, etc.)

Formula Best Practices

- All numbers should flow from assumptions
 - Change an assumption → entire model updates
 - No circular references
 - Use named ranges for key cells
 - Keep formulas simple and auditable
 - Add comments for complex calculations
-

Step-by-Step Modeling Workflow

Step 1: Extract Historical Financials

If historical financials are already extracted, skip to Step 2.

For Public Companies:

1. Download 10-K Filing

- Go to SEC EDGAR (<https://www.sec.gov/edgar/searchedgar/companysearch.html>)
- Search for company name or ticker
- Download latest 10-K (annual report)
- Navigate to Item 8: Financial Statements and Supplementary Data

2. Create Historical Financials Excel File

- File name: *[Company]_Historical_Financials_[Date].xlsx*
- This file will be the foundation for the model

3. Extract Income Statement (3-5 years)

- Create Sheet 1: “Historical Income Statement”
- Extract ALL line items for 3-5 years:
 - Revenue (total and by segment if disclosed)
 - Cost of revenue / COGS
 - Gross profit
 - Operating expenses (R&D, Sales & Marketing, G&A broken out)
 - EBITDA (calculate if not disclosed: EBIT + D&A)
 - EBIT / Operating income

- Interest expense/income
- Other income/expense
- Pre-tax income
- Income tax and tax rate
- Net income
- EPS (basic and diluted)
- Shares outstanding (basic and diluted)

4. Extract Cash Flow Statement (3-5 years)

- Create Sheet 2: “Historical Cash Flow”
- Extract ALL line items:
 - Operating activities (starting from net income)
 - Depreciation & amortization
 - Stock-based compensation
 - Changes in working capital (receivables, inventory, payables)
 - Cash from operations
 - Investing activities (CapEx, acquisitions)
 - Financing activities (debt issuance/repayment, equity, dividends)
 - Net change in cash
 - Beginning and ending cash

5. Extract Balance Sheet (3-5 years)

- Create Sheet 3: “Historical Balance Sheet”
- Extract ALL line items:
 - Current assets (cash, receivables, inventory, other)
 - Non-current assets (PP&E, intangibles, goodwill)
 - Total assets
 - Current liabilities (payables, accrued expenses, current debt)
 - Non-current liabilities (long-term debt, deferred taxes)
 - Total liabilities
 - Shareholders’ equity (common stock, retained earnings)
 - Total liabilities + equity

6. Calculate Historical Metrics

- Create Sheet 4: “Historical Metrics”
- Calculate from statements:
 - Revenue growth % (YoY)
 - Gross margin %
 - EBITDA margin %
 - Operating margin %
 - Net margin %

- Free cash flow (CFO - CapEx)
- FCF margin %
- ROIC (approximate: NOPAT / Invested Capital)
- Debt/Equity ratio
- Current ratio (Current Assets / Current Liabilities)

7. Document Sources and Notes

- Create Sheet 5: “Notes”
- Document:
 - 10-K filing date and fiscal year end
 - Any one-time items or adjustments noted
 - Non-GAAP vs GAAP differences
 - Segment breakdown (if revenue split by product/geography)
 - Data quality notes and limitations

For Private Companies:

1. Gather Available Data

- Financial statements (if available)
- Press releases with revenue figures
- Funding announcements
- Industry estimates or comparable company data

2. Create Simplified Historical File

- Estimated revenue (if available)
- Estimated margins (from comparables if needed)
- Key ratios and metrics
- Document all assumptions and sources

Verification:

- ☐ All 3 financial statements extracted (3-5 years)
- ☐ Numbers reconcile across statements (net income ties)
- ☐ Key metrics calculated correctly
- ☐ Excel file saved and can be opened
- ☐ Data sources documented (10-K dates, page numbers)

Foundation for projection model is now complete. Proceed to Step 2.

- Capital expenditures
- Working capital items
- Debt and interest expense

- Share count (basic and diluted)

1. Organize historical data for entry

- Prepare 3-5 years of actuals
- Will be entered directly into Income Statement, Cash Flow Statement, and Balance Sheet tabs
- Historical years in columns, projected years following

2. Calculate historical trends

- Revenue CAGR
- Margin progression
- OpEx leverage
- Working capital patterns
- CapEx as % of revenue
- These trends will inform projection assumptions

Note: Assumptions will be documented directly in each tab as blue text inputs, not in a separate tab.

Step 2: Model Revenue

CRITICAL: This is the most important and detailed part of the model.

A. Revenue by Product/Category (20-30 rows)

Create detailed table:

Structured Template

- 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- Product Category A
- Sub-product A1 XX XX XX XX XX XX XX XX XX
- Sub-product A2 XX XX XX XX XX XX XX XX XX
- Sub-product A3 XX XX XX XX XX XX XX XX XX
- Category A Total XX XX XX XX XX XX XX XX XX
- % of Total Rev X% X% X% X% X% X% X% X% X%
- YoY Growth % - X% X% X% X% X% X% X% X%
- Product Category B
- [Similar structure]
- [Continue for all product categories]
- Services Revenue XX XX XX XX XX XX XX XX XX
- Other Revenue XX XX XX XX XX XX XX XX XX

- TOTAL REVENUE XX XX XX XX XX XX XX XX XX
- Total Revenue Growth % - X% X% X% X% X% X% X% X%

Key Requirements:

- Show absolute revenue (\$M) for each category
- Calculate % of total revenue for each category
- Show YoY growth % for each category
- Must have granular sub-categories (not just 3-5 top-level categories)
- Show mix shift over time
- Link all projections to Assumptions tab

B. Revenue by Geography (15-20 rows)

Create detailed table:

Structured Template

- 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- North America
- United States XX XX XX XX XX XX XX XX XX
- Canada XX XX XX XX XX XX XX XX XX
- Mexico XX XX XX XX XX XX XX XX XX
- NA Total XX XX XX XX XX XX XX XX XX
- % of Total X% X% X% X% X% X% X% X%
- YoY Growth % - X% X% X% X% X% X% X% X%
- Europe
- UK XX XX XX XX XX XX XX XX XX
- Germany XX XX XX XX XX XX XX XX XX
- France XX XX XX XX XX XX XX XX XX
- Other Europe XX XX XX XX XX XX XX XX XX
- Europe Total XX XX XX XX XX XX XX XX XX
- % of Total X% X% X% X% X% X% X% X%
- YoY Growth % - X% X% X% X% X% X% X% X%
- Asia-Pacific
- [Similar structure]
- Rest of World

- [Similar structure]
- TOTAL REVENUE XX XX XX XX XX XX XX XX XX

Verification:

- Revenue by product total = Revenue by geography total = Total revenue
- All percentages sum to 100%
- Growth rates calculated correctly

C. Revenue by Channel (if applicable)

Structured Template

- 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- Direct Sales XX XX XX XX XX XX XX XX XX
- E-commerce/Online XX XX XX XX XX XX XX XX XX
- Wholesale/Partner XX XX XX XX XX XX XX XX XX
- Retail Stores
- Company-owned stores XX XX XX XX XX XX XX XX XX
- Store count XX XX XX XX XX XX XX XX XX
- Sales per store XX XX XX XX XX XX XX XX XX
- Other Channels XX XX XX XX XX XX XX XX XX
- TOTAL REVENUE XX XX XX XX XX XX XX XX XX

Step 3: Model Operating Expenses

A. Cost of Revenue

1. Break down COGS components

- Product costs (materials, manufacturing)
- Shipping and logistics
- Service delivery costs
- Other direct costs

2. Link to revenue

- Calculate COGS as % of revenue
- Model gross margin by year
- Link to Assumptions tab

B. R&D Expenses

Structured Template

- Research & Development 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- R&D Headcount XX XX XX XX XX XX XX XX XX
- R&D Comp per head XX XX XX XX XX XX XX XX XX
- R&D Personnel Costs XX XX XX XX XX XX XX XX XX
- R&D Other Costs XX XX XX XX XX XX XX XX XX
- Total R&D XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X% X%

C. Sales & Marketing Expenses

Structured Template

- Sales & Marketing 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- S&M Headcount XX XX XX XX XX XX XX XX XX
- S&M Comp per head XX XX XX XX XX XX XX XX XX
- S&M Personnel Costs XX XX XX XX XX XX XX XX XX
- Marketing Spend XX XX XX XX XX XX XX XX XX
- S&M Other Costs XX XX XX XX XX XX XX XX XX
- Total S&M XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X% X%

D. General & Administrative

Structured Template

- G&A 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- G&A Headcount XX XX XX XX XX XX XX XX XX
- G&A Comp per head XX XX XX XX XX XX XX XX XX
- G&A Personnel Costs XX XX XX XX XX XX XX XX XX
- G&A Other Costs XX XX XX XX XX XX XX XX XX
- Total G&A XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X% X%

E. Depreciation & Amortization

- Link to CapEx schedule
- Apply depreciation rates from Assumptions
- Calculate annual D&A

Step 4: Build Income Statement

Create full P&L with 40-50 line items:

Structured Template

- INCOME STATEMENT 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- REVENUE
- [Link to Revenue Model tab]
- Total Revenue XX XX XX XX XX XX XX XX XX
- YoY Growth % - X% X% X% X% X% X% X% X%
- COST OF REVENUE
- [Link to COGS breakdown]
- Total COGS XX XX XX XX XX XX XX XX XX
- GROSS PROFIT XX XX XX XX XX XX XX XX XX
- Gross Margin % X% X% X% X% X% X% X% X%
- OPERATING EXPENSES
- Total R&D XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X%
- Total S&M XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X%
- Total G&A XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X%
- Depreciation & Amort. XX XX XX XX XX XX XX XX XX
- Total Operating Exp. XX XX XX XX XX XX XX XX XX
- % of Revenue X% X% X% X% X% X% X% X%
- EBITDA XX XX XX XX XX XX XX XX XX
- EBITDA Margin % X% X% X% X% X% X% X% X%
- EBIT XX XX XX XX XX XX XX XX XX
- EBIT Margin % X% X% X% X% X% X% X% X%

- Interest expense (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Interest income XX XX XX XX XX XX XX XX XX
- Other income/(expense) XX XX XX XX XX XX XX XX XX
- Pre-tax income XX XX XX XX XX XX XX XX XX
- Income tax (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Tax rate % X% X% X% X% X% X% X% X% X%
- NET INCOME XX XX XX XX XX XX XX XX XX
- Net Margin % X% X% X% X% X% X% X% X% X%
- SHARES OUTSTANDING
- Basic shares (M) XX XX XX XX XX XX XX XX XX
- Diluted shares (M) XX XX XX XX XX XX XX XX XX
- EARNINGS PER SHARE
- Basic EPS \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX
- Diluted EPS \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX \$X.XX

Step 5: Build Cash Flow Statement

Structured Template

- CASH FLOW STATEMENT 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- OPERATING ACTIVITIES
- Net Income XX XX XX XX XX XX XX XX XX
- Adjustments:
- Depreciation & Amort. XX XX XX XX XX XX XX XX XX
- Stock-based comp XX XX XX XX XX XX XX XX XX
- Other non-cash XX XX XX XX XX XX XX XX XX
- Changes in WC:
- Accounts Receivable (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Inventory (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Accounts Payable XX XX XX XX XX XX XX XX XX
- Other working capital (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Cash from Operations XX XX XX XX XX XX XX XX XX
- INVESTING ACTIVITIES
- Capital Expenditures (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)

- Acquisitions (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Other investing XX XX XX XX XX XX XX XX XX
- Cash from Investing (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- FREE CASH FLOW XX XX XX XX XX XX XX XX XX
- FCF Margin % X% X% X% X% X% X% X% X% X%
- FINANCING ACTIVITIES
- Debt issuance XX XX XX XX XX XX XX XX XX
- Debt repayment (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Equity issuance XX XX XX XX XX XX XX XX XX
- Dividends paid (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Other financing XX XX XX XX XX XX XX XX XX
- Cash from Financing XX XX XX XX XX XX XX XX XX
- NET CHANGE IN CASH XX XX XX XX XX XX XX XX XX
- Beginning Cash XX XX XX XX XX XX XX XX XX
- Ending Cash XX XX XX XX XX XX XX XX XX

Step 6: Build Balance Sheet

Create full balance sheet with 35-45 line items:

Structured Template

- BALANCE SHEET 2021A 2022A 2023A 2024A 2025E 2026E 2027E 2028E 2029E
- ASSETS
- Current Assets:
- Cash & Equivalents XX XX XX XX XX XX XX XX XX
- Accounts Receivable XX XX XX XX XX XX XX XX XX
- Inventory XX XX XX XX XX XX XX XX XX
- Prepaid expenses XX XX XX XX XX XX XX XX XX
- Other current assets XX XX XX XX XX XX XX XX XX
- Total Current Assets XX XX XX XX XX XX XX XX XX
- Non-Current Assets:
- PP&E, gross XX XX XX XX XX XX XX XX XX
- Accumulated Depr. (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- PP&E, net XX XX XX XX XX XX XX XX XX

- Intangible assets XX XX XX XX XX XX XX XX XX
- Goodwill XX XX XX XX XX XX XX XX XX
- Other non-current XX XX XX XX XX XX XX XX XX
- Total Non-Current XX XX XX XX XX XX XX XX XX
- TOTAL ASSETS XX XX XX XX XX XX XX XX XX
- LIABILITIES
- Current Liabilities:
- Accounts Payable XX XX XX XX XX XX XX XX XX
- Accrued expenses XX XX XX XX XX XX XX XX XX
- Deferred revenue XX XX XX XX XX XX XX XX XX
- Current debt XX XX XX XX XX XX XX XX XX
- Other current liab. XX XX XX XX XX XX XX XX XX
- Total Current Liab. XX XX XX XX XX XX XX XX XX
- Non-Current Liabilities:
- Long-term debt XX XX XX XX XX XX XX XX XX
- Deferred taxes XX XX XX XX XX XX XX XX XX
- Other non-current XX XX XX XX XX XX XX XX XX
- Total Non-Current Liab. XX XX XX XX XX XX XX XX XX
- TOTAL LIABILITIES XX XX XX XX XX XX XX XX XX
- EQUITY
- Common stock XX XX XX XX XX XX XX XX XX
- Additional paid-in XX XX XX XX XX XX XX XX XX
- Retained earnings XX XX XX XX XX XX XX XX XX
- Treasury stock (XX) (XX) (XX) (XX) (XX) (XX) (XX) (XX)
- Other equity XX XX XX XX XX XX XX XX XX
- TOTAL EQUITY XX XX XX XX XX XX XX XX XX
- TOTAL LIAB + EQUITY XX XX XX XX XX XX XX XX XX
- BALANCE CHECK OK OK OK OK OK OK OK OK OK

Balance Check Formula:

- Total Assets must equal Total Liabilities + Equity for each year
- Flag any imbalances in red

Step 7: Build DCF Inputs Tab

Prepare inputs for valuation (Task 3):

Structured Template

- DCF INPUTS 2025E 2026E 2027E 2028E 2029E
- EBIT XX XX XX XX XX
- Tax Rate X% X% X% X% X%
- NOPAT XX XX XX XX XX
- Add: D&A XX XX XX XX XX
- Less: CapEx (XX) (XX) (XX) (XX) (XX)
- Less: Chg in NWC (XX) (XX) (XX) (XX) (XX)
- UNLEVERED FCF XX XX XX XX XX
- Terminal Year Metrics:
- 2029E Revenue \$X,XXX
- 2029E EBITDA \$XXX
- 2029E EBIT \$XXX
- 2029E Unlevered FCF \$XXX

Step 8: Build Scenarios Tab

Create three scenarios with different assumptions:

Scenario Assumptions Table

Structured Template

- Assumption Bull Base Bear
- Revenue CAGR (2025-2029) XX% XX% XX%
- Gross Margin 2029E XX% XX% XX%
- EBITDA Margin 2029E XX% XX% XX%
- CapEx as % of Revenue X% X% X%
- [Add other key assumptions]

Scenario Output Table

Structured Template

- Metric Bull Base Bear
- 2029E Revenue (\$M) \$X,XXX \$X,XXX \$X,XXX

- 2029E EBITDA (\$M) \$XXX \$XXX \$XXX
- 2029E EBITDA Margin XX% XX% XX%
- 2029E Net Income (\$M) \$XXX \$XXX \$XXX
- 2029E EPS \$X.XX \$X.XX \$X.XX
- 2029E FCF (\$M) \$XXX \$XXX \$XXX
- 2029E FCF Margin XX% XX% XX%
- Cumulative FCF 2025-2029 (\$M) \$XXX \$XXX \$XXX

Document scenario rationale:

- Bull case: [Describe optimistic but achievable assumptions]
- Base case: [Describe most likely scenario]
- Bear case: [Describe downside risks and triggers]

Step 9: Quality Check

Verify model integrity:

1. ☐ Test all formulas (spot check calculations)
 2. ☐ Change assumption → verify model updates correctly
 3. ☐ Test scenario switching
 4. ☐ Verify color coding (blue/black/green)
 5. ☐ Check balance sheet balances for all years
 6. ☐ Verify no circular references (Excel will flag)
 7. ☐ Check for hardcoded numbers in projections
 8. ☐ Verify all cross-sheet links work
 9. ☐ Test that revenue totals tie across all tabs
 10. ☐ Review formatting and presentation
-

Quality Standards

Model Integrity

- All formulas link properly across sheets
- No hardcoded numbers in projections (except in Assumptions tab)
- No circular references
- Balance sheet balances for all years
- Scenario switching works properly

Completeness

- All 6 essential tabs: Revenue Model, Income Statement, Cash Flow Statement, Balance Sheet, Scenarios, DCF Inputs
- 40-50 line items in Income Statement
- 20-30 rows in Revenue Model (product breakdown)
- 15-20 rows in Revenue Model (geography breakdown)
- Full cash flow and balance sheet with all line items
- Bull/Base/Bear scenarios complete

Professional Formatting

- Consistent color coding (blue/black/green)
- Clear headers and labels
- Proper borders and shading
- Named ranges for key cells
- Grouped rows for collapsibility
- Units clearly labeled (\$ thousands vs. \$ millions)

Documentation

- Assumptions documented with rationale (blue text cells with comments)
 - Data sources noted in cell comments or notes section within tabs
 - Complex calculations explained with comments
 - Methodology described
-

File Naming Convention

Save the financial model as: *[Company]_Financial_Model_[Date].xlsx*

Example: *Tesla_Financial_Model_2024-10-27.xlsx*

Success Criteria

A successful financial model should:

1. Have all 6 essential tabs (Revenue Model, Income Statement, Cash Flow Statement, Balance Sheet, Scenarios, DCF Inputs)
2. Be fully dynamic (change assumption → model updates)
3. Have no hardcoded numbers in projections

4. Include detailed revenue breakdowns (20-30 rows by product, 15-20 rows by geography)
 5. Contain 40-50 line items in Income Statement
 6. Include Bull/Base/Bear scenarios
 7. Be professionally formatted with color coding
 8. Balance properly (balance sheet, cash flows)
 9. Be auditable and easy to follow
 10. Support valuation analysis with proper FCF calculations
-

Common Model Types - Special Considerations

High-Growth Tech/SaaS

- Focus on ARR growth and net retention
- Model by product line and geography
- Heavy R&D and S&M spend
- Path to profitability timeline
- Unit economics (LTV/CAC)

E-commerce/Retail

- Revenue by product category and channel
- Store count and comp store growth (if applicable)
- Inventory turns and working capital
- Fulfillment costs
- Customer acquisition

Manufacturing/Industrial

- Production capacity utilization
 - Raw material costs and pricing
 - Gross margin bridge (volume/price/mix/cost)
 - CapEx heavy models
 - Working capital cycles
-

Next Steps

After completing Task 2, the financial model will be used for:

- **Task 3 (Valuation):** DCF inputs, projected financials
 - **Task 4 (Charts):** Data for revenue trends, margin charts, scenario comparisons
 - **Task 5 (Report Assembly):** Financial data for report tables and analysis
-

3.24 Skill Resource: [skills/initiating-coverage/references/task3-valuation.md](#)

Task 3: Valuation Analysis - Detailed Workflow

This document provides step-by-step instructions for executing Task 3 (Valuation Analysis) of the initiating-coverage skill.

Task Overview

Purpose: Perform comprehensive valuation using DCF, comparables, and precedent transactions.

Prerequisites: ⚠️ Verify before starting

- **Required:** Financial model from Task 2
 - Projected income statements
 - Projected cash flows
 - Revenue and EBITDA forecasts
 - DCF inputs (unlevered FCF)

⚠️ CRITICAL: DO NOT START THIS TASK UNLESS TASK 2 IS COMPLETE

This task requires the financial model from Task 2. Starting without it will result in incomplete work.

IF TASK 2 IS NOT COMPLETE: Stop immediately and inform the user that Task 2 (Financial Modeling) must be completed first. Do not attempt to proceed or create placeholder valuations.

Output: Valuation Analysis (4-6 pages + Excel tabs)

- DCF analysis with sensitivity tables
 - Comparable companies analysis
 - Precedent transactions (if applicable)
 - Valuation football field
 - Price target and recommendation
-

Input Verification

BEFORE STARTING - CHECK:

- ☐ Task 2 complete? (Financial model exists)
- ☐ Model file path/location known?
- ☐ Can access projected financials from model?

Required from model:

- ☐ Projected FCF (5 years)
- ☐ Revenue projections
- ☐ EBITDA projections
- ☐ Terminal year metrics
- ☐ Balance sheet data (debt, cash, shares)

IF VERIFICATION FAILS: Stop and complete Task 2 (Financial Modeling) before proceeding.

Detailed Methodology Reference

For deep dive on valuation methodologies, formulas, and theory, see: @@TOKEN0@@

This workflow document focuses on execution steps. Reference the methodology file for:

- DCF theory and formulas
 - WACC calculation details
 - Terminal value methods
 - Comparable companies theory
 - Precedent transactions theory
-

Step-by-Step Valuation Workflow

Step 1: Extract Data from Financial Model

From Task 2's financial model, extract:

1. Projected Financials (5 years)

- Revenue by year (2025E-2029E)
- EBITDA by year
- EBIT by year
- Tax rate
- D&A by year
- CapEx by year
- Change in NWC by year

2. **Unlevered Free Cash Flow** ““ Extract from DCF Inputs tab in financial model: 2025E 2026E 2027E 2028E 2029E EBIT \$XXX \$XXX \$XXX \$XXX \$XXX $\times (1 - \text{Tax Rate}) = \text{NOPAT}$ \$XXX \$XXX \$XXX \$XXX \$XXX
 - D&A \$XXX \$XXX \$XXX \$XXX \$XXX
 - CapEx (\$XX) (\$XX) (\$XX) (\$XX) (\$XX)
 - Chg in NWC (\$XX) (\$XX) (\$XX) (\$XX) (\$XX) = Unlevered FCF \$XXX \$XXX \$XXX \$XXX \$XXX ““
3. **Balance Sheet Data (current)**
 - Total debt
 - Cash & equivalents
 - Net debt (Debt - Cash)
 - Diluted shares outstanding
4. **Scenario Data**
 - Bull case revenue CAGR and terminal margin
 - Base case revenue CAGR and terminal margin
 - Bear case revenue CAGR and terminal margin

Step 2: Build DCF Analysis

A. Calculate WACC

1. Determine Risk-Free Rate

- Use 10-year Treasury yield (check current rate)
- Example: 4.0-4.5% as of late 2024

2. Determine Cost of Equity (CAPM)

Structured Template

- Cost of Equity = Risk-Free Rate + Beta $\times$ Equity Risk Premium
- Inputs:
 - - Risk-Free Rate: [Current 10-year Treasury, e.g., 4.2%]
 - - Beta: [Company beta from Bloomberg/FactSet or peer average]
 - - Equity Risk Premium: 5-6% (historical average)
- Example:
 - Cost of Equity = 4.2% + 1.3 $\times$ 5.5% = 11.35%

3. Determine Cost of Debt

Structured Template

- Cost of Debt = Current borrowing rate or implied yield on bonds

- For private companies:
- Cost of Debt = Risk-Free Rate + Credit Spread (based on rating)
- Example:
- Cost of Debt (pre-tax) = 6.5%
- Cost of Debt (after-tax) = $6.5\% \times (1 - 25\% \text{ tax rate}) = 4.875\%$

4. Determine Capital Structure

Structured Template

- Use market values (not book values):
- Market Value of Equity (E) = Share Price $\times$ Shares Outstanding
- Market Value of Debt (D) = Total Debt (use book value if bonds not traded)
- Total Value (V) = E + D
- Weight of Equity = E / V
- Weight of Debt = D / V
- Example:
- E = \$5,000M (90.9%)
- D = \$500M (9.1%)
- V = \$5,500M (100%)

5. Calculate WACC

Structured Template

- $WACC = (E/V \times \text{Cost of Equity}) + (D/V \times \text{Cost of Debt} \times (1 - \text{Tax Rate}))$
- Example:
- $WACC = (90.9\% \times 11.35\%) + (9.1\% \times 6.5\% \times (1 - 25\%))$
- $WACC = 10.32\% + 0.44\% = 10.76\%$
- Round to: 10.8% for base case

B. Calculate Terminal Value

Method 1: Perpetuity Growth (Preferred)

Structured Template

- Terminal Value = $FCF(2029) \times (1 + g) / (WACC - g)$
- Where:
- - FCF(2029) = Final year unlevered FCF from model
- - g = Perpetual growth rate (typically 2.0-3.0%)
- - Should not exceed long-term GDP growth

- - Use 2.5% as base case
- Example:
- $FCF(2029) = \$500M$
- $g = 2.5\%$
- $WACC = 10.8\%$
- $Terminal\ Value = \$500M \times (1.025) / (0.108 - 0.025)$
- $Terminal\ Value = \$512.5M / 0.083 = \$6,175M$

Method 2: Exit Multiple (Alternative)

Structured Template

- $Terminal\ Value = EBITDA(2029) \times Exit\ Multiple$
- Where:
- - Exit Multiple = Current peer trading median (e.g., 12-15x EBITDA)
- Example:
- $EBITDA(2029) = \$800M$
- Exit Multiple = 13x
- $Terminal\ Value = \$800M \times 13x = \$10,400M$

Choose one method or average both.

C. Discount Cash Flows to Present Value

Structured Template

- $PV\ of\ Projected\ FCF = \sum [FCF_t / (1 + WACC)^t]$ for $t = 1$ to 5
- Example:
- Year FCF Discount PV of FCF
- (\$M) Factor (\$M)
- 2025 \$250 $1/(1.108)^1 = 0.9026$ \$226
- 2026 \$320 $1/(1.108)^2 = 0.8147$ \$261
- 2027 \$390 $1/(1.108)^3 = 0.7353$ \$287
- 2028 \$450 $1/(1.108)^4 = 0.6636$ \$299
- 2029 \$500 $1/(1.108)^5 = 0.5988$ \$299
- Total PV: \$1,372M
- $PV\ of\ Terminal\ Value = Terminal\ Value / (1 + WACC)^5$
- $PV\ of\ Terminal\ Value = \$6,175M / (1.108)^5 = \$6,175M \times 0.5988 = \$3,697M$
- $Enterprise\ Value = \$1,372M + \$3,697M = \$5,069M$

D. Calculate Equity Value and Price Per Share

Structured Template

- Enterprise Value \$5,069M
- - Net Debt (Debt - Cash) (\$450M)
- + Non-operating Assets \$0M
- - Minority Interest \$0M
- - Preferred Stock \$0M
- = Equity Value \$4,619M
- Diluted Shares Outstanding 100M
- Price Per Share = $\$4,619\text{M} / 100\text{M} = \46.19
- Current Stock Price: \$42.00
- Implied Upside: 10.0%

E. DCF Sensitivity Analysis CRITICAL

Table 1: WACC vs. Terminal Growth Rate

Create 2-way sensitivity table:

Structured Template

- Price Per Share (\$) Terminal Growth Rate
- WACC 1.5% 2.0% 2.5% 3.0% 3.5%
- 9.0% \$52 \$55 \$59 \$63 \$68
- 9.5% \$48 \$51 \$54 \$58 \$62
- 10.0% \$45 \$48 \$51 \$54 \$57
- 10.5% \$42 \$45 \$47 \$50 \$53
- 11.0% \$40 \$42 \$44 \$47 \$50
- 11.5% \$38 \$40 \$42 \$44 \$47
- 12.0% \$36 \$38 \$40 \$42 \$44
- Base Case: WACC = 10.8%, $g = 2.5\% \rightarrow \$46$
- Format as heatmap: Green (high values) $\rightarrow$ Yellow $\rightarrow$ Red (low values)

Table 2: Revenue CAGR vs. Terminal EBITDA Margin

Structured Template

- Price Per Share (\$) Terminal EBITDA Margin (2029E)
- Revenue CAGR 28% 30% 32% 34% 36%
- 15% \$38 \$42 \$46 \$50 \$54

- 20% \$42 \$46 \$51 \$56 \$61
- 25% \$46 \$51 \$56 \$62 \$68
- 30% \$51 \$56 \$62 \$68 \$75
- 35% \$56 \$62 \$68 \$75 \$83
- Base Case: Rev CAGR = 25%, EBITDA Margin = 32% → \$56

Step 3: Comparable Companies Analysis

A. Select Comparable Companies

Selection Criteria:

- Same industry/sector (primary requirement)
- Similar business model
- Comparable size (market cap, revenue)
- Similar growth profile
- Similar geographies

Identify 5-10 peer companies:

1. [Peer 1] - Direct competitor
2. [Peer 2] - Direct competitor
3. [Peer 3] - Adjacent player
4. [Peer 4] - Similar business model
5. [Peer 5] - Regional competitor
6. [Add 3-5 more]

Document rationale for each peer selected.

B. Gather Peer Financial Data

For each comparable, gather:

- Current stock price
- Shares outstanding (diluted)
- Market capitalization
- Total debt and cash (for EV calculation)
- Enterprise value
- LTM (Last Twelve Months) financials:

- Revenue
- EBITDA
- EBIT
- Net Income
- NTM (Next Twelve Months) consensus estimates
- Revenue growth rate
- EBITDA margin

Data sources:

- FactSet, CapitalIQ, Bloomberg (preferred)
- Company 10-Ks/10-Qs for actuals
- Consensus estimates from Yahoo Finance, Seeking Alpha (if pro tools unavailable)

C. Calculate Valuation Multiples

For each peer, calculate:

Structured Template

- $EV/Revenue (LTM) = Enterprise\ Value / LTM\ Revenue$
- $EV/Revenue (NTM) = Enterprise\ Value / NTM\ Revenue\ (est.)$
- $EV/EBITDA (LTM) = Enterprise\ Value / LTM\ EBITDA$
- $EV/EBITDA (NTM) = Enterprise\ Value / NTM\ EBITDA\ (est.)$
- $P/E (NTM) = Market\ Cap / NTM\ Net\ Income\ (est.)$

D. Create Comparable Companies Table (MANDATORY FORMAT)

Structured Template

- COMPARABLE COMPANIES ANALYSIS
- Company Ticker Mkt Cap EV/Rev EV/Rev EV/EBITDA EV/EBITDA P/E Rev EBITDA
- (\$B) LTM NTM LTM NTM NTM Growth Margin
- Peer A PRA 45.2 3.5x 3.2x 15.2x 13.8x 25x 18% 23%
- Peer B PRB 32.8 3.2x 2.9x 14.1x 12.5x 22x 15% 23%
- Peer C PRC 28.5 2.8x 2.6x 12.8x 11.2x 20x 12% 22%
- Peer D PRD 52.1 4.1x 3.7x 17.5x 15.2x 29x 22% 23%
- Peer E PRE 38.9 3.6x 3.3x 15.8x 14.1x 25x 17% 23%
- Peer F PRF 41.2 3.7x 3.4x 16.1x 13.9x 26x 19% 23%
- Peer G PRG 35.5 3.3x 3.0x 14.5x 12.8x 23x 16% 22%

- [Target] TRGT 38.0 3.4x 3.1x 14.8x 13.0x 24x 17% 23%
- STATISTICAL SUMMARY
- Maximum 52.1 4.1x 3.7x 17.5x 15.2x 29x 22% 23%
- 75th Percentile 45.2 3.7x 3.4x 16.1x 14.1x 26x 19% 23%
- Median 38.9 3.5x 3.2x 15.2x 13.8x 25x 17% 23%
- 25th Percentile 32.8 3.2x 2.9x 14.1x 12.5x 22x 15% 22%
- Minimum 28.5 2.8x 2.6x 12.8x 11.2x 20x 12% 22%
- Note: Market data as of [Date]. LTM = Last Twelve Months. NTM = Next Twelve Months.
- Source: FactSet, company filings, [Analyst] estimates.

CRITICAL: The statistical summary (max/75th/median/25th/min) is MANDATORY.

E. Apply Multiples to Target Company

Choose primary multiple (typically EV/EBITDA for mature companies):

Structured Template

- Target Company NTM EBITDA = \$550M (from financial model)
- Apply Median Peer Multiple:
- Peer Median EV/EBITDA (NTM) = 13.8x
- Implied EV = $\$550\text{M} \times 13.8\text{x} = \$7,590\text{M}$
- Apply 25th Percentile (Conservative):
- 25th Percentile EV/EBITDA (NTM) = 12.5x
- Implied EV = $\$550\text{M} \times 12.5\text{x} = \$6,875\text{M}$
- Apply 75th Percentile (Optimistic):
- 75th Percentile EV/EBITDA (NTM) = 14.1x
- Implied EV = $\$550\text{M} \times 14.1\text{x} = \$7,755\text{M}$
- Valuation Range (Comps): \$6,875M - \$7,755M
- Midpoint: \$7,315M
- Convert to Equity Value:
- Implied EV (Median) \$7,590M
- - Net Debt (\$450M)
- = Implied Equity Value \$7,140M
- Shares Outstanding 100M
- Implied Price/Share \$71.40

Justify Premium/Discount:

- Target is growing 17% vs. peer median 17% → In-line
- Target EBITDA margin 23% vs. peer median 23% → In-line
- Target market position → [Justify premium/discount]
- **Conclusion:** Apply median multiple (no adjustment)

Step 4: Precedent Transactions (Optional)

Note: Only if M&A is relevant for this sector/company.

A. Identify Relevant Transactions

Search for 5-10 M&A deals:

- Same industry, last 3-5 years
- Similar size (0.5x to 2x target's size)
- Announced and closed deals

Example:

Structured Template

- PRECEDENT TRANSACTIONS ANALYSIS
- Date Target Acquirer Deal EV/Rev EV/EBITDA Premium Rationale
- Value(\$B) LTM LTM
- Q1 2024 Comp A Strategic \$5.2B 4.2x 16.5x 35% Consolidation
- Q3 2023 Comp B PE Firm \$3.8B 3.8x 14.2x 28% Platform
- Q4 2023 Comp C Strategic \$4.5B 4.0x 15.8x 32% Geographic
- Q2 2023 Comp D Strategic \$6.1B 4.5x 17.2x 38% Strategic fit
- Q1 2023 Comp E PE Firm \$3.2B 3.5x 13.5x 25% Carve-out
- Median 4.0x 15.8x 32%
- Source: CapitalIQ, company filings, press releases.

B. Apply to Target Company

Structured Template

- Target Company LTM EBITDA = \$500M
- Precedent Median EV/EBITDA (LTM) = 15.8x
- Implied EV (Precedent) = \$500M × 15.8x = \$7,900M
- Note: Precedent multiples typically 10-20% higher than trading comps
- due to control premium and synergies.

Step 5: Valuation Reconciliation

A. Create Valuation Summary Table

Structured Template

- VALUATION SUMMARY
- Method Low Base High Weight Weighted Value
- DCF Analysis \$42 \$46 \$51 50% \$23.00
- Trading Comps (NTM) \$64 \$71 \$78 40% \$28.40
- Precedent Trans. \$70 \$79 \$88 10% \$7.90
- Weighted Average Target 100% \$59.30
- Rounded Price Target: \$59.00
- Current Price (as of [Date]): \$42.00
- Upside to Target: 40% ($\$59.00 / \$42.00 - 1$)

B. Determine Weighting Rationale

Typical Weighting:

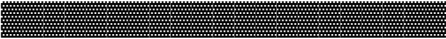
- DCF: 40-60% (higher when forecasts reliable)
- Trading Comps: 25-40% (reflects market sentiment)
- Precedent Trans: 10-25% (lower unless M&A likely)

For this example:

- DCF 50%: High confidence in projections
- Comps 40%: Robust peer set
- Precedent 10%: M&A unlikely near-term

C. Create Valuation Football Field Chart

Structured Template

- VALUATION FOOTBALL FIELD
- Method Low ◀ Range ▶ High
- DCF Analysis \$42  \$51
- Trading Comps (NTM) \$64  \$78
- Precedent Trans. \$70  \$88
- ↑
- Current: \$42

- Valuation Range \$42 \$88
- Price Target: \$59 (weighted average)
- Color code:
- - DCF: Blue
- - Trading Comps: Green
- - Precedent Trans: Orange
- - Vertical line at current price: Red dashed
- - Vertical line at target: Black solid

D. Scenario-Based Valuations

Structured Template

- VALUATION BY SCENARIO
- Scenario Probability Revenue EBITDA DCF Comps Weighted
- CAGR Margin Value Multiple Avg
- Bear Case 20% 18% 28% \$38 11.5x \$42
- Base Case 60% 25% 32% \$46 13.8x \$59
- Bull Case 20% 32% 36% \$58 16.0x \$82
- Expected Value (probability-weighted): \$59

Step 6: Final Price Target & Recommendation

Structured Template

- INVESTMENT RECOMMENDATION
- Current Price: \$42.00 (as of [Date])
- Price Target: \$59.00 (12-month)
- Upside/(Downside): +40.5%
- Rating: BUY / OUTPERFORM
- Valuation Methodology: Based on weighted average of DCF (50%),
- trading comparables (40%), and precedent
- transactions (10%).
- Time Horizon: 12 months
- KEY INVESTMENT CATALYSTS
- 1. New Product Launch (Q2 2025)
- - Expected to drive 15-20% revenue acceleration

- - Already seeing strong pre-orders
 - 2. Margin Expansion (FY2025-2026)
 - - Operating leverage from scale
 - - Path to 35% EBITDA margin (from current 28%)
 - 3. Market Share Gains (Ongoing)
 - - Taking share from legacy competitors
 - - Net Promoter Score improvement
 - 4. International Expansion (H2 2025)
 - - Entry into European markets
 - - Potential \$200M incremental revenue opportunity
 - 5. Potential M&A Target (12-18 months)
 - - Strategic fit for larger players
 - - Precedent transactions suggest 30-40% premium
 - KEY RISKS TO PRICE TARGET
 - Downside Risks:
 - 1. Competitive Pressure (High probability, -15% impact)
 - - New entrant launched competing product
 - - Could pressure pricing and market share
 - 2. Execution Risk (Medium probability, -10% impact)
 - - New product launch delays or underperformance
 - - Management turnover
 - 3. Macro Slowdown (Medium probability, -20% impact)
 - - Economic recession would impact customer spending
 - - Operating leverage would reverse
 - 4. Regulatory Risk (Low probability, -25% impact)
 - - Potential new regulations in key market
 - - Would increase compliance costs
 - Upside Risks:
 - 1. M&A Bid (Low probability, +35% impact)
 - - Strategic acquirer pays control premium
 - 2. Beat-and-Raise (Medium probability, +10% impact)
 - - Consistent outperformance vs. estimates
-

Quality Standards

DCF Quality Checks

- ☐ WACC properly calculated with documented components
- ☐ Terminal value reasonable ($< 70\%$ of total enterprise value)
- ☐ Sensitivity analysis covers realistic ranges (± 200 - 300 bps for WACC, ± 100 bps for terminal growth)
- ☐ Unlevered FCF properly calculated from EBIT
- ☐ Enterprise to equity value bridge correct
- ☐ Share count is diluted shares, not basic

Comparables Quality Checks

- ☐ 5-10 comparable companies selected
- ☐ Peer selection defensible (document why each peer was chosen)
- ☐ Statistical summary included (max/75th/median/25th/min) - MANDATORY
- ☐ Multiple selection appropriate (EV/EBITDA for mature, EV/Revenue for high-growth)
- ☐ Premium/discount justified with specific factors
- ☐ Data sourced properly with dates noted

Overall Valuation Quality Checks

- ☐ At least 2 valuation methods used (DCF + Comps minimum)
 - ☐ Weighting explained and appropriate
 - ☐ Valuation range provided (low/base/high), not just point estimate
 - ☐ Scenarios analyzed (Bull/Base/Bear)
 - ☐ Sanity checks performed (see below)
 - ☐ All assumptions documented with rationale
-

Sanity Checks

Always perform these validation checks:

1. Historical Multiple Check

- Is implied multiple in line with company's historical trading range?
- If not, explain why

2. Peer Comparison

- Is premium/discount vs. peers justified by fundamentals?
- Check: growth, margins, market position

3. Implied Growth Check

- What growth is market pricing in at current price?
- Is that reasonable given company trajectory?

4. Market Cap Reasonableness

- Does total market cap make sense given company size and peers?
- Would company be too large/small relative to industry?

5. Terminal Value Check

- Is terminal value < 60-70% of total enterprise value?
- If > 70%, projections may not be long enough

6. WACC Reasonableness

- Is WACC 8-14% range for typical companies?
- Tech/high-growth: 10-14%
- Mature/stable: 7-10%

7. Implied Returns Check

- What IRR from current price to target over 12 months?
- Is that consistent with recommendation rating?

Output Files

Create the following deliverables:

1. Valuation Analysis Document

File: *[Company]_Valuation_Analysis_[Date].md* (written analysis)

Contents (4-6 pages):

- Executive summary with price target
- DCF analysis (1 page) with sensitivity table
- Comparable companies analysis (1 page) with statistical summary
- Precedent transactions (0.5 page) if applicable
- Valuation summary and football field (0.5 page)
- Investment recommendation (1 page)
- Key catalysts and risks (1 page)

2. Excel Valuation Tabs

Add to Task 2's financial model file: *[Company]_Financial_Model_[Date].xlsx*

IMPORTANT: Do NOT create a separate Excel file. Add these tabs to the existing financial model from Task 2. This keeps all quantitative data in one place.

Tabs to add:

- DCF tab with full calculations
 - Sensitivity analysis tab
 - Comps tab with peer data
 - Precedent transactions tab (if applicable)
 - Valuation summary tab
-

Success Criteria

A successful valuation analysis should:

1. Use at least 2 methods (DCF + Comps minimum)
 2. Include comprehensive DCF sensitivity analysis (2-way tables)
 3. Include statistical summary in comps (max/75th/median/25th/min)
 4. Provide valuation range (low/base/high), not point estimate
 5. Document all key assumptions with clear rationale
 6. Perform sanity checks
 7. Arrive at defensible price target
 8. Provide clear buy/hold/sell recommendation
 9. Identify 3-5 key catalysts
 10. Identify 3-5 key risks
 11. Be auditable and transparent
-

Next Steps

After completing Task 3, the valuation analysis will be used for:

- **Task 4 (Charts):** Create DCF sensitivity heatmaps, valuation football field, scenario comparison charts
- **Task 5 (Report Assembly):** Integrate valuation analysis into final report

The price target and recommendation are the foundation of the final investment recommendation in the equity research report.

3.25 Skill Resource: [skills/initiating-coverage/references/task4-chart-generation.md](#)

Task 4: Chart Generation - Detailed Workflow

This document provides step-by-step instructions for executing Task 4 (Chart Generation) of the initiating-coverage skill.

Task Overview

Purpose: Generate 25-35 professional financial charts for the report.

Prerequisites: ⚠️ Verify before starting

- **Required:** Company research from Task 1
 - Company history, milestones (for timeline charts)
 - Management team, org structure (for org charts)
 - Product portfolio (for product charts)
 - Customer segmentation (for customer charts)
 - Competitive landscape (for competitive positioning charts)
 - TAM analysis (for market size charts)
- **Required:** Financial model from Task 2
 - Revenue by product/geography data
 - Margin trends
 - Scenario comparison data
- **Required:** Valuation analysis from Task 3
 - DCF sensitivity table
 - Comparable companies data
 - Valuation ranges
- **Required:** External market data
 - Historical stock price data (Yahoo Finance, Bloomberg)
 - Historical valuation multiples (optional for chart 34)

⚠️ **CRITICAL: DO NOT START THIS TASK UNLESS TASKS 1, 2, AND 3 ARE COMPLETE**

This task requires outputs from all three previous tasks. Starting without them will result in incomplete charts.

IF ANY OF TASKS 1, 2, OR 3 ARE NOT COMPLETE: Stop immediately and inform the user which tasks need to be completed first. The specific requirements are:

- Task 1: Company research document (for 9 charts)
- Task 2: Financial model with all 6 tabs (for 8 charts)
- Task 3: Valuation tabs added to the model (for 6 charts)
- External data access (for 2 charts)

Do not attempt to create placeholder charts or skip charts due to missing data.

Output: 25-35 Professional Chart Files (PNG/JPG, 300 DPI)

Input Verification

BEFORE STARTING - CHECK ALL PREREQUISITES:

Task 1 Verification (Company Research)

- ☐ Task 1 complete? (Company research document exists)
- ☐ Company history and milestones documented? (for charts 05, 06)
- ☐ Management team and org structure described? (for chart 07)
- ☐ Product portfolio detailed? (for chart 08)
- ☐ Customer segmentation analyzed? (for chart 09)
- ☐ Competitive landscape mapped? (for charts 16, 17, 18)
- ☐ TAM sizing completed? (for chart 15)

Task 2 Verification (Financial Model)

- ☐ Task 2 complete? (Financial model Excel file exists)
- ☐ Revenue by product breakdown available? (for chart 03 ★)
- ☐ Revenue by geography breakdown available? (for chart 04 ★)
- ☐ Historical + projected financials complete? (for charts 02, 10, 11, 12)
- ☐ Scenario analysis (Bull/Base/Bear) complete? (for chart 14)
- ☐ Operating metrics available? (for chart 13)

Task 3 Verification (Valuation)

- ☐ Task 3 complete? (Valuation tabs added to model)
- ☐ DCF sensitivity matrix exists? (for chart 28 ★)
- ☐ DCF calculation details available? (for chart 29)
- ☐ Comparable companies data collected? (for charts 30, 31)
- ☐ Valuation ranges calculated? (for chart 32 ★)

External Data Verification

- ☐ Can access historical stock price data? (Yahoo Finance, Bloomberg for chart 01)
- ☐ Can access historical valuation data? (Optional, for chart 34)

IF ANY VERIFICATION FAILS:

- Missing Task 1? → Complete Task 1 (Company Research) first
 - Missing Task 2? → Complete Task 2 (Financial Modeling) first
 - Missing Task 3? → Complete Task 3 (Valuation Analysis) first
 - Missing external data? → Gather from Yahoo Finance, Bloomberg, or similar sources
-

Chart Requirements: 25 Required + 10 Optional

IMPORTANT: Task 5 (Report Assembly) will embed **ALL charts created** throughout the report. The report requires dense visual content (1 chart every 200-300 words), so create comprehensive chart coverage.

4 MANDATORY Charts (Non-Negotiable) ★

These 4 charts are critical visualizations that **MUST** be present:

1. **chart_03:** Revenue by Product/Segment - Stacked Area Chart ★
2. **chart_04:** Revenue by Geography - Stacked Bar Chart ★
3. **chart_28:** DCF Sensitivity Analysis - 2-Way Heatmap ★
4. **chart_32:** Valuation Football Field - Horizontal Bar Chart ★

25 REQUIRED Charts (Complete Set)

Create all 25 of these charts. Each has a specific purpose in Task 5:

Investment Summary Section (1 chart):

- chart_01: Stock Price Performance (12-24 months)

Financial Performance Section (6 charts):

- chart_02: Revenue Growth Trajectory
- chart_03: Revenue by Product - Stacked Area ★ MANDATORY
- chart_04: Revenue by Geography - Stacked Bar ★ MANDATORY
- chart_10: Gross Margin Evolution
- chart_11: EBITDA Margin Progression
- chart_12: Free Cash Flow Trend

Company 101 Section (7 charts):

- chart_05: Company Overview/Timeline
- chart_06: Key Milestones Timeline
- chart_07: Organizational Structure

- chart_08: Product Portfolio Overview
- chart_09: Customer Segmentation
- chart_15: Market Size Evolution (TAM)
- chart_16: Competitive Positioning Matrix

Competitive & Market Section (2 charts):

- chart_17: Market Share Breakdown
- chart_18: Competitive Benchmarking

Scenario Analysis Section (2 charts):

- chart_13: Operating Metrics Dashboard
- chart_14: Scenario Comparison (Bull/Base/Bear)

Valuation Section (7 charts):

- chart_28: DCF Sensitivity Heatmap ★ MANDATORY
- chart_29: DCF Valuation Waterfall
- chart_30: Trading Comps Scatter Plot
- chart_31: Peer Multiples Comparison
- chart_32: Valuation Football Field ★ MANDATORY
- chart_33: Price Target Scenarios
- chart_34: Historical Valuation Multiples

Total: 25 Required Charts

10 OPTIONAL Charts (For 30-35 Range)

Add these for greater visual density and storytelling (reach 26-35 total):

- chart_19: Customer Acquisition Trends
- chart_20: Unit Economics Evolution
- chart_21: Product Roadmap Timeline
- chart_22: Geographic Expansion Map
- chart_23: R&D Investment Trends
- chart_24: Sales & Marketing Efficiency
- chart_25: Working Capital Trends
- chart_26: Debt Maturity Schedule
- chart_27: Ownership Structure

- chart_35: Analyst Price Target Distribution

Total Range: 25-35 Charts (25 required + 0-10 optional)

Data Source Mapping for Required Charts

Understanding where each chart's data comes from:

From Task 1 (Company Research) - 9 charts

- chart_05: Company Overview → Task 1: Company Overview section
- chart_06: Key Milestones → Task 1: Company History section
- chart_07: Org Structure → Task 1: Management Team section
- chart_08: Product Portfolio → Task 1: Products & Services section
- chart_09: Customer Segmentation → Task 1: Customers & Go-to-Market section
- chart_15: Market Size Evolution → Task 1: Market Opportunity (TAM) section
- chart_16: Competitive Positioning → Task 1: Competitive Landscape section
- chart_17: Market Share → Task 1: Competitive Landscape section
- chart_18: Competitive Benchmarking → Task 1: Competitive Landscape section

From Task 2 (Financial Model) - 8 charts

- chart_02: Revenue Growth → Income Statement tab (Revenue row)
- chart_03: Revenue by Product ★ → Revenue Model tab (Product breakdown)
- chart_04: Revenue by Geography ★ → Revenue Model tab (Geography breakdown)
- chart_10: Gross Margin → Income Statement tab (Gross Profit / Revenue)
- chart_11: EBITDA Margin → Income Statement tab (EBITDA / Revenue)
- chart_12: Free Cash Flow → Cash Flow Statement tab (CFO - CapEx)
- chart_13: Operating Metrics → Multiple tabs (Income Statement, Cash Flow)
- chart_14: Scenario Comparison → Scenarios tab (Bull/Base/Bear)

From Task 3 (Valuation) - 6 charts

- chart_28: DCF Sensitivity ★ → Sensitivity Analysis tab
- chart_29: DCF Waterfall → DCF tab (Enterprise Value components)
- chart_30: Trading Comps Scatter → Comparable Companies tab
- chart_31: Peer Multiples → Comparable Companies tab

- chart_32: Valuation Football Field ★ → Valuation Summary tab
- chart_33: Price Target Scenarios → Valuation Summary tab (or calculate from scenarios)

From External Sources - 2 charts

- chart_01: Stock Price Performance → Yahoo Finance, Bloomberg, Alpha Vantage
- chart_34: Historical Valuation Multiples → Yahoo Finance, Bloomberg (historical P/E, EV/EBITDA)

IMPORTANT: Require ALL three tasks (1, 2, 3) complete PLUS external data access to create all 25 required charts.

Step-by-Step Chart Generation Workflow

Step 1: Set Up Environment

Install required libraries:

Structured Template

- pip install matplotlib seaborn pandas numpy plotly

Create Python script header:

Structured Template

- import matplotlib.pyplot as plt
- import seaborn as sns
- import pandas as pd
- import numpy as np
- from matplotlib.patches import Rectangle
- import warnings
- warnings.filterwarnings('ignore')
- # Set global style
- plt.style.use('seaborn-v0_8-darkgrid')
- sns.set_palette("husl")
- # Global settings
- DPI = 300
- FIGURE_WIDTH = 10
- FIGURE_HEIGHT = 6
- TITLE_FONT_SIZE = 14
- AXIS_FONT_SIZE = 12
- LABEL_FONT_SIZE = 10

Step 2: Extract Data from Model and Valuation

A. Extract Revenue Data

Structured Template

- # Revenue by Product (from Task 2 model)
- years = [2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029]
- # Extract from Excel or define manually from model
- product_a = [100, 120, 145, 175, 210, 252, 302, 363, 435, 522]
- product_b = [80, 95, 115, 138, 165, 198, 238, 285, 342, 411]
- product_c = [50, 62, 78, 98, 122, 153, 191, 239, 299, 374]
- product_d = [30, 38, 48, 61, 77, 97, 122, 153, 191, 239]
- # Revenue by Geography
- north_america = [150, 180, 220, 265, 320, 384, 461, 553, 664, 797]
- europe = [80, 95, 115, 140, 170, 204, 245, 294, 353, 423]
- asia_pacific = [40, 50, 63, 80, 101, 127, 159, 199, 249, 311]
- rest_of_world = [20, 25, 32, 40, 51, 64, 80, 100, 125, 156]

B. Extract Margin Data

Structured Template

- # Margin evolution
- gross_margin = [58.0, 59.2, 60.5, 61.8, 63.0, 64.5, 66.0, 67.0, 67.5, 68.0]
- ebitda_margin = [12.0, 15.5, 18.8, 22.0, 25.0, 28.0, 30.5, 32.0, 33.0, 34.0]
- fcf_margin = [8.0, 11.0, 14.5, 18.0, 21.0, 24.0, 26.5, 28.0, 29.0, 30.0]

C. Extract DCF Sensitivity Data

Structured Template

- # DCF Sensitivity (from Task 3 valuation)
- wacc_values = [7.0, 8.0, 9.0, 10.0, 11.0, 12.0]
- terminal_growth = [1.5, 2.0, 2.5, 3.0, 3.5]
- # Price per share matrix (rows = WACC, columns = terminal growth)
- dcf_sensitivity = np.array([
- [66, 71, 76, 82, 89],
- [58, 62, 67, 72, 78],

- [52, 55, 59, 63, 68],
- [47, 50, 53, 56, 60],
- [42, 45, 48, 51, 54],
- [39, 41, 44, 46, 49]
-])

D. Extract Valuation Ranges

Structured Template

- # Valuation Football Field (from Task 3)
- valuation_methods = ['DCF Analysis', 'Trading Comps\n(NTM)', 'Precedent\nTransactions']
- valuation_low = [48, 45, 52]
- valuation_high = [62, 57, 66]
- current_price = 50
- target_price = 55

Step 3: Create Mandatory Charts

Chart 1: Revenue by Product - Stacked Area ★ MANDATORY

Structured Template

- def create_revenue_by_product_chart():
- """Create revenue by product stacked area chart"""
- fig, ax = plt.subplots(figsize=(10, 6))
- # Create stacked area chart
- ax.stackplot(years, product_a, product_b, product_c, product_d,
- labels=['Product A', 'Product B', 'Product C', 'Product D'],
- colors=['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728'],
- alpha=0.8)
- # Formatting
- ax.set_xlabel('Year', fontsize=12, fontweight='bold')
- ax.set_ylabel('Revenue (\$M)', fontsize=12, fontweight='bold')
- ax.set_title('Figure 3 - Revenue by Product/Segment (2020-2029E)',
- fontsize=14, fontweight='bold', pad=20)
- # Legend

- `ax.legend(loc='upper left', frameon=False, fontsize=10)`
- `# Grid`
- `ax.grid(axis='y', alpha=0.3, linestyle='--')`
- `ax.set_axisbelow(True)`
- `# Remove top and right spines`
- `ax.spines['top'].set_visible(False)`
- `ax.spines['right'].set_visible(False)`
- `# Add vertical line to separate historical from projected`
- `ax.axvline(x=2024, color='gray', linestyle='--', linewidth=1, alpha=0.5)`
- `ax.text(2024.2, ax.get_ylim()[1]*0.95, 'Projected →',`
- `fontsize=9, color='gray', ha='left')`
- `# Source line`
- `fig.text(0.12, 0.02, 'Source: Company data, [Firm] estimates',`
- `fontsize=9, style='italic', color='gray')`
- `# Save`
- `plt.tight_layout()`
- `plt.savefig('chart_03_revenue_by_product_stacked_area.png',`
- `dpi=300, bbox_inches='tight', facecolor='white')`
- `plt.close()`
- `print("✓ Created: chart_03_revenue_by_product_stacked_area.png")`
- `create_revenue_by_product_chart()`

Chart 2: Revenue by Geography - Stacked Bar ★ MANDATORY

Structured Template

- `def create_revenue_by_geography_chart():`
- `"""Create revenue by geography stacked bar chart"""`
- `years_labels = ['2020', '2021', '2022', '2023', '2024',`
- `'2025E', '2026E', '2027E', '2028E', '2029E']`
- `fig, ax = plt.subplots(figsize=(10, 6))`
- `# Create stacked bar chart`
- `width = 0.6`
- `x = np.arange(len(years_labels))`

- `p1 = ax.bar(x, north_america, width, label='North America', color='#1f77b4')`
- `p2 = ax.bar(x, europe, width, bottom=north_america,`
- `label='Europe', color='#ff7f0e')`
- `p3 = ax.bar(x, asia_pacific, width,`
- `bottom=np.array(north_america) + np.array(europe),`
- `label='Asia-Pacific', color='#2ca02c')`
- `p4 = ax.bar(x, rest_of_world, width,`
- `bottom=np.array(north_america) + np.array(europe) + np.array(asia_pacific),`
- `label='Rest of World', color='#d62728')`
- `# Formatting`
- `ax.set_xlabel('Year', fontsize=12, fontweight='bold')`
- `ax.set_ylabel('Revenue ($M)', fontsize=12, fontweight='bold')`
- `ax.set_title('Figure 4 - Revenue by Geography (2020-2029E)',`
- `fontsize=14, fontweight='bold', pad=20)`
- `ax.set_xticks(x)`
- `ax.set_xticklabels(years_labels, rotation=45, ha='right')`
- `# Legend`
- `ax.legend(loc='upper left', frameon=False, fontsize=10)`
- `# Grid`
- `ax.grid(axis='y', alpha=0.3, linestyle='--')`
- `ax.set_axisbelow(True)`
- `# Remove top and right spines`
- `ax.spines['top'].set_visible(False)`
- `ax.spines['right'].set_visible(False)`
- `# Source line`
- `fig.text(0.12, 0.02, 'Source: Company data, [Firm] estimates',`
- `fontsize=9, style='italic', color='gray')`
- `# Save`
- `plt.tight_layout()`
- `plt.savefig('chart_04_revenue_by_geography_stacked_bar.png',`
- `dpi=300, bbox_inches='tight', facecolor='white')`
- `plt.close()`

- `print("✓ Created: chart_04_revenue_by_geography_stacked_bar.png")`
- `create_revenue_by_geography_chart()`

Chart 3: DCF Sensitivity - Heatmap ★ MANDATORY

Structured Template

- `def create_dcf_sensitivity_heatmap():`
- `"""Create DCF sensitivity analysis heatmap"""`
- `# Create DataFrame`
- `df = pd.DataFrame(dcf_sensitivity,`
- `index=[f'{w}%' for w in wacc_values],`
- `columns=[f'{g}%' for g in terminal_growth])`
- `fig, ax = plt.subplots(figsize=(8, 6))`
- `# Create heatmap`
- `sns.heatmap(df, annot=True, fmt='d', cmap='RdYlGn',`
- `cbar_kws={'label': 'Price per Share ($)'},`
- `linewidths=0.5, linecolor='white',`
- `ax=ax, vmin=35, vmax=95)`
- `# Formatting`
- `ax.set_xlabel('Terminal Growth Rate', fontsize=12, fontweight='bold')`
- `ax.set_ylabel('WACC', fontsize=12, fontweight='bold')`
- `ax.set_title('Figure 28 - DCF Sensitivity Analysis ($/share)',`
- `fontsize=14, fontweight='bold', pad=20)`
- `# Rotate y-axis labels`
- `plt.yticks(rotation=0)`
- `# Source line`
- `fig.text(0.12, 0.02, 'Source: [Firm] estimates',`
- `fontsize=9, style='italic', color='gray')`
- `# Save`
- `plt.tight_layout()`
- `plt.savefig('chart_28_dcf_sensitivity_heatmap.png',`
- `dpi=300, bbox_inches='tight', facecolor='white')`
- `plt.close()`

- print("✓ Created: chart_28_dcf_sensitivity_heatmap.png")
- create_dcf_sensitivity_heatmap()

Chart 4: Valuation Football Field ★ MANDATORY

Structured Template

- def create_valuation_football_field():
- """Create valuation football field chart"""
- fig, ax = plt.subplots(figsize=(10, 5))
- # Create horizontal bars
- y_positions = np.arange(len(valuation_methods))
- colors = ['#1f77b4', '#ff7f0e', '#2ca02c']
- for i, (method, low, high, color) in enumerate(
- zip(valuation_methods, valuation_low, valuation_high, colors)):
- ax.barh(i, high - low, left=low, height=0.6,
- color=color, alpha=0.7, label=method)
- # Add value labels at ends
- ax.text(low - 1, i, f'\${low}', va='center', ha='right', fontsize=10)
- ax.text(high + 1, i, f'\${high}', va='center', ha='left', fontsize=10)
- # Add current price line
- ax.axvline(x=current_price, color='red', linestyle='-', linewidth=2,
- label=f'Current: \${current_price}', alpha=0.7)
- # Add target price line
- ax.axvline(x=target_price, color='black', linestyle='-', linewidth=2,
- label=f'Target: \${target_price}')
- # Formatting
- ax.set_yticks(y_positions)
- ax.set_yticklabels(valuation_methods, fontsize=11)
- ax.set_xlabel('Price Per Share (\$)', fontsize=12, fontweight='bold')
- ax.set_title('Figure 32 - Valuation Football Field',
- fontsize=14, fontweight='bold', pad=20)
- # Set x-axis limits
- ax.set_xlim(40, 70)

- # Remove spines
- ax.spines['top'].set_visible(False)
- ax.spines['right'].set_visible(False)
- ax.spines['left'].set_visible(False)
- # Grid
- ax.grid(axis='x', alpha=0.3, linestyle='--')
- ax.set_axisbelow(True)
- # Legend
- ax.legend(loc='upper right', frameon=False, fontsize=9)
- # Source line
- fig.text(0.12, 0.02, 'Source: [Firm] estimates',
- fontsize=9, style='italic', color='gray')
- # Save
- plt.tight_layout()
- plt.savefig('chart_32_valuation_football_field.png',
- dpi=300, bbox_inches='tight', facecolor='white')
- plt.close()
- print("✓ Created: chart_32_valuation_football_field.png")
- create_valuation_football_field()

Step 4: Create Remaining Required Charts (Charts 1-34)

Complete the 25 REQUIRED charts by creating all remaining charts from the required list. Each chart has a specific purpose in Task 5.

Investment Summary (1 chart)

Structured Template

- # chart_01: Stock Price Performance (12-24 months)
- # - Line chart showing stock price over time vs. market index
- # - Used on Page 1 of final report

Financial Performance (5 additional charts beyond chart_03 and chart_04)

Structured Template

- # chart_02: Revenue Growth Trajectory

- # chart_10: Gross Margin Evolution
- # chart_11: EBITDA Margin Progression
- # chart_12: Free Cash Flow Trend
- # chart_14: Scenario Comparison (Bull/Base/Bear)

Company 101 Section (7 charts)

Structured Template

- # chart_05: Company Overview/Timeline
- # chart_06: Key Milestones Timeline
- # chart_07: Organizational Structure
- # chart_08: Product Portfolio Overview
- # chart_09: Customer Segmentation
- # chart_15: Market Size Evolution (TAM)
- # chart_16: Competitive Positioning Matrix

Competitive & Market (2 charts)

Structured Template

- # chart_17: Market Share Breakdown
- # chart_18: Competitive Benchmarking

Scenario Analysis (1 chart)

Structured Template

- # chart_13: Operating Metrics Dashboard

Valuation Section (6 additional charts beyond chart_28 and chart_32)

Structured Template

- # chart_29: DCF Valuation Waterfall
- # chart_30: Trading Comps Scatter Plot
- # chart_31: Peer Multiples Comparison
- # chart_33: Price Target Scenarios
- # chart_34: Historical Valuation Multiples

Use consistent formatting across all charts:

- 300 DPI resolution

- Professional color scheme
- Clear labels, legends, and titles
- Figure numbers (e.g., “Figure 5 - Company Timeline”)
- Source citations at bottom

Step 4B: Create Optional Charts (For 26-35 Total)

Optional: Add 1-10 additional charts from this list for greater visual density:

Structured Template

- # chart_19: Customer Acquisition Trends
- # chart_20: Unit Economics Evolution
- # chart_21: Product Roadmap Timeline
- # chart_22: Geographic Expansion Map
- # chart_23: R&D Investment Trends
- # chart_24: Sales & Marketing Efficiency
- # chart_25: Working Capital Trends
- # chart_26: Debt Maturity Schedule
- # chart_27: Ownership Structure
- # chart_35: Analyst Price Target Distribution

These optional charts provide additional visual storytelling and help achieve the “1 chart per 200-300 words” density target in Task 5.

Step 5: Create Chart Index

Create a text file documenting all charts:

Structured Template

- `def create_chart_index():`
- `“”“Create index of all charts”””`
- `# 25 REQUIRED CHARTS`
- `required_charts = [`
- `“chart_01_stock_price_performance.png - Stock Price Performance (12-24M)”,`
- `“chart_02_revenue_growth_trajectory.png - Revenue Growth Trajectory”,`
- `“chart_03_revenue_by_product_stacked_area.png - Revenue by Product [MANDATORY]”,`
- `“chart_04_revenue_by_geography_stacked_bar.png - Revenue by Geography [MANDATORY]”,`
- `“chart_05_company_overview.png - Company Overview/Timeline”,`

- “chart_06_key_milestones_timeline.png - Key Milestones Timeline”,
- “chart_07_organizational_structure.png - Organizational Structure”,
- “chart_08_product_portfolio.png - Product Portfolio Overview”,
- “chart_09_customer_segmentation.png - Customer Segmentation”,
- “chart_10_gross_margin_evolution.png - Gross Margin Evolution”,
- “chart_11_ebitda_margin_progression.png - EBITDA Margin Progression”,
- “chart_12_free_cash_flow_trend.png - Free Cash Flow Trend”,
- “chart_13_operating_metrics_dashboard.png - Operating Metrics Dashboard”,
- “chart_14_scenario_comparison.png - Scenario Comparison (Bull/Base/Bear)”,
- “chart_15_market_size_evolution.png - Market Size Evolution (TAM)”,
- “chart_16_competitive_positioning.png - Competitive Positioning Matrix”,
- “chart_17_market_share.png - Market Share Breakdown”,
- “chart_18_competitive_benchmarking.png - Competitive Benchmarking”,
- “chart_28_dcf_sensitivity_heatmap.png - DCF Sensitivity Heatmap [MANDATORY]”,
- “chart_29_dcf_waterfall.png - DCF Valuation Waterfall”,
- “chart_30_trading_comps_scatter.png - Trading Comps Scatter Plot”,
- “chart_31_peer_multiples_comparison.png - Peer Multiples Comparison”,
- “chart_32_valuation_football_field.png - Valuation Football Field [MANDATORY]”,
- “chart_33_price_target_scenarios.png - Price Target Scenarios”,
- “chart_34_historical_valuation_multiples.png - Historical Valuation Multiples”,
-]
- # 10 OPTIONAL CHARTS (for 26-35 range)
- optional_charts = [
- “chart_19_customer_acquisition_trends.png - Customer Acquisition Trends [OPTIONAL]”,
- “chart_20_unit_economics_evolution.png - Unit Economics Evolution [OPTIONAL]”,
- “chart_21_product_roadmap_timeline.png - Product Roadmap Timeline [OPTIONAL]”,
- “chart_22_geographic_expansion_map.png - Geographic Expansion Map [OPTIONAL]”,
- “chart_23_rd_investment_trends.png - R&D Investment Trends [OPTIONAL]”,
- “chart_24_sales_marketing_efficiency.png - Sales & Marketing Efficiency [OPTIONAL]”,
- “chart_25_working_capital_trends.png - Working Capital Trends [OPTIONAL]”,
- “chart_26_debt_maturity_schedule.png - Debt Maturity Schedule [OPTIONAL]”,
- “chart_27_ownership_structure.png - Ownership Structure [OPTIONAL]”,

- “chart_35_analyst_price_targets.png - Analyst Price Target Distribution [OPTIONAL]”,
-]
- with open('chart_index.txt', 'w') as f:
- f.write(“CHART INDEX FOR [COMPANY] EQUITY RESEARCH REPORT\n”)
- f.write(“=” * 60 + “\n\n”)
- f.write(“4 MANDATORY CHARTS (Must be present):\n”)
- f.write(“- chart_03: Revenue by Product (Stacked Area) ★\n”)
- f.write(“- chart_04: Revenue by Geography (Stacked Bar) ★\n”)
- f.write(“- chart_28: DCF Sensitivity (Heatmap) ★\n”)
- f.write(“- chart_32: Valuation Football Field ★\n\n”)
- f.write(“25 REQUIRED CHARTS:\n”)
- for chart in required_charts:
- f.write(f“ {chart}\n”)
- f.write(“\n10 OPTIONAL CHARTS (for 26-35 total):\n”)
- for chart in optional_charts:
- f.write(f“ {chart}\n”)
- f.write(“\n” + “=” * 60 + “\n”)
- f.write(“NOTE: Task 5 will embed ALL charts created (25-35) throughout\n”)
- f.write(“the report for visual density (1 chart every 200-300 words).\n”)
- print(“✓ Created: chart_index.txt”)
- create_chart_index()

Step 6: Quality Check

Run verification checks:

Structured Template

- import os
- def verify_charts():
- “”“Verify all charts were created successfully””“
- mandatory_charts = [
- 'chart_03_revenue_by_product_stacked_area.png',
- 'chart_04_revenue_by_geography_stacked_bar.png',
- 'chart_28_dcf_sensitivity_heatmap.png',

- 'chart_32_valuation_football_field.png'
-]
- print("\n" + "="*60)
- print("CHART GENERATION VERIFICATION")
- print("="*60)
- # Check mandatory charts
- print("\n1. MANDATORY CHARTS:")
- all_mandatory_present = True
- for chart in mandatory_charts:
- if os.path.exists(chart):
- size = os.path.getsize(chart) / 1024 # KB
- print(f'✓ {chart} ({size:.1f} KB)')
- else:
- print(f'✗ MISSING: {chart}')
- all_mandatory_present = False
- # Count total charts
- chart_files = [f for f in os.listdir('.') if f.startswith('chart_') and f.endswith('.png')]
- print(f"\n2. TOTAL CHARTS: {len(chart_files)}")
- print(f'Target: 25-35 charts')
- print(f'Status: {'✓ PASS' if 25 <= len(chart_files) <= 35 else '⚠ WARNING'})
- # Check file sizes (should be > 50KB for 300 DPI)
- print("\n3. FILE SIZE CHECK:")
- small_files = []
- for chart in chart_files[:5]: # Sample first 5
- size = os.path.getsize(chart) / 1024
- if size < 50:
- small_files.append(chart)
- print(f'{chart}: {size:.1f} KB')
- if small_files:
- print(f'⚠ WARNING: {len(small_files)} files may be low resolution')
- else:
- print(f'✓ All sampled files have adequate size')

- # Final verdict
 - `print("\n" + "="*60)`
 - `if all_mandatory_present and 25 <= len(chart_files) <= 35:`
 - `print("✓ VERIFICATION PASSED - Ready for Task 5")`
 - `else:`
 - `print("✗ VERIFICATION FAILED - Review missing charts")`
 - `print("="*60 + "\n")`
 - `verify_charts()`
-

Quality Standards

Visual Quality

- ☐ High resolution (300 DPI minimum)
- ☐ Professional color scheme (consistent across all charts)
- ☐ Clear, legible text (no fonts smaller than 9pt)
- ☐ Proper aspect ratio (no distortion)
- ☐ No pixelation or artifacts

Data Accuracy

- ☐ Data matches source (financial model and valuation)
- ☐ Proper units and labels (\$ millions, percentages, etc.)
- ☐ Appropriate scale and range
- ☐ Consistent time periods across charts
- ☐ Verified calculations

Formatting Quality

- ☐ Consistent styling across all charts
- ☐ Proper figure numbering (sequential)
- ☐ Clear titles and captions
- ☐ Source citations on every chart
- ☐ Professional appearance

Completeness

- ☐ All 4 mandatory charts created
 - ☐ 25-35 total charts
 - ☐ Proper file naming (chart_01, chart_02, etc.)
 - ☐ Chart index created
 - ☐ Ready for embedding in Word
-

Chart Types Reference

When to Use Each Chart Type

Line Charts: Time series trends (revenue, margins, stock price)

Stacked Area: Revenue by product ★, market size composition

Stacked Bar: Revenue by geography ★, quarterly breakdowns

Heatmap: DCF sensitivity ★, correlation matrices

Horizontal Bar: Valuation football field ★, peer rankings

Waterfall: Revenue bridges, margin analysis, DCF build-up

Scatter/Bubble: Growth vs. valuation, competitive positioning

2×2 Matrix: Competitive positioning, product portfolio

File Naming Convention

Always use this format:

Structured Template

- chart_[NUMBER]_[DESCRIPTION].png
- Examples:
- chart_01_stock_price_performance.png
- chart_03_revenue_by_product_stacked_area.png
- chart_28_dcf_sensitivity_heatmap.png

Number charts sequentially based on their position in the report, not creation order.

Common Chart Generation Issues

Issue 1: Low Resolution

Problem: Chart looks pixelated **Solution:** Ensure *dpi=300* in *plt.savefig()*

Issue 2: Text Cutoff

Problem: Labels or titles cut off at edges **Solution:** Use `bbox_inches='tight'` in `plt.savefig()`

Issue 3: Poor Colors

Problem: Colors don't look professional **Solution:** Use established palettes like Tableau10 or define custom corporate colors

Issue 4: Overlapping Labels

Problem: Axis labels overlap **Solution:** Rotate labels (e.g., `rotation=45`) or reduce font size

Issue 5: White Space

Problem: Too much white space around chart **Solution:** Use `plt.tight_layout()` before saving

Success Criteria

A successful chart package should:

1. **Include all 4 mandatory charts** (verified) ★
 - chart_03: Revenue by Product
 - chart_04: Revenue by Geography
 - chart_28: DCF Sensitivity
 - chart_32: Valuation Football Field
2. **Create 25 required charts minimum** (verified)
3. **Optional: 1-10 additional charts** for 26-35 total
4. Have consistent professional styling across all charts
5. Be high resolution (300 DPI) for print quality
6. Have clear labels, legends, and titles on every chart
7. Include proper figure numbers and source citations
8. Be ready for immediate embedding in Word
9. Cover all key financial metrics and analyses
10. Tell a visual story complementing the written analysis
11. Be accurate and auditable to source data (model/valuation)
12. All charts packaged in zip file with chart index

Remember: Task 5 will embed ALL charts created (25-35) throughout the report for visual density.

Output Files

After completing Task 4, deliverables include:

25 REQUIRED Chart Files (Minimum):

1. chart_01_stock_price_performance.png
2. chart_02_revenue_growth_trajectory.png
3. chart_03_revenue_by_product_stacked_area.png ★ MANDATORY
4. chart_04_revenue_by_geography_stacked_bar.png ★ MANDATORY
5. chart_05_company_overview.png
6. chart_06_key_milestones_timeline.png
7. chart_07_organizational_structure.png
8. chart_08_product_portfolio.png
9. chart_09_customer_segmentation.png
10. chart_10_gross_margin_evolution.png
11. chart_11_ebitda_margin_progression.png
12. chart_12_free_cash_flow_trend.png
13. chart_13_operating_metrics_dashboard.png
14. chart_14_scenario_comparison.png
15. chart_15_market_size_evolution.png
16. chart_16_competitive_positioning.png
17. chart_17_market_share.png
18. chart_18_competitive_benchmarking.png

19-27. *Reserved for optional charts if created*

1. chart_28_dcf_sensitivity_heatmap.png ★ MANDATORY
2. chart_29_dcf_waterfall.png
3. chart_30_trading_comps_scatter.png
4. chart_31_peer_multiples_comparison.png
5. chart_32_valuation_football_field.png ★ MANDATORY
6. chart_33_price_target_scenarios.png
7. chart_34_historical_valuation_multiples.png
8. *Reserved for optional chart if created*

10 OPTIONAL Chart Files (For 26-35 Total):

- chart_19 through chart_27, chart_35 (if created)

Chart Index (1 text file):

- chart_index.txt (listing all charts with descriptions and categories)

All chart files must be:

- 300 DPI resolution (print quality)
- 6-10 inches wide (standard Word embedding size)
- White background (professional appearance)
- PNG format (lossless quality)
- Ready for immediate Word embedding

Final Step: Package All Charts

Create a zip file containing all chart files and the chart index:

Structured Template

- [Company]_Charts_[Date].zip
- chart_01_stock_price_performance.png
- chart_02_revenue_growth_trajectory.png
- chart_03_revenue_by_product_stacked_area.png ★
- chart_04_revenue_by_geography_stacked_bar.png ★
- chart_05_company_overview.png
- ... (all 25-35 chart files)
- chart_28_dcf_sensitivity_heatmap.png ★
- chart_32_valuation_football_field.png ★
- chart_34_historical_valuation_multiples.png
- chart_index.txt

Example: *Tesla_Charts_2024-10-28.zip*

Why this matters: Task 5 will embed ALL charts created (25-35) throughout the report. The report requires visual density (1 chart per 200-300 words), so all charts serve a purpose—either for specific analytical sections or for visual storytelling and page density.

- Verify all 25-35 charts are present
- Extract charts for Task 5 (Report Assembly)

Next Steps

After completing Task 4, the zip file will be used for:

- **Task 5 (Report Assembly):** Extract charts and embed all into the final DOCX report at appropriate locations throughout the document

The 4 mandatory charts are critical for the valuation and financial analysis sections of the report.

3.26 Skill Resource: [skills/initiating-coverage/references/task5-report-assembly.md](#)

Task 5: Report Assembly - Detailed Workflow

This document provides step-by-step instructions for executing Task 5 (Report Assembly) of the initiating-coverage skill.

Task Overview

Purpose: Write and assemble the comprehensive final DOCX report.

Prerequisites: ⚠️ Verify before starting - ALL PREVIOUS TASKS REQUIRED

- **Required:** Company research from Task 1
- **Required:** Financial model from Task 2
- **Required:** Valuation analysis from Task 3
- **Required:** Chart files from Task 4

⚠️ **CRITICAL: DO NOT START THIS TASK UNLESS ALL TASKS 1-4 ARE COMPLETE**

This is the final assembly task. It cannot be completed without all previous work products.

IF ANY OF TASKS 1, 2, 3, OR 4 ARE NOT COMPLETE: Stop immediately and inform the user which tasks need to be completed first. The specific requirements are:

- Task 1: Company research document (6-8K words)
- Task 2: Financial model with all 6 tabs
- Task 3: Valuation analysis with price target and recommendation
- Task 4: Charts zip file with 25-35 charts

Do not attempt to create placeholder content, substitute missing sections, or assemble an incomplete report. The report requires ALL inputs to be publication-ready.

Output: Comprehensive Equity Research Report (.docx)

- Length: 30-50 pages (MINIMUM 30)
- Word count: 10,000-15,000 words (MINIMUM 10,000)
- Charts: 25-35 embedded images

- Tables: 12-20 comprehensive tables
-

CRITICAL INSTRUCTION: SPARE NO TOKENS OR EFFORT

THIS IS THE FINAL DELIVERABLE. GO ALL OUT. NO SHORTCUTS. NO ABBREVIATIONS.

After completing 4 previous tasks, this final task assembles everything into publication-ready institutional research. **This must be PERFECT.**

Absolute Requirements

DO:

-  **Use ENTIRE token budget if needed** - This is what it's for
-  **Write EVERY section in FULL** - Not summaries, not placeholders, FULL CONTENT
-  **Include ALL 25-35 charts** - Embed every single chart from Task 4 throughout the document
-  **Create ALL 12-20 tables** - Extract every financial table from Excel, don't skip any
-  **Copy ALL 6-8K words from Task 1** - Use Company 101 content verbatim (40-50% of report)
-  **Write 2,000-3,000 words on Projection Assumptions** - Product-by-product, region-by-region detail
-  **Write 1,500-2,000 words on Scenario Analysis** - Specific Bull/Base/Bear parameters
-  **Achieve 10,000-15,000 total words** - This is a MINIMUM, not a suggestion
-  **Produce 30-50 pages minimum** - Text-dense with charts every 200-300 words
-  **Professional institutional quality** - Indistinguishable from JPMorgan/Goldman Sachs

NEVER:

-  **"This section would include..."** - WRITE THE ACTUAL SECTION
-  **"Charts would be inserted here..."** - INSERT THE ACTUAL CHARTS
-  **"See financial model for details..."** - EXTRACT AND WRITE THE DETAILS
-  **"For brevity, we'll summarize..."** - NO SUMMARIZING, WRITE IN FULL
-  **Skip sections to conserve tokens** - USE WHATEVER TOKENS ARE NEEDED
-  **Create abbreviated versions** - EVERY SECTION MUST BE COMPLETE
-  **Reference external files instead of including content** - INCLUDE EVERYTHING

Quality Standard

This report will be read by institutional investors making million-dollar decisions.

It must be:

- **Complete:** Every section written in full with no placeholders

- **Comprehensive:** All data extracted and included, all charts embedded
- **Professional:** Proper formatting, citations, tables, charts throughout
- **Thorough:** Deep analysis with specific numbers, detailed assumptions, complete scenarios
- **Dense:** 60-80% page coverage with text and visuals on every page

Creating the final work product of a 6-10 hour equity research process. Make it count.

Input Verification (CRITICAL)

BEFORE STARTING - ALL TASKS MUST BE COMPLETE:

Task 1 Verification:

- ☐ Company research document exists? (6-8K words)
- ☐ Management bios complete? (300-400 words × 3-4 execs)
- ☐ Competitive analysis complete? (5-10 competitors)
- ☐ Risk assessment complete? (8-12 risks)

Task 2 Verification:

- ☐ Financial model exists and can be opened?
- ☐ Model has projections (5 years)?
- ☐ Scenarios exist (Bull/Base/Bear)?
- ☐ Revenue by product table complete (20-30 rows)?
- ☐ Revenue by geography table complete (15-20 rows)?

Task 3 Verification:

- ☐ Valuation analysis complete?
- ☐ Price target determined?
- ☐ Recommendation set? (BUY/HOLD/SELL)
- ☐ DCF analysis complete with sensitivity table?
- ☐ Comparable companies analysis complete with statistical summary?

Task 4 Verification:

- ☐ 25-35 chart files exist?
- ☐ All 4 mandatory charts present?
 - ☐ Revenue by product (stacked area)

- ☐ Revenue by geography (stacked bar)
- ☐ DCF sensitivity (heatmap)
- ☐ Valuation football field
- ☐ Chart files accessible and can be opened?
- ☐ Chart index created?

IF ANY VERIFICATION FAILS: Stop and complete missing task first.

Report Specifications

Length Requirements

- **Pages:** 30-50 (MINIMUM 30 pages)
- **Word Count:** 10,000-15,000 words (MINIMUM 10,000 words)
- **Charts:** 25-35 embedded PNG/JPG images
- **Tables:** 12-20 comprehensive financial tables
- **Density:** 60-80% page coverage

Critical Sections with Word Counts

Section	Minimum	Target	Critical?
Investment Summary (Page 1)	500	700	
Investment Thesis	800	1,200	
Risk Factors	600	900	
Company Description	800	1,200	
Management Bios	1,000	1,400	
Products & Services	700	1,000	
Projection Assumptions	2,000	3,000	★ YES
Scenario Analysis	1,500	2,000	★ YES
Financial Analysis	1,200	1,800	
Valuation Methodology	800	1,200	

Total: 10,000-15,000 words

Report Structure

Page 1: Investment Summary (CRITICAL PAGE)

This is the most important page. Must have:

1. **“INITIATING COVERAGE” header** (NOT “Company Update”)
2. **Thesis-focused title** (e.g., “AI Platform Leader Positioned for 40% CAGR”)
3. **Rating box** with:
 - Rating (BUY/OUTPERFORM/HOLD/UNDERPERFORM/SELL)
 - Current price
 - Target price
 - 52-week range
 - Market cap
 - Enterprise value
4. **Research analyst information** with credentials
5. **Stock price performance chart** (Figure 1)
6. **3-4 detailed investment bullets** with ■ character
 - Each bullet has **bold topic header** + 3-5 sentences
 - Lead with key numbers
7. **Financial summary table** (2-3 years historical + 2-3 years projected)
 - Years noted as “A” for actual, “E” for estimate

Bullet Format Example:

Structured Template

- ■ **Vertical SaaS leadership and regulatory moat should enable \$50bn+ TAM by 2030.**
- Deep domain expertise in healthcare IT, strong customer retention (95%+ net revenue retention),
- and cross-sell capabilities have driven Acme Health’s market expansion. With the healthcare IT
- market expected to reach \$50bn+ by 2030, Acme Health is well-positioned to capture share given
- its regulatory moat and high switching costs. Management has indicated that 70% of current
- revenue comes from enterprise hospital systems, suggesting strong product-market fit.

Pages 2-5: Investment Thesis & Risks

Investment Thesis (800-1,200 words)

- 3-5 key thesis pillars
- Each pillar: 200-300 words
- Lead with key statistic
- Quantify financial impact
- Include timeline

Risk Assessment (600-900 words)

- 8-12 identified risks
- Organized by category:
 - Company-specific risks (4-6)
 - Industry/market risks (3-4)
 - Financial risks (2-3)
 - Macroeconomic risks (2-3)
- Each risk: 50-100 word description

Pages 6-17: Company 101

Company Description (800-1,200 words)

- What the company does (plain English)
- Business model and monetization
- Geographic presence
- Scale metrics

Company History (800-1,200 words)

- Founding story
- Timeline of major milestones
- Strategic pivots
- Recent developments

Management Team (1,000-1,400 words)

- 300-400 word bio for each of 3-4 key executives
- Include: role, background, accomplishments, education
- Governance structure

Products & Services (700-1,000 words)

- Detailed product portfolio
- Features and differentiation
- Target customers
- Pricing models

Customers & Go-to-Market (500-700 words)

- Customer segments
- Distribution channels

- Sales strategy
- Key partnerships

Industry Overview (800-1,200 words)

- Industry definition and scope
- Market size and growth
- Key trends
- Regulatory environment

Competitive Landscape (700-1,000 words)

- 5-10 key competitors
- Market positioning
- Competitive advantages
- Market share analysis

TAM Analysis (500-700 words)

- Total addressable market sizing
- Market growth projections
- Company's serviceable market

Pages 18-30: Financial Analysis

Historical Financial Analysis (1,200-1,800 words)

- Revenue trends and drivers
- Margin evolution
- Cash flow analysis
- Key metrics trajectory
- Historical context

Projection Assumptions (2,000-3,000 words) ★ CRITICAL

MUST be extremely detailed. Structure:

A. Revenue by Product Assumptions (1,000-1,500 words)

For EACH major product category:

Structured Template

- [Product Category A] Revenue Assumptions
- We project [Product A] revenue to grow from \$XXM in 2024A to \$XXM in 2029E,

- representing a XX% CAGR. This growth is driven by:
- 1. [Driver 1 with specific quantification]
- - Specific metric: from XX to XX
- - Timeline: achieving YY by 2026E
- - Basis: [source or rationale]
- 2. [Driver 2 with specific quantification]
- 3. [Driver 3 with specific quantification]
- [... 8-12 detailed points total for this product ...]
- Specific assumptions by year:
- - 2025E: XX% growth driven by [specific factors]
- - 2026E: XX% growth as [specific factors]
- - 2027-2029E: XX% CAGR as [longer-term factors]
- Key risks to these assumptions include [specific risks].

Repeat for EACH major product category.

B. Geographic Revenue Assumptions (500-800 words)

For EACH major region:

Structured Template

- [Region] Revenue Assumptions
- We project [Region] revenue to grow XX% CAGR from 2024-2029E, reaching \$XXM, driven by:
- 1. [Market dynamic with quantification]
- 2. [Distribution expansion with specifics]
- 3. [Competitive positioning]
- [... 6-8 detailed points total for this region ...]

Repeat for EACH major geographic region.

C. Other Key Assumptions (500-700 words)

- Gross margin evolution (with specific drivers and bridge)
- Operating expense assumptions (R&D, S&M, G&A as % of revenue)
- Working capital assumptions (DSO, DIO, DPO with specific days)
- CapEx as % of sales (with justification)
- Tax rate assumptions

Scenario Analysis (1,500-2,000 words) ★ CRITICAL

MUST have specific parameters for each scenario. Structure:

Bull Case (500-700 words)

Structured Template

- Bull Case: [Title describing key optimistic scenario]
- Probability: XX%
- Key Assumptions:
 - - Revenue CAGR (2024-2029E): XX% (vs. XX% base case)
 - - 2029E Revenue: \$X,XXXm (vs. \$X,XXXm base)
 - - 2029E EBITDA Margin: XX% (vs. XX% base)
 - - Key product growth: XX% CAGR (vs. XX% base)
 - - Geographic expansion: [specific milestones and timeline]
 - - Market share: XX% by 2029E (vs. XX% base)
- Catalysts Required for Bull Case:
 - 1. [Specific catalyst] - Expected timing: [date/quarter]
 - 2. [Specific catalyst] - Expected timing: [date/quarter]
 - 3. [Specific catalyst] - Expected timing: [date/quarter]
- Detailed Rationale:
 - [200-300 words explaining what needs to happen for bull case to materialize.
 - Be specific about product launches, market conditions, competitive dynamics, etc.]
- Valuation Implications:
 - - DCF Value: \$XX per share (XX% upside from current)
 - - Trading Comps: XX.Xx EV/EBITDA implies \$XX per share
 - - Bull Case Target: \$XX per share

Base Case (300-500 words)

Structured Template

- Base Case: [Title describing most likely scenario]
- Probability: XX%
- Key Assumptions:
 - [Similar structure to Bull Case with base assumptions]
- Rationale:
 - [Explain why this is most likely scenario]
- Valuation:
 - - DCF Value: \$XX per share

- - Trading Comps: \$XX per share
- - Base Case Target: \$XX per share (weighted average)

Bear Case (500-700 words)

Structured Template

- Bear Case: [Title describing downside scenario]
- Probability: XX%
- Key Assumptions:
- [Similar structure with downside parameters]
- Downside Triggers:
- 1. [Specific risk event] - Likelihood: [%]
- 2. [Specific risk event] - Likelihood: [%]
- 3. [Specific risk event] - Likelihood: [%]
- Rationale:
- [200-300 words on what would cause bear case]
- Valuation Implications:
- - DCF Value: \$XX per share (XX% downside from current)
- - Trading Comps: \$XX per share
- - Bear Case Target: \$XX per share

Scenario Comparison (200-300 words)

- Comprehensive comparison table with key metrics
- Analysis of probability-weighted outcomes
- Risk/reward assessment
- Path dependency discussion

Growth Drivers (800-1,200 words)

- 3-5 key growth drivers
- Each quantified with specific opportunity size
- Timeline and milestones
- Supporting data from model

Pages 31-40: Valuation Analysis

Valuation Methodology (800-1,200 words)

DCF Analysis (300-400 words)

- Methodology explanation
- Key assumptions:
 - WACC: X.X% (calculation breakdown)
 - Terminal growth: X.X% (rationale)
 - Terminal margin: XX% (justification)
- Sensitivity analysis discussion
- DCF value: \$XX per share

Comparable Companies (300-400 words)

- Peer selection rationale (why these 5-10 companies)
- Statistical summary (max/75th/median/25th/min)
- Multiple selection (why EV/EBITDA vs. EV/Revenue vs. P/E)
- Premium/discount justification (why target deserves premium/discount)
- Comparable companies value: \$XX per share

Precedent Transactions (200-300 words, if applicable)

- Transaction relevance
- Control premium analysis
- Precedent transactions value: \$XX per share

Valuation Reconciliation (200-300 words)

- Weighting rationale (e.g., DCF 50%, Comps 40%, Precedent 10%)
- Weighted average calculation
- Valuation range (low/base/high)
- Final price target: \$XX

Price Target & Recommendation (300-500 words)

- Final recommendation (BUY/OUTPERFORM/HOLD/UNDERPERFORM/SELL)
- Price target: \$XX (XX% upside from current \$XX)
- Time horizon: 12 months
- Key catalysts (3-5 with specific timeframes)
- Key risks to price target (3-5 with impact quantification)

Pages 41-50: Appendices

Data Sources & References

- All sources listed with dates
- Organized by category:
 - SEC Filings (with EDGAR links)
 - Earnings Calls (with transcript links)
 - Company Materials
 - Industry Reports
 - News Articles
- **ALL URLs must be clickable hyperlinks**

Detailed Financial Model Assumptions

- Comprehensive assumptions detail
- Calculation methodologies
- Data sources for historical figures

Additional Supporting Tables

- Extended financial projections
 - Detailed comparable companies data
 - Sensitivity analyses
-

Report Assembly Philosophy

CRITICAL PRINCIPLE 1: A good equity research report is **text-dense with lots of illustrating images**.

Target density: 60-80% page coverage

- Every page should have BOTH text AND visuals
- Charts should be interspersed throughout text, not grouped
- Average 1 chart per page minimum (30-50 pages = 25-35+ charts)
- Tables should break up large text blocks

CRITICAL PRINCIPLE 2: Use Claude's DOCX and XLSX skills to programmatically create the report.

REQUIRED TOOLS (Claude has built-in skills for these):

- **DOCX skill** - To create and manipulate Word documents
 - Read Task 1 .md file → Convert to Word formatting
 - Insert images from Task 4 chart files
 - Create tables
 - Format text, headers, footers, page numbers
 - Add hyperlinks

- **XLSX skill** - To read data from Excel files
 - Extract tables from Task 2 financial model
 - Read Task 3 valuation tabs
 - Pull historical financials from Task 1
- **Direct file operations** - Work with actual files
 - Read: *[Company]_Research_Document_[Date].md*
 - Read: *[Company]_Financial_Model_[Date].xlsx*
 - Read: *chart_01.png, chart_02.png, etc.*
 - Write: *[Company]_Initiation_Report_[Date].docx*

DO NOT: Manually copy/paste or describe what should be done **DO:** Use Claude's DOCX and XLSX skills to open files, extract data, and create the DOCX report

Content Reuse Strategy:

- **Task 1 content (40-50% of report):** Read .md file → Convert to Word format → Add charts
- **Task 2/3 data (30-40% of report):** Read .xlsx file → Extract tables → Write interpretation
- **Original writing (10-20% of report):** Investment thesis, projection assumptions, scenario analysis

This approach:

- Maximizes efficiency (no rewriting 6-8K words that are already good)
 - Maintains quality (Task 1 content is substantive, professional analysis)
 - Focuses effort on value-add (quantitative interpretation and investment thesis)
 - Uses actual files programmatically (not manual work)
-

Step-by-Step Report Assembly Workflow

Step 1: Organize All Inputs and Verify Files

Verify all input files exist:

Use Claude's file operations to check:

- *[Company]_Research_Document_[Date].md* (Task 1)
- *[Company]_Historical_Financials_[Date].xlsx* (Task 1)
- *[Company]_Financial_Model_[Date].xlsx* (Task 2 with Task 3 tabs)
- *[Company]_Valuation_Analysis_[Date].md* (Task 3)
- *[Company]_Charts_[Date].zip* (Task 4) - **Extract this first**

Step 1a: Extract Charts from Zip File

Before proceeding, extract all chart files from the Task 4 zip:

- Locate *[Company]_Charts_[Date].zip*
- Extract all contents to a working directory (e.g., *task4_charts/*)
- Verify 25-35 PNG files were extracted
- Verify *chart_index.txt* is present

Expected folder structure after extraction:

Structured Template

- *[Company]_Report_Working/*
- *[Company]_Research_Document_[Date].md*
- *[Company]_Historical_Financials_[Date].xlsx*
- *[Company]_Financial_Model_[Date].xlsx* (includes Task 3 valuation tabs)
- [Task 2 tabs: Revenue Model, Income Statement, Scenarios, etc.]
- [Task 3 tabs: DCF, Sensitivity, Comps, Valuation Summary]
- *[Company]_Valuation_Analysis_[Date].md*
- *[Company]_Charts_[Date].zip*
- *task4_charts/* (extracted from zip)
- *chart_01_stock_price.png*
- *chart_02_revenue_growth.png*
- *chart_03_revenue_by_product.png* ★
- *chart_04_revenue_by_geography.png* ★
- ... (21-31 more charts)
- *chart_28_dcf_sensitivity.png* ★
- *chart_32_valuation_football_field.png* ★
- *chart_index.txt*
- *sources_and_urls.txt*

Open and inspect files using Claude skills:

1. **Read Task 1 markdown file** - Use Read tool to view content
2. **Open Task 2/3 Excel file** - Use XLSX skill to inspect tabs:
 - Verify required tabs exist: Revenue Model, Income Statement, Scenarios, DCF, Sensitivity Analysis, Comparable Companies
3. **Read Task 3 markdown file** - Use Read tool to view valuation analysis
4. **Check chart files** - Verify all 25-35 PNG files present

Note: Task 2's financial model file now contains both the original modeling tabs (from Task 2) AND the valuation tabs (added by Task 3). This single Excel file contains all quantitative data needed for report assembly.

Step 2: Extract Tables from Excel Using XLSX Skill

Use Claude's XLSX skill to extract data from Excel files:

Table 1: Page 1 Summary Financials

Use XLSX skill to:

1. Open *[Company]_Financial_Model_[Date].xlsx*
2. Read from *Income Statement* tab
3. Extract key rows: Revenue, Gross Profit, EBITDA, Net Income, EPS, FCF
4. Extract years: 2022A, 2023A, 2024A, 2025E, 2026E, 2027E
5. Create summary table with growth rates and margins

Table 2: Full Income Statement (40-50 line items)

Use XLSX skill to:

1. Open *[Company]_Financial_Model_[Date].xlsx*
2. Read entire *Income Statement* tab
3. Extract all line items (40-50 rows)
4. Extract columns for historical (2020A-2024A) + projected years (2025E-2029E)
5. Include all margins and growth rates

Table 3: Revenue by Product (20-30 rows)

Use XLSX skill to:

1. Open *[Company]_Financial_Model_[Date].xlsx*
2. Read from *Revenue Model* tab
3. Navigate to product section (typically starts ~row 5)
4. Extract 20-30 rows showing each product category
5. Include columns: Product name, historical years, projected years, % of Total, YoY Growth

Table 4: Revenue by Geography (15-20 rows)

Use XLSX skill to:

1. Open *[Company]_Financial_Model_[Date].xlsx*
2. Read from *Revenue Model* tab
3. Navigate to geography section (typically starts ~row 40)
4. Extract 15-20 rows showing each geographic region
5. Include columns: Region, historical years, projected years, % of Total, YoY Growth

Table 5: Comparable Companies

Extract from: Task 3 valuation tabs in Task 2's financial model (*Comparable Companies* tab)

Use XLSX skill to:

1. Open *[Company]_Financial_Model_[Date].xlsx*
2. Read from *Comparable Companies* tab (added by Task 3)
3. Extract full table with company names as row headers
4. **CRITICAL:** Verify statistical summary rows are present at bottom:
 - Maximum
 - 75th Percentile
 - Median
 - 25th Percentile
 - Minimum
5. If statistical summary is missing, report ERROR

Expected format:

Structured Template

- Company Ticker Mkt Cap EV/Rev EV/Rev EV/EBITDA EV/EBITDA P/E Rev EBITDA
- (\$B) LTM NTM LTM NTM NTM Growth Margin
- [5-10 peers plus target, then statistical summary]

Additional Tables (7-15 more)

Extract from Task 2 financial model (with Task 3 tabs):

Use XLSX skill to extract these tables:

DCF Assumptions Table (Task 3 *DCF* tab)

- Open *[Company]_Financial_Model_[Date].xlsx*
- Read from DCF tab
- Extract columns A-C (Assumption, Value, Source)
- Extract first 20 rows

DCF Sensitivity Matrix (Task 3 *Sensitivity Analysis* tab)

- Read from Sensitivity Analysis tab
- Extract full sensitivity matrix
- WACC values as row headers
- Terminal growth rates as column headers

Scenario Comparison Table (Task 2 *Scenarios* tab)

- Read from Scenarios tab
- Extract full scenario table
- Metrics as row headers (Revenue, EBITDA, Margins, etc.)
- Columns: Bull, Base, Bear

Other supporting tables to extract:

- Cash flow statement
- Balance sheet highlights
- Key metrics dashboard
- Margin bridge
- Working capital schedule
- TAM sizing table
- Market share table

Create all 12-20 tables with proper formatting.

Step 3: Write Quantitative Sections

These sections interpret the financial model.

Write in this order:

A. Financial Analysis (1,200-1,800 words)

- Analyze historical performance from model
- Discuss trends in revenue, margins, cash flow
- Reference specific charts and tables
- Lead with numbers

B. Projection Assumptions (2,000-3,000 words) ★ CRITICAL

- Follow detailed structure from Report Structure section above
- Must be product-by-product (8-12 points per product)
- Must be region-by-region (6-8 points per region)
- Must include margin, opex, capex, working capital assumptions
- **This section separates amateur from professional analysis**

C. Scenario Analysis (1,500-2,000 words) ★ CRITICAL

- Follow detailed structure from Report Structure section above
- Bull case: specific parameters, catalysts, probability, valuation
- Base case: most likely scenario with rationale
- Bear case: downside triggers and parameters
- Comparison table and analysis
- **Must have specific quantified parameters for each scenario**

D. Growth Drivers (800-1,200 words)

- 3-5 key drivers with quantified opportunities
- Timeline and milestones
- Evidence from model

E. Valuation Methodology (800-1,200 words)

- DCF explanation with assumptions
- Comparables rationale
- Precedent transactions (if applicable)
- Reconciliation and weighting
- Price target derivation

Step 4: Write Synthesis Sections

Write in this order:

A. Investment Thesis (800-1,200 words)

- 3-5 key pillars
- Each pillar: 200-300 words
- Lead with key statistic
- Quantify financial impact
- Include timeline

B. Risk Assessment (600-900 words)

- Pull from Task 1 research document
- Organize into 4 categories

- 8-12 risks total
- Each risk: 50-100 words

C. Price Target & Recommendation (300-500 words)

- Final recommendation
- Price target with upside %
- Key catalysts with timeframes
- Key risks to target

D. Investment Summary (500-700 words) - WRITE LAST

- Page 1 content
- 3-4 detailed bullets with bold headers
- Complete synthesis of all findings
- **Write this section LAST after full analysis complete**

Step 5: Integrate Company Content from Task 1

CRITICAL INSTRUCTION: Use Task 1 research document almost verbatim. DO NOT rewrite.

The company research from Task 1 (6-8K words) is already professional, substantive analysis. Objective:

1. **Reformat for Word** - Convert markdown to DOCX formatting
2. **Insert charts inline** - Add relevant charts from Task 4 throughout the text
3. **Minor style adjustments** - Ensure consistent formatting with rest of report

Extract these sections from Task 1 research document:

- Company description (800-1,200 words) → **Use verbatim, insert company overview charts**
- Company history (800-1,200 words) → **Use verbatim, insert timeline chart**
- Management bios (1,000-1,400 words) → **Use verbatim, insert org chart if available**
- Products & services (700-1,000 words) → **Use verbatim, insert product portfolio charts**
- Customers & GTM (500-700 words) → **Use verbatim, insert customer segmentation charts**
- Industry overview (800-1,200 words) → **Use verbatim, insert market size evolution charts**
- Competitive landscape (700-1,000 words) → **Use verbatim, insert competitive positioning charts**
- TAM analysis (500-700 words) → **Use verbatim, insert TAM sizing charts**
- Risk assessment (600-900 words) → **Use verbatim, format as Investment Thesis & Risks section**

Chart Integration Strategy:

- Every 200-300 words of text → Insert 1 chart
- Company 101 section (pages 6-17) should have 8-12 charts interspersed
- Place charts immediately after the paragraph that discusses the topic
- **Result:** Dense, visually rich pages (60-80% coverage)

Step 6: Assemble DOCX Report

CRITICAL: Create actual DOCX file, NOT markdown.

Assembly Order (Most Efficient):

Phase A: Create Structure & Add Page 1

1. Create DOCX document
2. Set up professional styling (fonts, headers, footers)
3. Create Page 1 - Investment Summary (write this LAST after all analysis complete)
4. Add Table of Contents placeholder

Phase B: Copy Task 1 Content + Insert Charts

This is 40-50% of the report - mostly copy/paste + chart insertion

Use Claude's DOCX skill to:

1. Initialize new DOCX document

- Create new Word document
- Set professional styling (fonts, margins)

2. Read Task 1 markdown file

- Use Read tool: *[Company]_Research_Document_[Date].md*
- Identify sections by markdown headers (**##** Section Title)

3. Extract and convert each section from Task 1 to Word format:

SECTION 1: Investment Thesis & Risks

- Add heading: 'Investment Thesis & Risks' (level 1)
- Extract 'Risk Assessment' section from Task 1 markdown
- Convert markdown formatting to Word formatting (remove **##**, ******, etc.)
- Add paragraphs to Word document (split by blank lines)
- Add new 'Investment Thesis' heading (level 2)
- Write new investment thesis content (800-1,200 words based on all analysis)

SECTION 2: Company 101 (Pages 6-17) Copy each section from Task 1 verbatim with formatting conversion:

- **Company Overview**
 - Add heading: 'Company Overview' (level 1)
 - Extract 'Company Overview' section from Task 1
 - Convert to Word paragraphs
 - Insert chart: *task4_charts/chart_05_company_overview.png* (6 inches wide)
- **Company History**
 - Add heading: 'Company History' (level 1)
 - Extract 'Company History' section from Task 1
 - Convert to Word paragraphs
 - Insert chart: *task4_charts/chart_06_company_timeline.png* (6 inches wide)
- **Management Team**
 - Add heading: 'Management Team' (level 1)
 - Extract 'Management Team' section from Task 1
 - Convert to Word paragraphs
 - Insert chart: *task4_charts/chart_07_org_structure.png* (5 inches wide)
- **Products & Services**
 - Add heading: 'Products & Services' (level 1)
 - Extract 'Products & Services' section from Task 1
 - Add first paragraph
 - Insert chart: *task4_charts/chart_08_product_portfolio.png* (6 inches wide)
 - Add remaining paragraphs
- **Customers & Go-to-Market**
 - Add heading: 'Customers & Go-to-Market' (level 1)
 - Extract section from Task 1
 - Convert to Word paragraphs
 - Insert chart: *task4_charts/chart_09_customer_segments.png* (6 inches wide)
- **Industry Overview**
 - Add heading: 'Industry Overview' (level 1)
 - Extract section from Task 1
 - Add first paragraph
 - Insert chart: *task4_charts/chart_10_market_size_evolution.png* (6 inches wide)
 - Add remaining paragraphs
 - Insert chart: *task4_charts/chart_11_industry_trends.png* (6 inches wide)
- **Competitive Landscape**
 - Add heading: 'Competitive Landscape' (level 1)

- Extract section from Task 1
- Add first paragraph
- Insert chart: *task4_charts/chart_16_competitive_positioning.png* (6 inches wide)
- Add remaining paragraphs
- Insert chart: *task4_charts/chart_17_market_share.png* (5 inches wide)
- **Market Opportunity**
 - Add heading: 'Market Opportunity' (level 1)
 - Extract 'Market Opportunity' section from Task 1
 - Convert to Word paragraphs
 - Insert chart: *task4_charts/chart_15_TAM_sizing.png* (6 inches wide)

Result after Phase B: Pages 6-17 complete (~12 pages, 6-8K words, 8-12 charts embedded)

Key Point: Use DOCX skill to READ from Task 1's .md file and INSERT actual image files. No manual copy/paste required.

Phase C: Add Financial Analysis with Data from Task 2

This requires NEW WRITING interpreting quantitative data

Use Claude's DOCX and XLSX skills to:

SECTION 3: Financial Analysis (Pages 18-30)

1. **Add section heading: 'Financial Analysis' (level 1)**
2. **Historical Financial Analysis (1,200-1,800 words) - NEW WRITING**
 - Add heading: 'Historical Performance' (level 2)
 - Use XLSX skill to open *[Company]_Financial_Model_[Date].xlsx*
 - Read *Income Statement* tab to extract historical data
 - Read *Revenue Model* tab to extract revenue trends
 - Calculate key metrics (e.g., Revenue CAGR from 2020-2024)
 - Write analytical paragraphs interpreting the trends (1,200-1,800 words)
 - Lead with specific numbers: "Revenue grew from \$XXM in 2020 to \$XXM in 2024, representing a XX% CAGR. This growth was driven by..."
 - Insert chart: *task4_charts/chart_02_revenue_growth_trajectory.png* (6 inches wide)
3. **Create Table: Full Income Statement**
 - Add heading: 'Historical Income Statement' (level 3)
 - Use XLSX skill to extract entire Income Statement tab (40-50 rows)
 - Create Word table with all columns (historical years 2020A-2024A + projected 2025E-2029E)
 - Include all line items: Revenue, COGS, Gross Profit, Operating Expenses, EBITDA, Net Income, etc.
4. **Add mandatory charts and tables for Revenue breakdown:**

- Insert chart: *task4_charts/chart_03_revenue_by_product.png* (6.5 inches wide) ★ MANDATORY
 - **Create Table: Revenue by Product (20-30 rows)**
 - Use XLSX skill to extract from Revenue Model tab (product section, typically rows 5-35)
 - Create Word table showing each product category with historical and projected years
 - Include columns: Product name, historical years, projected years, % of Total, YoY Growth
 - Insert chart: *task4_charts/chart_04_revenue_by_geography.png* (6.5 inches wide) ★ MANDATORY
 - **Create Table: Revenue by Geography (15-20 rows)**
 - Use XLSX skill to extract from Revenue Model tab (geography section, typically rows 40-60)
 - Create Word table showing each geographic region with historical and projected years
5. **Add additional financial charts:**
- Insert chart: *task4_charts/chart_10_gross_margin_evolution.png* (6 inches wide)
 - Insert chart: *task4_charts/chart_11_ebitda_margin_progression.png* (6 inches wide)
 - Insert chart: *task4_charts/chart_12_free_cash_flow_trend.png* (6 inches wide)
6. **Projection Assumptions (2,000-3,000 words) ★ CRITICAL - NEW WRITING**
- Add heading: 'Projection Assumptions' (level 2)
 - Use XLSX skill to read Scenarios tab to inform assumptions
 - Use XLSX skill to read Revenue Model tab for specific product/geography projections
 - Add heading: 'Revenue Assumptions by Product' (level 3)
 - Write detailed product-by-product assumptions (8-12 points per major product)
 - Write detailed region-by-region assumptions (6-8 points per major region)
 - Include margin, opex, capex, working capital assumptions
 - **Total: 2,000-3,000 words of specific, quantified assumptions**
7. **Scenario Analysis (1,500-2,000 words) ★ CRITICAL - NEW WRITING**
- Add heading: 'Scenario Analysis' (level 2)
 - Use XLSX skill to extract scenario data from Scenarios tab
 - Extract Bull/Base/Bear parameters for key metrics (2029E Revenue, EBITDA Margin, etc.)
 - Write Bull Case (500-700 words): specific parameters, catalysts, probability, valuation
 - Write Base Case (300-500 words): most likely scenario with rationale
 - Write Bear Case (500-700 words): downside triggers, parameters, probability, valuation
 - Write Scenario Comparison (200-300 words)
 - Insert chart: *task4_charts/chart_14_scenario_comparison.png* (6 inches wide)
 - **Create Table: Scenario Comparison**
 - Use XLSX skill to extract from Scenarios tab
 - Create Word table with Bull/Base/Bear columns and key metrics as rows
8. **Growth Drivers (800-1,200 words) - NEW WRITING**
- Add heading: 'Key Growth Drivers' (level 2)

- Write 3-5 key drivers with specific quantified opportunities
- Include timelines and milestones
- Reference specific data from financial model

Result after Phase C: Pages 18-30 complete (~13 pages, 5-7K words, 7-8 charts, 6-8 tables)

Key Point: Use XLSX skill to READ data from Task 2's Excel file, use the data to inform NEW analytical writing, and use DOCX skill to create Word tables from Excel data.

Phase D: Add Valuation Analysis from Task 3

Mix of copying Task 3 analysis + inserting data from Excel

Use Claude's DOCX and XLSX skills to:

SECTION 4: Valuation Analysis (Pages 31-40)

1. Add section heading: 'Valuation Analysis' (level 1)

2. Read Task 3 markdown file

- Use Read tool: *[Company]_Valuation_Analysis_[Date].md*
- Identify sections by markdown headers: DCF Analysis, Comparable Companies, Price Target

3. DCF Analysis section

- Add heading: 'DCF Analysis' (level 2)
- Extract 'DCF Analysis' section from Task 3 markdown
- Convert markdown to Word paragraphs
- Insert chart: *task4_charts/chart_28_dcf_sensitivity_heatmap.png* (6 inches wide) ★ MANDATORY
- **Create Table: DCF Key Assumptions**
 - Add heading: 'DCF Key Assumptions' (level 3)
 - Use XLSX skill to open *[Company]_Financial_Model_[Date].xlsx*
 - Read DCF tab (columns A-C, first 20 rows: Assumption, Value, Source)
 - Create Word table from extracted data
- **Create Table: DCF Sensitivity Matrix**
 - Use XLSX skill to read Sensitivity Analysis tab
 - Extract full sensitivity matrix (WACC values as rows, terminal growth as columns)
 - Create Word table showing valuation at different parameter combinations
- Insert chart: *task4_charts/chart_29_dcf_waterfall.png* (6 inches wide)

4. Comparable Companies section

- Add heading: 'Comparable Companies Analysis' (level 2)
- Extract 'Comparable Companies' section from Task 3 markdown
- Convert markdown to Word paragraphs
- **Create Table: Comparable Companies** ★ CRITICAL
 - Add heading: 'Comparable Companies' (level 3)

- Use XLSX skill to read Comparable Companies tab
- Extract full table including:
 - * 5-10 peer companies plus target company
 - * Statistical summary rows (Maximum, 75th Percentile, Median, 25th Percentile, Minimum)
- Create Word table with all columns: Ticker, Market Cap, EV/Revenue (LTM & NTM), EV/EBITDA (LTM & NTM), P/E (NTM), Revenue Growth, EBITDA Margin
- **Verify statistical summary is included in table**
- Insert chart: *task4_charts/chart_31_peer_multiples_comparison.png* (6 inches wide)

5. Valuation Summary

- Insert chart: *task4_charts/chart_32_valuation_football_field.png* (6.5 inches wide) ★ MANDATORY
- **Create Table: Valuation Summary**
 - Use XLSX skill to read Valuation Summary tab
 - Extract valuation methods (DCF, Comps, Precedent Transactions if applicable)
 - Create Word table showing: Method, Low Case, Base Case, High Case, Weight, Weighted Value

6. Price Target & Recommendation

- Add heading: 'Price Target and Recommendation' (level 2)
- Extract 'Price Target' section from Task 3 markdown
- Convert markdown to Word paragraphs
- Should include: Final recommendation (BUY/HOLD/SELL), price target with % upside, key catalysts, key risks

Result after Phase D: Pages 31-40 complete (~10 pages, 3-4K words, 5-6 charts, 4-5 tables)

Key Point: Use Read tool for Task 3's .md file to get written analysis, and use XLSX skill to READ from Task 3's Excel tabs (which were added to Task 2's model file) to create quantitative tables.

Phase E: Add Appendices & Finalize

Use Claude's DOCX skill to:

SECTION 5: Appendices (Pages 41-50)

1. Data Sources & References

- Add heading: 'Data Sources & References' (level 1)
- List all sources used throughout the report
- Organize by category:
 - SEC Filings (10-K, 10-Q, DEF 14A, 8-K with EDGAR links)
 - Earnings Calls (with transcript links)
 - Company Materials (investor presentations, press releases)
 - Industry Reports (Gartner, Forrester, etc.)
 - News Articles
- **CRITICAL:** All URLs must be clickable hyperlinks (not plain text)

- Include dates for all sources

2. Additional Tables

- Add heading: 'Additional Tables' (level 1)
- Add extended financial projections
- Add detailed assumptions tables
- Add any supporting tables that didn't fit in main sections

Phase F: Write Page 1 Investment Summary

NOW write Page 1 - after all analysis complete

- INITIATING COVERAGE header
- Rating box
- 3-4 detailed bullets synthesizing entire report
- Financial summary table
- Stock price chart

Phase G: Add Table of Contents & Page Numbers

- Auto-generate TOC
- Add page numbers to all pages

Key formatting requirements:

- Professional fonts (Calibri, Arial, or similar)
- Proper headers and footers with page numbers
- Section breaks between major sections
- Embed all 25-35 charts inline throughout text
- Insert all 12-20 tables inline with text
- **All URLs as clickable hyperlinks** (NOT plain text)
- **60-80% page density** - Every page has text AND visuals

Visual Density Strategy:

Structured Template

- Good page layout example:
- Section Header
- Text paragraph (200 words)
- [Chart embedded]

- Text paragraph (200 words)
- [Table embedded]
- Text paragraph (200 words)
- [Chart embedded]
- BAD - Avoid:
 - - Full page with only one chart
 - - Multiple pages of pure text
 - - Charts grouped at end of sections

Result: 30-50 page report that is text-dense with illustrating images throughout

File Operations Summary

Throughout the entire assembly process, use Claude's DOCX and XLSX skills with actual file operations:

Reading Input Files:

- ✓ Use Read tool: *[Company]_Research_Document_[Date].md* - Read Task 1 research
- ✓ Use XLSX skill: Open *[Company]_Financial_Model_[Date].xlsx* and read tabs - Extract tables from Task 2/3
- ✓ Use Read tool: *[Company]_Valuation_Analysis_[Date].md* - Read Task 3 analysis
- ✓ Use DOCX skill: Insert images from *task4_charts/chart_XX.png* files

Writing Output File:

- ✓ Use DOCX skill: Create new Word document
- ✓ Use DOCX skill: Add paragraphs (text from input .md files)
- ✓ Use DOCX skill: Create tables (data from Excel files read via XLSX skill)
- ✓ Use DOCX skill: Insert images (chart .png files)
- ✓ Use DOCX skill: Save final file as *[Company]_Initiation_Report_[Date].docx*

Do NOT manually copy/paste. Use Claude's built-in skills to:

1. Read from .md files (Task 1, Task 3) using Read tool
2. Read from .xlsx files (Task 2 with Task 3 tabs) using XLSX skill
3. Read from .png files (Task 4) as image files
4. Write to .docx file (Task 5 output) using DOCX skill

This approach is efficient, reproducible, and ensures all data flows correctly from source files to final report.

Step 7: Quality Check

Run comprehensive verification:

Structured Template

- REPORT QUALITY CHECKLIST
- LENGTH REQUIREMENTS:
 - - ☐ Report is 30-50 pages (count: ____ pages)
 - - ☐ Word count is 10,000-15,000 (count: ____ words)
 - - ☐ 25-35 charts embedded (count: ____ charts)
 - - ☐ 12-20 tables included (count: ____ tables)
- PAGE 1 FORMAT:
 - - ☐ “INITIATING COVERAGE” header present
 - - ☐ Thesis-focused title (not generic)
 - - ☐ Rating box complete with all elements
 - - ☐ Stock price chart (Figure 1) embedded
 - - ☐ 3-4 detailed bullets with ■ character
 - - ☐ Each bullet has **bold header** + 3-5 sentences
 - - ☐ Financial summary table included
 - - ☐ Years noted as “A” (actual) and “E” (estimate)
- SECTION WORD COUNTS:
 - - ☐ Investment Thesis: 800-1,200 words ✓
 - - ☐ Risk Assessment: 600-900 words ✓
 - - ☐ Company Description: 800-1,200 words ✓
 - - ☐ Management Bios: 1,000-1,400 words (300-400 per exec for 3-4 execs) ✓
 - - ☐ Products & Services: 700-1,000 words ✓
 - - ☐ Financial Analysis: 1,200-1,800 words ✓
 - - ☐ **Projection Assumptions: 2,000-3,000 words ✓ ★ CRITICAL**
 - - ☐ **Scenario Analysis: 1,500-2,000 words ✓ ★ CRITICAL**
 - - ☐ Growth Drivers: 800-1,200 words ✓
 - - ☐ Valuation Methodology: 800-1,200 words ✓
- MANDATORY CHARTS (4 TOTAL):
 - - ☐ Revenue by Product (stacked area) embedded ★
 - - ☐ Revenue by Geography (stacked bar) embedded ★

- - [] DCF Sensitivity (heatmap) embedded ★
- - [] Valuation Football Field embedded ★
- MANDATORY TABLES:
 - - [] Page 1 financial summary table
 - - [] Full income statement (40-50 line items)
 - - [] Revenue by product table (20-30 rows)
 - - [] Revenue by geography table (15-20 rows)
 - - [] Comparable companies table with statistical summary ★
 - - [] DCF assumptions table
 - - [] Scenario comparison table
 - - [] Additional 5-13 tables
- CITATIONS & HYPERLINKS:
 - - [] All figures have source lines
 - - [] All tables have source lines
 - - [] All URLs are clickable hyperlinks (NOT plain text)
 - - [] Test 5-10 random hyperlinks to verify they work
 - - [] Data Sources & References page included
 - - [] All sources have dates
- DATA ACCURACY:
 - - [] All numbers match financial model exactly
 - - [] Revenue figures consistent across all tables/text
 - - [] Price target matches valuation analysis
 - - [] All growth rates calculated correctly
 - - [] All percentages sum to 100% where applicable
- CONTENT REUSE (CRITICAL):
 - - [] Task 1 content used almost verbatim (not rewritten)
 - - [] Company 101 sections (pages 6-17) copied from Task 1 with only formatting changes
 - - [] Writing effort focused on quantitative sections (financial analysis, projections, scenarios)
- VISUAL DENSITY (CRITICAL):
 - - [] Every page has BOTH text AND visuals (not pure text pages)
 - - [] Charts interspersed throughout (not grouped at end)
 - - [] Average 1+ chart per page (30-50 pages = 25-35+ charts)

- - [] Charts appear every 200-300 words of text
- - [] 60-80% page density achieved across entire report
- **FORMATTING:**
- - [] No markdown syntax visible (no #, ##, **, etc.)
- - [] Professional fonts throughout
- - [] Headers and footers present
- - [] Page numbers present
- - [] Section breaks appropriate
- - [] Charts embedded (not just file paths)
- - [] Tables formatted professionally
- **WRITING QUALITY:**
- - [] Lead with numbers (not generic statements)
- - [] Use “vs.” not “versus”
- - [] Quantify everything
- - [] Professional tone throughout
- - [] No typos or grammatical errors
- - [] Specific examples (not vague statements)
- **FINAL VERIFICATION**
- IF ALL ITEMS CHECKED: ✓ READY FOR DELIVERY
- IF ANY ITEMS UNCHECKED: ✗ FIX BEFORE DELIVERY

IF ANY ITEM FAILS, DO NOT DELIVER. Fix before proceeding.

Writing Style Guidelines

Lead with Numbers (CRITICAL)

✓ **CORRECT:** “Revenue increased 150% YoY to \$250M in Q4 2024, driven by...” ✗ **INCORRECT:** “The company saw strong revenue growth this quarter...”

✓ **CORRECT:** “EBITDA margin expanded 500bps to 30% vs. 25% in FY2023” ✗ **INCORRECT:** “EBITDA margin expanded versus the prior year”

✓ **CORRECT:** “Market share increased 3 percentage points to 18% vs. 15% in 2023” ✗ **INCORRECT:** “Market share increased compared to last year”

✓ **CORRECT:** “Management expects 40-50% revenue growth in FY2025E” ✗ **INCORRECT:** “Management expects strong revenue growth”

Professional Writing Standards

- **Front-load:** Most important information first
- **Data-driven:** Lead with numbers and metrics
- **Specific:** Concrete examples, not generic statements
- **Objective:** Present facts, acknowledge risks
- **Confident:** State views clearly with supporting evidence
- **Active voice:** “We estimate revenue will reach \$500M”
- **Precise:** Avoid “might”, “could”, “possibly”

Number Formatting

Consistency:

- Billions: \$X.XB (e.g., “\$2.5B”)
- Millions: \$XXXM (e.g., “\$250M”)
- Always specify: YoY, QoQ, CAGR
- Basis points for small margin changes: “500bps”
- Year format: “2024A” (actual), “2025E” (estimate)

Use “vs.” not “versus”

✓ **CORRECT:** “Gross margin of 65% vs. 60% in prior year” ✗ **INCORRECT:** “Gross margin of 65% versus 60%”

Common Pitfalls to Avoid

⚠ **MOST COMMON MISTAKE: TAKING SHORTCUTS DUE TO LENGTH**

Many reports fail because they use placeholders like “details would be included here” or “see model for data” instead of actually writing/extracting the content. **DO NOT DO THIS.** Write every section in full. Extract every table. Embed every chart. Use whatever tokens are needed.

1. **Rewriting Task 1 content:** DO NOT rewrite the 6-8K words from Task 1. Use almost verbatim - just reformat and add charts. Focus writing effort on quantitative sections (projections, scenarios, valuation).
2. **Sparse pages:** Every page must have BOTH text AND visuals. Target 60-80% page density. Insert charts every 200-300 words.
3. **Grouping charts at end:** Charts must be interspersed throughout text, not grouped. Place chart immediately after paragraph discussing that topic.
4. **Writing in markdown:** Use DOCX format, NOT markdown
5. **Skipping Page 1 format:** Must follow exact institutional format

6. **Generic bullets:** Page 1 bullets need bold headers + specific data
 7. **Short sections:** Must meet minimum word counts
 8. **Thin assumptions:** Projection Assumptions MUST be 2,000-3,000 words with product-by-product and region-by-region detail
 9. **Vague scenarios:** Must have specific parameters for Bull/Base/Bear
 10. **Plain text URLs:** All citations must be clickable hyperlinks
 11. **Missing statistical summary:** Comps table must have max/75th/median/25th/min
 12. **Charts not embedded:** All 25-35 charts must be IN document, not just referenced
 13. **Numbers don't match model:** Verify all figures against source
 14. **Skipping verification:** Quality check is NOT optional
-

Success Criteria

A successful equity research report should:

1. **Meet all length requirements**
 - 30-50 pages (MINIMUM 30)
 - 10,000-15,000 words (MINIMUM 10,000)
 - 25-35 charts embedded
 - 12-20 tables included
2. **Have properly formatted Page 1**
 - “INITIATING COVERAGE” header
 - Rating box, analyst info, chart, bullets, table
3. **Meet all section word count minimums**
 - Especially Projection Assumptions (2,000-3,000) ★
 - And Scenario Analysis (1,500-2,000) ★
4. **Include all 4 mandatory charts**
 - Revenue by product (stacked area) ★
 - Revenue by geography (stacked bar) ★
 - DCF sensitivity (heatmap) ★
 - Valuation football field ★
5. **Have management bios**
 - 300-400 words each for 3-4 key executives
6. **Include comprehensive comps table**
 - With statistical summary (max/75th/median/25th/min)

7. **Have all citations as clickable hyperlinks**
 - Test multiple links to verify they work
 8. **Be professionally formatted**
 - 60-80% page density
 - No markdown syntax visible
 - Charts and tables embedded properly
 9. **Have numbers matching model exactly**
 - Verify all figures against Excel model
 10. **Enable informed investment decision**
 - Client should understand company, valuation, risks
 - Should be indistinguishable from JPM/GS/MS research
-

Output Files

Primary Deliverable: *[Company]_Initiation_Report_[Date].docx*

Example: *Tesla_Initiation_Report_2024-10-27.docx*

Supporting Deliverable: *[Company]_Financial_Model_[Date].xlsx* (from Task 2)

Both files should be packaged together for final delivery.

Final Note

This is the culmination of all equity research work from Tasks 1-4. The output should be:

- **Comprehensive:** 30-50 pages covering all aspects
- **Professional:** Indistinguishable from major investment bank research
- **Actionable:** Enables reader to make informed investment decision
- **Publication-ready:** Can be delivered directly to clients

Standard: JPMorgan, Goldman Sachs, Morgan Stanley institutional equity research.

Quality bar: Client-ready initiation report suitable for publication.

FINAL REMINDER: NO SHORTCUTS, NO COMPROMISES

Use whatever tokens are needed to deliver a complete, professional report.

This is not a draft. This is not a summary. This is not an outline. This is the **FINAL PUBLICATION-READY REPORT**.

- Write every section in full (10,000-15,000 words minimum)

- Embed every chart (all 25-35 charts throughout)
- Extract and include every table (12-20 tables minimum)
- Copy all Company 101 content from Task 1 verbatim (6-8K words)
- Write detailed projection assumptions (2,000-3,000 words)
- Write comprehensive scenario analysis (1,500-2,000 words)
- Achieve 30-50 pages minimum with 60-80% page density

If running low on tokens, that's expected and acceptable for this task. Keep going.

This represents the complete professional work product. Deliver institutional-quality research worthy of a \$1M+ investment decision.

3.27 Skill Resource: skills/initiating-coverage/references/valuation-methodologies.md

Valuation Methodologies for Equity Research

This reference document provides comprehensive guidance on the three primary valuation methodologies used in equity research: Discounted Cash Flow (DCF), Trading Comparables, and Precedent Transactions.

Table of Contents

1. [Discounted Cash Flow \(DCF\) Analysis](#)
 2. [Trading Comparables Analysis](#)
 3. [Precedent Transactions Analysis](#)
 4. [Valuation Reconciliation](#)
-

Discounted Cash Flow (DCF) Analysis

Overview

DCF analysis values a company based on the present value of its projected future cash flows. This is considered the most theoretically sound valuation method as it's based on fundamental value creation.

Step-by-Step DCF Process

1. Historical Financial Analysis

- Collect 3-5 years of historical financials
- Calculate historical FCF = $EBIT(1 - \text{Tax Rate}) + D\&A - \text{CapEx} - \text{Change in NWC}$
- Analyze historical growth rates and margins
- Identify trends and cyclicalities

2. Build Revenue Projections (5-10 years)

Approaches:

- **Top-down:** Start with market size (TAM) → Market share → Revenue
- **Bottom-up:** Units sold × Price per unit
- **Hybrid:** Combine multiple drivers

Key Considerations:

- Management guidance and historical growth
- Industry growth rates and market trends
- Competitive dynamics and market share evolution
- Product pipeline and new market opportunities
- Macroeconomic factors

3. Project Operating Expenses

- **COGS:** As % of revenue (analyze historical margins)
- **SG&A:** Often semi-fixed; model as % of revenue with scale effects
- **R&D:** Critical for tech/pharma; model as % of revenue
- **D&A:** Based on CapEx assumptions

Calculate EBIT = Revenue - COGS - Operating Expenses

4. Calculate Unlevered Free Cash Flow

Structured Template

- EBIT
- × (1 - Tax Rate)
- = NOPAT (Net Operating Profit After Tax)
- + Depreciation & Amortization
- - Capital Expenditures
- - Increase in Net Working Capital
- = Unlevered Free Cash Flow (UFCF)

CapEx Assumptions:

- Maintenance CapEx: Required to maintain current operations (typically 2-4% of revenue)
- Growth CapEx: Required for expansion
- Consider industry benchmarks and company guidance

Net Working Capital:

- $NWC = (\text{Accounts Receivable} + \text{Inventory}) - \text{Accounts Payable}$
- Model as % of revenue or days (DSO, DIO, DPO)
- An increase in NWC is a use of cash

5. Determine Terminal Value**Method A: Perpetuity Growth Method****Structured Template**

- $\text{Terminal Value} = FCF(\text{final year}) \times (1 + g) / (WACC - g)$
- g = perpetual growth rate (typically 2-3%, not exceeding GDP growth)
- Use when company has reached stable, mature growth

Method B: Exit Multiple Method**Structured Template**

- $\text{Terminal Value} = EBITDA(\text{final year}) \times \text{Exit Multiple}$
- Exit multiple based on current trading comps
- More appropriate for cyclical businesses

6. Calculate Weighted Average Cost of Capital (WACC)**Structured Template**

- $WACC = (E/V \times \text{Cost of Equity}) + (D/V \times \text{Cost of Debt} \times (1 - \text{Tax Rate}))$

Cost of Equity (using CAPM):**Structured Template**

- $\text{Cost of Equity} = \text{Risk-Free Rate} + \text{Beta} \times \text{Equity Risk Premium}$
- Risk-Free Rate: 10-year Treasury yield
- Beta: Regression of stock returns vs. market (or use comparable beta)
- Equity Risk Premium: Historical average ~5-6%

Cost of Debt:**Structured Template**

- $\text{Cost of Debt} = \text{Risk-Free Rate} + \text{Credit Spread}$
- Use company's current borrowing rate or implied from bonds
- Adjust for credit rating if no bonds outstanding

Capital Structure:

- $E/V = \text{Market value of equity} / \text{Total value}$
- $D/V = \text{Market value of debt} / \text{Total value}$
- Use target capital structure, not current (if significantly different)

7. Discount Cash Flows to Present Value

Structured Template

- $PV = \sum [FCF_t / (1 + WACC)^t] + [\text{Terminal Value} / (1 + WACC)^n]$

8. Calculate Enterprise Value and Equity Value

Structured Template

- Enterprise Value = PV of Projected FCF + PV of Terminal Value
- Less: Net Debt (Total Debt - Cash)
- Plus: Non-operating Assets
- Less: Minority Interest
- Less: Preferred Stock
- = Equity Value
- Price Per Share = Equity Value / Diluted Shares Outstanding

DCF Sensitivity Analysis

Always perform sensitivity analysis on key variables:

1. **Two-way sensitivity table:** WACC vs. Terminal Growth Rate
2. **Revenue growth scenarios:** Base / Bull / Bear cases
3. **Margin assumptions:** Operating leverage scenarios
4. **Terminal multiple sensitivity:** If using exit multiple method

Example Sensitivity Table:

Structured Template

- Terminal Growth Rate
- WACC 2.0% 2.5% 3.0%
- 8.0% \$45 \$48 \$52
- 9.0% \$40 \$43 \$46
- 10.0% \$36 \$39 \$41

Common DCF Pitfalls to Avoid

1. **Double-counting growth:** Don't project high growth without corresponding investment (CapEx, NWC)
 2. **Unrealistic terminal growth:** Should not exceed long-term GDP growth
 3. **Ignoring cyclical:** Normalize earnings for cyclical businesses
 4. **Wrong cash flow definition:** Use unlevered FCF, not net income
 5. **Inconsistent assumptions:** Match discount rate to cash flows (unlevered FCF → WACC)
-

Trading Comparables Analysis

Overview

Trading comps values a company based on how similar companies are valued in the public markets. This reflects current market sentiment and relative valuation.

Step-by-Step Comps Process

1. Select Comparable Companies

Selection Criteria:

- Same industry/sector (primary criterion)
- Similar business model and revenue streams
- Comparable size (market cap, revenue)
- Similar growth profile and margins
- Similar end markets and geographies

Typical Universe:

- Start with 8-15 companies
- Remove companies with unique circumstances
- Final set of 5-10 companies

2. Gather Financial Information

Required Data:

- Current stock price and shares outstanding
- Latest fiscal year financial statements
- Next-year (NTM) estimates from consensus
- Historical growth rates

Calculate Market Metrics:

- $\text{Market Cap} = \text{Share Price} \times \text{Shares Outstanding}$
- $\text{Enterprise Value} = \text{Market Cap} + \text{Debt} + \text{Minority Interest} + \text{Preferred} - \text{Cash}$
- $\text{Net Debt} = \text{Total Debt} - \text{Cash \& Equivalents}$

3. Calculate Valuation Multiples

Enterprise Value Multiples:

- **EV/Revenue:** Good for early-stage/high-growth companies
- **EV/EBITDA:** Most common; good for capital-intensive businesses
- **EV/EBIT:** Useful when D&A varies significantly

Equity Value Multiples:

- **P/E (Price/Earnings):** Most widely used
- **P/B (Price/Book):** Good for financial institutions
- **P/S (Price/Sales):** For unprofitable companies

Calculate for:

- Last Twelve Months (LTM) - historical
- Next Twelve Months (NTM) - forward-looking (preferred)

4. Analyze and Select Multiples

Create Comparable Company Table:

Company	Market Cap	EV/ Revenue	EV/ EBITDA	EV/ EBIT	P/ E (NTM)	Revenue Growth	EBITDA Margin
Comp A	\$10B	3.5x	12.0x	18.0x	22.0x	15%	28%
Comp B	\$8B	3.0x	10.5x	16.0x	19.0x	12%	27%
...	...	...	...	...	...	...	...
Median	-	3.2x	11.0x	17.0x	20.5x	13%	27.5%

Adjustments:

- Remove outliers (typically >2 standard deviations)
- Consider using median instead of mean (less affected by outliers)
- Weight multiples if some comps are more comparable
- Adjust for differences in growth, margins, risk

5. Apply Multiples to Target Company

Example Calculation:

Structured Template

- Target Company NTM EBITDA = \$500M
- Selected EV/EBITDA Multiple = 11.0x
- Implied Enterprise Value = $\$500\text{M} \times 11.0\text{x} = \$5,500\text{M}$
- Less: Net Debt = \$1,000M
- Equity Value = \$4,500M
- Shares Outstanding = 100M
- Implied Price Per Share = \$45.00

6. Select Appropriate Multiple

Choose based on:

- **EV/Revenue:** High-growth, unprofitable companies (tech, biotech pre-profit)
- **EV/EBITDA:** Most common; capital-intensive industries (manufacturing, telecom)
- **P/E:** Profitable companies with stable cap structure (consumer, retail)
- **Sector-specific:** P/B for banks, EV/Production for oil & gas, EV/Subscriber for media

Premium/Discount Analysis

Apply premiums or discounts based on:

- **Growth premium:** Higher growth → higher multiple
 - **Profitability:** Higher margins → higher multiple
 - **Size:** Larger companies typically trade at premium (liquidity)
 - **Market position:** Market leaders → premium
 - **Geographic:** Developed vs. emerging markets
-

Precedent Transactions Analysis

Overview

Precedent transactions values a company based on prices paid for similar companies in M&A transactions. This reflects control premiums and strategic value.

Step-by-Step Process

1. Identify Relevant Transactions

Selection Criteria:

- Same or similar industry
- Similar size (within 0.5x to 2x target's size)
- Similar business characteristics
- Recent transactions (last 3-5 years preferred)
- Announced and closed deals (avoid withdrawn deals)

Typical Universe:

- 5-10 transactions minimum
- Focus on recent deals (weight more recent higher)

2. Gather Transaction Details

Required Information:

- Transaction date (announcement and close)
- Acquisition price and structure (cash, stock, mixed)
- Target's financials at time of transaction
- Strategic rationale and synergies
- Control premium paid

Sources:

- SEC filings (S-4, 8-K, proxy statements)
- Press releases and investor presentations
- M&A databases (CapIQ, FactSet, Bloomberg)

3. Calculate Transaction Multiples

Same multiples as trading comps, but based on transaction value:

Structured Template

- Transaction Value = Equity Purchase Price + Assumed Debt - Cash Acquired
- EV/Revenue (LTM) = Transaction Value / Target's LTM Revenue
- EV/EBITDA (LTM) = Transaction Value / Target's LTM EBITDA
- EV/EBIT (LTM) = Transaction Value / Target's LTM EBIT

Calculate Control Premium:

Structured Template

- Control Premium = (Offer Price - Unaffected Price) / Unaffected Price
- Unaffected Price = Target's stock price 1-2 days before announcement
- Typical range: 20-40%

4. Analyze Precedent Transactions

Create Precedent Transactions Table:

Date	Target	Acquirer	Deal Value	EV/ Revenue	EV/ EBITDA	Premium	Rationale
Q1'24	CompX	BuyerA	\$5.0B	4.0x	14.0x	35%	Market consolidation
Q3'23	CompY	BuyerB	\$3.5B	3.5x	12.5x	28%	Strategic fit
Median	-	-	-	3.8x	13.0x	31%	-

5. Apply to Target Company

Important Considerations:

- Precedent multiples typically higher than trading comps (include control premium)
- Adjust for differences in transaction rationale
- Consider market conditions at time of transactions vs. current
- Weight recent transactions more heavily

Example Calculation:

Structured Template

- Target Company LTM EBITDA = \$450M
- Selected EV/EBITDA Multiple = 13.0x (precedent)
- vs Trading Comps Multiple = 11.0x
- Implied EV (Precedent) = $\$450M \times 13.0x = \$5,850M$
- Implied EV (Trading) = $\$450M \times 11.0x = \$4,950M$
- Implied Control Premium = $\$5,850M / \$4,950M - 1 = 18\%$

Adjustments to Transaction Multiples

Consider adjusting for:

- **Market conditions:** Bull vs. bear market (M&A activity levels)
- **Deal structure:** Strategic vs. financial buyer
- **Synergies:** Transactions with high synergies command premiums
- **Competitive dynamics:** Single vs. multiple bidders

- **Time value:** Older transactions less relevant

Valuation Reconciliation

Creating a Valuation Bridge

Present all three methods in a single framework:

Example Valuation Summary:

Method	Enterprise Value	Equity Value	Price/ Share	Weight	Implied Value
DCF Analysis	\$5,200M	\$4,200M	\$42.00	50%	\$21.00
Trading Comps	\$5,500M	\$4,500M	\$45.00	30%	\$13.50
Precedent Trans.	\$5,850M	\$4,850M	\$48.50	20%	\$9.70
Weighted Avg	-	-	\$44.20	-	\$44.20

Weighting the Methods

Typical Weighting:

- **DCF:** 40-60% (fundamental value, but assumes accuracy of projections)
- **Trading Comps:** 25-40% (reflects current market sentiment)
- **Precedent Trans.:** 15-25% (less relevant unless M&A likely)

Adjust weights based on:

- **Confidence in forecasts:** Higher confidence → higher DCF weight
- **Market conditions:** Bull/bear market affects comps reliability
- **M&A likelihood:** Higher if company in play or industry consolidating
- **Company maturity:** Mature companies → higher weight on comps; growth → higher DCF weight

Valuation Range

Always present a valuation range, not a point estimate:

Approach:

- **Base Case:** Most likely scenario
- **Bull Case:** Optimistic assumptions (revenue growth, margins)
- **Bear Case:** Conservative assumptions

Example:

Structured Template

- Bear Case: \$38 - \$40

- Base Case: \$42 - \$46
- Bull Case: \$48 - \$52
- Recommendation: BUY with target price of \$45 (midpoint of base case)

Sanity Checks

Cross-check valuation with:

1. **Historical multiples:** Is current valuation in line with history?
 2. **Peer comparison:** Justified premium/discount vs. peers?
 3. **Implied growth:** What growth is market pricing in?
 4. **Implied returns:** IRR from current price to target price
 5. **Market cap analysis:** Does total market cap make sense?
-

Conclusion

Using all three valuation methods provides a robust framework for determining fair value:

- **DCF** provides intrinsic value based on fundamentals
- **Trading Comps** reflects current market valuation
- **Precedent Transactions** indicates M&A value and control premium

The key is to understand the assumptions driving each method and to present a well-reasoned valuation range that considers multiple scenarios and methodologies.

3.28 Skill: model-update

Model Update

description: Update financial models with new data — quarterly earnings, management guidance, macro changes, or revised assumptions. Adjusts estimates, recalculates valuation, and flags material changes. Use after earnings, guidance updates, or when assumptions need refreshing. Triggers on “update model”, “plug earnings”, “refresh estimates”, “update numbers for [company]”, “new guidance”, or “revise estimates”.

Workflow

Step 1: Identify What Changed

Determine the update trigger:

- **Earnings release:** New quarterly actuals to plug in
- **Guidance change:** Company updated forward outlook

- **Estimate revision:** Analyst changing assumptions based on new data
- **Macro update:** Interest rates, FX, commodity prices changed
- **Event-driven:** M&A, restructuring, new product, management change

Step 2: Plug New Data

After Earnings

Update the model with reported actuals:

Line Item	Prior Estimate	Actual	Delta	Notes
Revenue				
Gross Margin				
Operating Expenses				
EBITDA				
EPS				
[Key metric 1]				
[Key metric 2]				

Segment Detail (if applicable):

- Update each segment's revenue and margin
- Note any segment mix shifts

Balance Sheet / Cash Flow Updates:

- Cash and debt balances
- Share count (buybacks, dilution)
- Capex actual vs. estimate
- Working capital changes

Step 3: Revise Forward Estimates

Based on the new data, adjust forward estimates:

	Old FY Est	New FY Est	Change	Old Next FY	New Next FY	Change
Revenue						
EBITDA						
EPS						

Key Assumption Changes:

- What assumptions are you changing and why?

- Revenue growth rate: old → new (reason)
- Margin assumption: old → new (reason)
- Any new items (restructuring charges, one-time gains, etc.)

Step 4: Valuation Impact

Recalculate valuation with updated estimates:

Valuation Method	Prior	Updated	Change
DCF fair value			
P/ E (NTM EPS × target multiple)			
EV/ EBITDA (NTM EBITDA × target multiple)			
Price Target			

Step 5: Summary & Action

Estimate Change Summary:

- One paragraph: what changed, why, and what it means for the stock
- Is this a thesis-changing event or noise?

Rating / Price Target:

- Maintain or change rating?
- New price target (if changed) with methodology
- Upside/downside to current price

Step 6: Output

- Updated Excel model (if user provides the existing model)
- Estimate change summary (markdown or Word)
- Updated price target derivation

Important Notes

- Always reconcile your estimates to the company's reported figures before projecting forward
- Note any non-recurring items and whether your estimates are GAAP or adjusted
- Track your estimate revision history — it shows your analytical progression
- If the quarter was noisy, separate signal from noise in your estimate changes
- Check consensus after updating — how do your revised estimates compare to the Street?

- Share count matters — dilution from stock comp, converts, or buybacks can materially affect EPS
-

3.29 Skill: morning-note

Morning Note

description: Draft concise morning meeting notes summarizing overnight developments, trade ideas, and key events for coverage stocks. Designed for the 7am morning meeting format — tight, opinionated, actionable. Triggers on “morning note”, “morning meeting”, “what happened overnight”, “trade idea”, “morning call prep”, or “daily note”.

Workflow

Step 1: Overnight Developments

Scan for relevant events across coverage universe:

Earnings & Guidance

- Any coverage companies reporting overnight or pre-market?
- Earnings surprises (beat/miss on revenue, EPS, key metrics)
- Guidance changes (raised, lowered, maintained)

News & Events

- M&A announcements or rumors
- Management changes
- Product launches or regulatory decisions
- Analyst upgrades/downgrades from competitors
- Macro data or policy changes affecting the sector

Market Context

- Overnight futures / pre-market moves
- Sector ETF performance
- Relevant commodity or currency moves
- Key economic data releases today

Step 2: Morning Note Format

Keep it tight — a morning note should be readable in 2 minutes:

[Date] Morning Note — [Analyst Name] [Sector Coverage]

Top Call: [Headline — the one thing PMs need to hear]

- 2-3 sentences on the key development and why it matters
- Stock impact: price target, rating reiteration/change

Overnight/Pre-Market Developments

- [Company A]: One-line summary of earnings/news + our take
- [Company B]: One-line summary + our take
- [Sector/Macro]: Relevant sector-wide development

Key Events Today

- [Time]: [Company] earnings call
- [Time]: Economic data release (expectations vs. our view)
- [Time]: Conference or investor day

Trade Ideas (if any)

- [Long/Short] [Company]: 1-2 sentence thesis + catalyst
- Risk: What would make this wrong

Step 3: Quick Takes on Earnings

If a coverage company reported, provide a quick reaction:

Metric	Consensus	Actual	Beat/ Miss
Revenue			
EPS			
[Key metric]			
Guidance			

Our Take: 2-3 sentences — is this good or bad for the stock? Does it change our thesis?

Action: Maintain / Upgrade / Downgrade rating? Adjust price target?

Step 4: Output

- Markdown text for email/Slack distribution
- Word document if formal distribution is needed
- Keep to 1 page max — PMs and traders won't read more

Important Notes

- Be opinionated — morning notes that just summarize news without a view are useless

- Lead with the most important thing — don't bury the headline
 - “No news” is a valid morning note — say “nothing material overnight, maintaining positioning”
 - Distinguish between actionable events (earnings, M&A) and noise (minor analyst notes, non-events)
 - Time-stamp your takes — if you're writing at 6am, note that pre-market may change by open
 - If you're wrong, own it in the next morning note — credibility matters more than being right every time
-

3.30 Skill: sector-overview

Sector Overview

description: Create comprehensive industry and sector landscape reports covering market dynamics, competitive positioning, key players, and thematic trends. Use for client requests, sector initiations, thematic research pieces, or internal knowledge building. Triggers on “sector overview”, “industry report”, “market landscape”, “sector analysis”, “industry deep dive”, or “thematic research”.

Workflow

Step 1: Define Scope

- **Sector / subsector:** What industry and how narrowly defined?
- **Purpose:** Client report, internal research, pitch material, idea generation
- **Depth:** High-level overview (5-10 pages) or deep dive (20-30 pages)
- **Angle:** Neutral landscape vs. thematic thesis (e.g., “AI infrastructure buildout”)
- **Universe:** Public companies only, or include private?

Step 2: Market Overview

Market Size & Growth

- Total addressable market (TAM) with source
- Historical growth rate (5-year CAGR)
- Forecast growth rate and key assumptions
- Market segmentation (by product, geography, end market, customer type)

Industry Structure

- Fragmented vs. consolidated — top 5 market share
- Value chain map — where does value accrue?
- Business model types (subscription, transaction, licensing, services)
- Barriers to entry (capital, regulatory, technical, network effects)

Key Trends & Drivers

- Secular tailwinds (3-5 major trends)
- Headwinds and risks
- Technology disruption vectors
- Regulatory developments
- M&A activity and consolidation trends

Step 3: Competitive Landscape

Company Profiles (for top 5-10 players):

Company	Revenue	Growth	EBITDA Margin	Market Share	Key Differentiator

For each company, brief profile:

- Business description (2-3 sentences)
- Strategic positioning and moat
- Recent developments (earnings, M&A, product launches)
- Valuation snapshot (P/E, EV/EBITDA, EV/Revenue)

Competitive Dynamics

- How do companies compete? (price, product, service, distribution)
- Who is gaining/losing share and why?
- Disruption risk from new entrants or adjacent players

Step 4: Valuation Context

- Sector trading multiples (current and historical range)
- Premium/discount drivers (growth, margins, market position)
- Recent M&A transaction multiples
- How does the sector compare to the broader market?

Step 5: Investment Implications

- Where are the best risk/reward opportunities?
- What thematic bets can be expressed through this sector?
- Key debates in the sector (bull vs. bear arguments)
- Catalysts that could change the sector narrative

Step 6: Output

- Word document or PowerPoint with:
 - Market overview and sizing
 - Competitive landscape map
 - Company comparison table
 - Valuation summary
 - Key charts: market growth, share trends, valuation history
- Excel appendix with detailed company data

Important Notes

- Source all market size data — cite the research firm or methodology
 - Distinguish between TAM hype and realistic addressable market
 - Sector overviews age fast — note the date and flag data that may be stale
 - Charts are essential — market size waterfall, competitive positioning matrix, valuation scatter plot
 - If for a client, tailor the “so what” to their specific situation (M&A target identification, competitive positioning, market entry)
-

3.31 Skill: thesis-tracker

Thesis Tracker

description: Maintain and update investment theses for portfolio positions and watchlist names. Track key data points, catalysts, and thesis milestones over time. Use when updating a thesis with new information, reviewing position rationale, or checking if a thesis is still intact. Triggers on “update thesis for [company]”, “is my thesis still intact”, “thesis check”, “add data point to [company]”, or “review my positions”.

Workflow

Step 1: Define or Load Thesis

If creating a new thesis:

- **Company:** Name and ticker
- **Position:** Long or Short
- **Thesis statement:** 1-2 sentence core thesis (e.g., “Long ACME — margin expansion from pricing power + operating leverage as mix shifts to software”)
- **Key pillars:** 3-5 supporting arguments
- **Key risks:** 3-5 risks that would invalidate the thesis

- **Catalysts:** Upcoming events that could prove/disprove the thesis (earnings, product launches, regulatory decisions)
- **Target price / valuation:** What's it worth if the thesis plays out
- **Stop-loss trigger:** What would make you exit

If updating an existing thesis, ask the user for the new data point or development.

Step 2: Update Log

For each new data point or development:

- **Date:** When this happened
- **Data point:** What changed (earnings beat, management departure, competitor move, etc.)
- **Thesis impact:** Does this strengthen, weaken, or neutralize a specific pillar?
- **Action:** No change / Increase position / Trim / Exit
- **Updated conviction:** High / Medium / Low

Step 3: Thesis Scorecard

Maintain a running scorecard:

Pillar	Original Expectation	Current Status	Trend
Revenue growth >20%	On track	Q3 was 22%	Stable
Margin expansion	Behind	Margins flat YoY	Concerning
New product launch	Pending	Delayed to Q2	Watch

Step 4: Catalyst Calendar

Track upcoming catalysts:

Date	Event	Expected Impact	Notes

Step 5: Output

Thesis summary suitable for:

- Morning meeting discussion
- Portfolio review
- Risk committee presentation

Format: Concise markdown or Word doc with the scorecard, recent updates, and current conviction level.

Important Notes

- A thesis should be falsifiable — if nothing could disprove it, it's not a thesis
 - Track disconfirming evidence as rigorously as confirming evidence
 - Review theses at least quarterly, even when nothing dramatic has happened
 - If the user manages multiple positions, offer to do a full portfolio thesis review
 - Store thesis data in a structured format so it can be referenced across sessions
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